Machine Learning Assisted Continuous-Variable Quantum Communication

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Machine Learning Assisted Continuous-Variable Quantum Communication

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A thesis in fulfilment of the requirements for the degree of

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Abstract

This thesis investigates how machine learning (ML) can assist state-of-the-art continuous-variable quantum key distribution (CV-QKD) protocols to advance a global Quantum Internet. Relative wavefront errors between multiplexed classical reference pulses and quantum signals pose a significant challenge to the practical implementation of a satellite-Earth CV-QKD network across turbulent atmospheric channels, potentially leading to the destruction of secure key transfer. ML is proposed as a solution, where ML-based wavefront correction algorithms are designed to mitigate the potentially deleterious effects of relative wavefront errors on CV-QKD, as a result of excess noise affecting the quantum signals. This thesis provides the first analysis of two-dimensional spatial relative wavefront errors. The potential causes of relative wavefront errors are explored, their evolution across satellite-Earth channels is simulated, and their effect on achievable secure key rates is evaluated. ML-based wavefront correction algorithms are designed to facilitate enhanced key rates by encoding the wavefront corrections onto a real local oscillator at the receiver. The analysis of relative wavefront errors presented in this thesis culminates in a series of experimental campaigns involving the transmission of multiplexed stronger reference signals and weaker signals across a 2.4 km free-space optical link, providing the first experimental evidence of relative wavefront errors in the context of CV-QKD. Modified ML-based wavefront correction algorithms are then applied to the experimental data, using phase retrieval, resulting in up to a 2/3 reduction in relative phase error variance. Overall, it is shown that ML algorithms can augment state-of-the-art CV-QKD protocols across turbulent atmospheric channels, thereby progressing the practical feasibility of developing a truly global CV-QKD network.

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Publications

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- [3] **Long, Nathan K**, Malaney, Robert, and Grant, Kenneth J, "Machine learning for phase estimation in satellite-to-Earth quantum communication," in *International Conference on Quantum Communications, Networking, and Computing*, IEEE, 2025, pp. 475–480.

- [4] **Long, Nathan K**, Wallis, John, Dix-Matthews, Benjamin P, Frost, Alex, Schediwy, Sascha W, Grant, Kenneth J, and Malaney, Robert, "Relative wavefront errors in continuous-variable quantum communication," in *International Conference on Quantum Computing and Engineering*, IEEE, 2025, pp. 510-511.

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Abbreviations

AM	Amplitude modulator
AOM	Acousto-optic modulator
BS	Beamsplitter
CCR	Corner cube retroreflector
Ch	Channel
CNN	Convolutional neural network
CV-QKD	Continuous-variable quantum key distribution
DFT	Discrete Fourier transformation
DL	Delay line
DM	Deformable mirror
DV-QKD	Discrete-variable quantum key distribution
EOM	Electro-optic modulator
FSO	Free-space optical
GMCS	Gaussian modulated coherent state
Het	Heterodyne detector
HG	Hermite-Gaussian
Hom	Homodyne detector
HWP	Half-waveplate
ICRAR	International Centre for Radio Astronomy Research
KNN	K-nearest neighbors
LDPC	Low-density parity-check
LO	Local oscillator
LS	Laser source
LSTM	Long short-term memory
MC	Mode cleaner
ML	Machine learning

MLP	Multilayer perceptron
MPLC	Multi-plane light converter
MSE	Mean squared error
NN	Neural network
NW	Nichrome wire
OA	Optical attenuator
OAM	Orbital angular momentum
PBS	Polarized beamsplitter
PD	Photodetector
PDF	Probability density function
PM	Phase modulator
Pol	Polarizer
PS	Phase shifter
PSK	Phase-shift keying
QAM	Quadrature amplitude modulation
QCRB	Quantum Cramér-Rao bound
QKD	Quantum key distribution
QWP	Quarter waveplate
RLO	Real local oscillator
Rx	Receiver
RQNN	Recurrent quantum neural network
SI	Scintillation index
SMF	Single-mode fiber
SNR	Signal-to-noise ratio
SNU	Shot noise unit
TLO	Transmitted local oscillator
TNN	Transformer neural network
Tx	Transmitter
UWA	University of Western Australia
WFE	Wavefront error

Introduction

The actualization of fault-tolerant quantum computing could render current encryption systems vulnerable to attack [7], demanding a new paradigm of encryption systems to overcome the security vulnerabilities introduced by quantum algorithms. Quantum key distribution (QKD) utilizes the principles of quantum mechanics to distribute secure keys between parties, offering unconditional information-theoretic security for encryption (e.g. [8]). Secure key information can be prepared using either discrete-variable quantum states of light — discrete-variable QKD (DV-QKD) — or continuous-variable quantum states of light — continuous-variable QKD (CV-QKD). While DV-QKD has been shown to provide improved robustness across high-noise channels (e.g. [9]), single photon detection in DV-QKD is highly susceptible to stray background photons, reducing its flexibility in the development of QKD networks across free-space optical (FSO) channels. CV-QKD, on the contrary, can be implemented using classical optical hardware, as information can be encoded on the phase and amplitude of the electric field quadratures of very weak optical signals [10]. However, CV-QKD is highly susceptible to phase noise, one of the biggest challenges in establishing an FSO CV-QKD network. As such, the reduction of phase noise in CV-QKD systems across turbulent atmospheric channels is the primary focus of this thesis. In particular, machine learning (ML) has shown potential as a tool to assist in phase noise reduction (e.g. [11], [12]), motivating further exploration in this thesis. The practical implementation of QKD across a satellite-Earth network could facilitate worldwide cyber protection via a global Quantum Internet.

1.1 Thesis Motivation

FSO CV-QKD often involves encoding secure key information on the electric field quadratures of weak quantum signals, before transmission across turbulent atmospheric channels. In order to measure the encoded information, a strong transmitted local oscillator (TLO) is commonly multiplexed with the quantum signals, such that they can be measured using balanced homodyne/heterodyne detection at the receiver [13]. However, the TLO can be manipulated by an eavesdropper, rendering secure key transmission insecure and potentially destroying the secure key information [14]. As such, common state-of-the-art CV-QKD protocols multiplex weaker reference pulses with the quantum signals, which are demultiplexed at the receiver, and the reference pulse phase error is measured. The reference pulse phase error is then encoded onto a real local oscillator (RLO) generated at the receiver, which is used to coherently measure the quantum signals [15], [16].

A first approximation assumes that the reference pulse phase errors caused by the channel are the same as the quantum signal phase errors, such that the quantum signal phase errors are annulled during coherent measurement using the RLO. However, previous works have shown that the reference pulses and quantum signals can experience relative phase errors between them, reducing their coherence during measurement [15], [17]. Relative phase errors have the potential to reduce, or even destroy, secure key rate transmission by contributing to phase noise affecting the quantum signal uncertainty. This thesis primarily focuses on the problem of relative phase errors between reference pulses and quantum signals in FSO CV-QKD across turbulent atmospheric channels. ML-based algorithms are proposed as a solution to estimate corrections to apply to an RLO at the receiver, thereby increasing coherence between the RLO and quantum signals. By increasing the coherence between the RLO and quantum signals, phase noise is reduced, thereby increasing the robustness of CV-QKD systems and improving the practical feasibility of establishing a global QKD network.

1.2 Thesis Structure

The remainder of this thesis is structured as follows.

In Chapter 2, we first provide an overview of CV-QKD, and an analysis of a variety of ML algorithms and methods, before exploring how ML has shown promise in alleviating the challenges involved in CV-QKD implementation. ML has been applied to almost every stage of CV-QKD protocols, including ML-assisted phase error estimation and correction, excess noise estimation, state discrimination, parameter estimation and optimization, key sifting, and key rate estimation. We provide a comprehensive analysis of the current literature on ML-assisted CV-QKD, compare the ML algorithms assisting CV-QKD with the traditional algorithms they aim to augment, and identify the key research gaps. A portion of the content in Chapter 2 is published as [1]:

- **Long, Nathan K**, Malaney, Robert, and Grant, Kenneth J, "A survey of machine learning assisted continuous-variable quantum key distribution," *Information: Quantum Information Processing and Machine Learning*, vol. 14, no. 10, 2023.

In Chapter 3, we explore an alternative to the traditional practice of measuring the reference pulse wavefront to apply a correction to an RLO, with the objective of reducing CV-QKD complexity in equipment requirements and deployment. We introduce a new method for estimating wavefront corrections using only intensity measurements from reference pulses, used as the input to a convolutional neural network, mitigating completely the need to use complex wavefront sensing hardware. We show that the wavefront correction accuracy needed to provide for non-zero secure key rates through satellite-to-Earth channels is achieved by the intensity-only measurements. Chapter 3 shows, for the first time, how ML algorithms can replace wavefront-measuring equipment in the context of CV-QKD delivered from space, thereby delivering an alternate deployment paradigm for this global quantum-communication application. A portion of the content in Chapter 3 is published as [2]:

- **Long, Nathan K**, Malaney, Robert, and Grant, Kenneth J, "Phase correction using deep learning for satellite-to-ground CV-QKD," in *Photonics for Quantum*, SPIE, vol. 13106, 2024.

In Chapter 4, we explore the speed and accuracy requirements of ML-based signal phase error estimation algorithms, with a focus on correcting temporal relative phase errors between reference pulses and quantum signals. Chapter 4 provides a framework to analyze long short-term memory neural network architecture parameterization, with respect to the quantum Cramér-Rao uncertainty bound of the signal phase error estimation, with a motivation of reducing the model complexity. More specifically, we demonstrate that signal phase error estimation can be achieved using low-complexity neural network architectures, without significantly sacrificing accuracy. The results could improve the real-time performance of practical CV-QKD systems deployed over satellite-to-Earth channels. A portion of the content in Chapter 4 is published as [3]:

- **Long, Nathan K**, Malaney, Robert, and Grant, Kenneth J, "Machine learning for phase estimation in satellite-to-Earth quantum communication," in *International Conference on Quantum Communications, Networking, and Computing*, IEEE, 2025, pp. 475–480.

In Chapter 5, we propose that spatial relative wavefront errors (WFEs) between reference pulses and quantum signals can exist, outline potential sources of relative WFEs, then examine several cases of relative WFEs via simulation. We introduce novel ML-based intelligent wavefront correction algorithms using transformer neural networks. We utilize multi-plane light conversion for decomposition of the reference pulses and quantum signals into the Hermite-Gaussian basis, then estimate the difference in Hermite-Gaussian mode phase measurements using the algorithms, reducing the impact of this problem. Through detailed simulations of the satellite-to-Earth channel, we demonstrate that the intelligent wavefront correction algorithms can identify and compensate for any relative WFEs that may exist, whilst causing no harm when WFEs are similar across both the reference pulses and quantum signals. We quantify the gains available in the algorithm in terms of the CV-QKD secure key rate. We show channels where positive secure key rates are obtained using our algorithms, while information loss without wavefront correction would result in null key rates. A portion of the content in Chapter 5 is published as [5]:

- **Long, Nathan K**, Wang, Ziqing, Dix-Matthews, Benjamin P, Frost, Alex, Wallis, John, Grant, Kenneth J, and Malaney, Robert, "Quantum wavefront correction via machine learning for satellite-to-Earth CV-QKD," *Advanced Quantum Technologies*, vol 9, no. 2, e00700, 2026.

In Chapter 6, we report on a series of experiments, in which strong and weak optical signals are polarization-multiplexed, simultaneously transmitted across a 2.4 km FSO link, and measured at the receiver using a multi-plane light converter. We find the presence of relative WFEs for all experimental campaigns, then develop ML-based intelligent wavefront correction algorithms, designed to estimate a correction to apply to an RLO at the receiver. The corrections result in up to a 2/3 reduction in relative phase error variance. We then discuss the implications of the findings on CV-QKD, where we analyze theoretical excess noise contributions from the relative WFEs in the context of CV-QKD. A portion of the content in Chapter 6 is published as [6]:

- **Long, Nathan K**, Dix-Matthews, Benjamin P, Frost, Alex, Wallis, John, Wang, Ziqing, Grant, Kenneth J, and Malaney, Robert,, "Relative wavefront error correction over a 2.4 km free-space optical link via machine learning," *Optics Letters*, vol. 51, no. 8, pp. 2092-2095, 2026.

Additionally, the following information is provided in the Appendices. In Appendix A, we outline the fundamental quantum optics theory, which underpins quantum information encoding in Gaussian modulated coherent state CV-QKD protocols. In Appendix B, we validate the phase screen model implemented in Chapters 3 to 5 against experimental results. In Appendix C, we include our initial lab-based experimental results on relative WFEs, where we show preliminary evidence of the effects of turbulence on relative WFEs. Note that a portion of Appendix C is published as [4]:

- **Long, Nathan K**, Wallis, John, Dix-Matthews, Benjamin P, Frost, Alex, Schediwy, Sascha W, Grant, Kenneth J, and Malaney, Robert, "Relative wavefront errors in continuous-variable quantum communication," in *International Conference on Quantum Computing and Engineering*, IEEE, 2025, pp. 510-511.

1.3 Thesis Contributions

Here, we summarize the contributions of this thesis to state-of-the-art CV-QKD.

1. A comprehensive survey on the literature on ML-assisted CV-QKD, which encompasses almost every step of state-of-the-art CV-QKD protocols, including a detailed discussion on the contemporary literature, and identification of research gaps, some of which are addressed in this thesis [1].

2. A method for estimating wavefront corrections to apply to an RLO, where a convolutional neural network architecture is designed to estimate reference pulse wavefronts using only reference pulse intensity information, thereby reducing hardware complexities involved in FSO CV-QKD implementation [2].

3. An investigation into neural network model complexity requirements for temporal signal phase error estimation, analyzing how long short-term memory neural network parameterization affects the neural network scale and estimation performance, using the quantum Cramér-Rao uncertainty bound as a benchmark [3].

4. An analysis of possible sources of spatial relative WFEs between reference pulses and quantum signals, in the context of CV-QKD across turbulent atmospheric channels, their potential effects on secure key rates, and the development of transformer neural network-based intelligent wavefront correction algorithms to correct for them [5].

5. An experimental campaign, where an optical reference beacon and weaker optical signal are polarization-multiplexed and transmitted across a 2.4 km FSO link, providing experimental evidence of the existence of relative WFEs between them. Transformer neural network-based intelligent wavefront correction algorithms are designed to correct for the relative WFEs, utilizing phase retrieval. The theoretical effects of the relative WFEs on excess noise are explored, in the context of CV-QKD across turbulent atmospheric channels. [6]

The primary conclusion of the work in this thesis is that ML is a powerful tool for assisting the correction of relative phase errors between reference pulses and quantum signals, after transmission across turbulent atmospheric channels — a fundamental challenge in implementing a global CV-QKD network. Further, we provide initial experimental evidence of the existence of relative WFEs in FSO communication, where we show that relative WFEs can have deleterious effects on secure key transfer in CV-QKD. We identify that these deleterious effects can be mitigated, where we propose ML-based intelligent wavefront correction algorithms as a solution. Together, the chapters in this thesis contribute to the advancement of ML-assisted CV-QKD, thereby advancing the feasibility of deploying CV-QKD over turbulent atmospheric channels, helping to lay the groundwork for a truly global and secure Quantum Internet.

Chapter 2

Background

This chapter outlines background information and methodologies on continuous-variable quantum key distribution protocols, as well as providing an overview of machine learning methods and algorithms. A comprehensive review of the literature on machine learning assisted continuous-variable quantum key distribution follows, covering Gaussian modulated coherent state continuous-variable quantum key distribution; discretely modulated continuous-variable quantum key distribution; parameter estimation and optimization; key sifting, and key rate estimation. Research gaps are identified, and a detailed critical discussion of the literature is presented. Note that a more in-depth mathematical description of quantum optics theory is provided in Appendix A, and that the nomenclature used in this chapter applies solely to the content within this chapter.

2.1 Introduction

Quantum key distribution (QKD) utilizes the principles of quantum mechanics to share a pair of secure keys between parties to encrypt sensitive data with information-theoretic unconditional security (when using a one-time pad). In continuous-variable QKD (CV-QKD), information is normally encoded on the quadratures of an electric field, which can be achieved using classical optical hardware and digital signal processing [7], making its implementation simpler than the alternative discrete-variable QKD. For example, the coherent measurement of CV quantum signals is possible using homodyne or heterodyne detectors [8].

Practical implementation of real-time CV-QKD is highly complex, with three major factors limiting its widespread implementation [18], [19]. Firstly, signal distortion due to the excess noise introduced by the channel and detector leads to a loss of information, threatening to reduce secure key rates below the null key threshold. Secondly, in fluctuating channels, effective parameter estimation and optimization need to be implemented if positive key rates are to be achieved. Thirdly, the computational time and power required to measure signals, filter out excess noise, and perform information reconciliation reduce the real-time capability of CV-QKD.

While many traditional approaches have been suggested to overcome the limitations on CV-QKD, machine learning (ML) has recently been shown to have advantages in terms of relative phase error estimation and excess noise filtering [11], [12], [20]–[33], state discrimination [34]–[37], parameter estimation and optimization [38]–[41], key sifting [42], and key rate estimation [43], [44]. ML-based phase error estimation and noise filtering algorithms offer the potential of improved filtering capabilities due to their ability to map complex relationships between inputs and outputs based on the data alone, without being based on idealistic models that may not represent reality. Likewise, the ability of ML models to adapt to different system conditions, without reliance on assumption-based traditional models, offers improved system parameter estimation and optimization capabilities. Further, traditional algorithms often rely on numerical searches to parameterize models, which can be more computationally complex than their ML-based counterparts.

This chapter aims to provide a background on CV-QKD and ML, as well as to present a comprehensive overview of the current literature on the use of ML algorithms to assist in achieving practical CV-QKD. Emphasis is placed on comparing the performance of the ML algorithms to their traditional counterparts when possible, such that the advantage of ML in assisting CV-QKD is highlighted.

In the following, ML algorithms adopted for different aspects of CV-QKD are contrasted with one another, and the differences in ML approaches to Gaussian modulated coherent state (GMCS) CV-QKD are compared with those of discrete-modulated (DM) CV-QKD. Beyond this, an analysis of the ML algorithms used for the two main channel types, free-space optical (FSO) and optical fiber, is undertaken, and how various common assumptions could affect the ML-based approaches is outlined.

Research on ML-assisted quantum attacks [45], [46], attack detection and prevention [41], [47]–[54], and methods for hacking ML-based attack prevention strategies [55], [56] is not included in the literature review in this chapter. Huang, Liu, and Zhang [57] have previously reviewed the literature in this area. We note that [46], [50]–[56] were published after [57], so were not included in their review. An analysis of literature on ML for information reconciliation in CV-QKD was undertaken in [58], and is not included in this chapter. A work [59] addressing ML-assisted quantum memory storage for CV-QKD is also not included in this literature review. Further, this chapter does not include works on ML for quantum communication in general, only addressing works specifically on the application of CV-QKD. The reader is referred to [60]–[64] for additional information on ML for CV quantum communication. Quantum attacks, attack detection, ML-based attack prevention, information reconciliation, quantum memory requirements, and quantum ML techniques are considered beyond the scope of this thesis.

The rest of this chapter is organized as follows: An overview of CV-QKD is provided in Section 2.2, followed by an introduction to ML techniques in Section 2.3. The application of ML to GMCS CV-QKD is outlined in Section 2.4; DM CV-QKD in Section 2.5; parameter estimation and optimization in Section 2.6; and key sifting and key rate estimation in Section 2.7. A detailed discussion is given in Section 2.8, and concluding remarks are provided in Section 2.9.

2.2 CV-QKD Overview

CV-QKD involves two legitimate parties: Alice (represented by subscript A), who prepares quantum signals at a transmitter, and Bob (represented by subscript B), who measures the signals at a receiver. A third party, Eve, is considered malicious with the objective to eavesdrop on the quantum channel and/or detector.

In the CV domain, quantum information is usually encoded on the electric field quadratures of light, where the quadrature values X and P represent the in-phase and out-of-phase components of the electric field. The coherent state, which can be labeled $|\alpha\rangle = |X + iP\rangle$, is usually adopted for the encoding. This state is defined as [65],

$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \tag{2.1}$$

in the basis of Fock states $|n\rangle$, where α is the amplitude and n is the photon number. The encoded quantum signal (henceforth referred to simply as the "signal") can be measured using homodyne or heterodyne detection at the receiver. Homodyne detection involves measurement of either the X or P quadrature, while heterodyne detection involves measuring both the X and P quadratures simultaneously. In what is to follow, unless otherwise stated, we will assume homodyne detection is used. The consideration of heterodyne measurements follows a similar path to what is given, with the simultaneous measurement of both quadratures incurring an additional noise penalty [10]. Please refer to Appendix A for a more in-depth derivation of coherent states, starting from the quantization of light.

Figure 2.1(a) illustrates the so-called phase-space diagram for encoding in coherent states, where we have defined the angle θ and amplitude α of the states as $\theta = \arctan(P/X)$ and $\alpha = (X^2 + P^2)^{1/2}$. We caution that the development of a "phase operator" in quantum optics has a somewhat complex history, and care must be used in interpreting the angle θ in regard to the classical phase ϕ of the electromagnetic wave. A phase operator can be developed that shows that $\theta \to \phi$ in the large photon limit (classical limit). When we later discuss the use of phase measurements on reference pulses in this chapter, we can consider that we are measuring ϕ of those pulses, with $\phi = 0$ being defined as the reference pulse phase *at the transmitter*. For further details on the relationship between θ and ϕ, the reader is referred to [65].

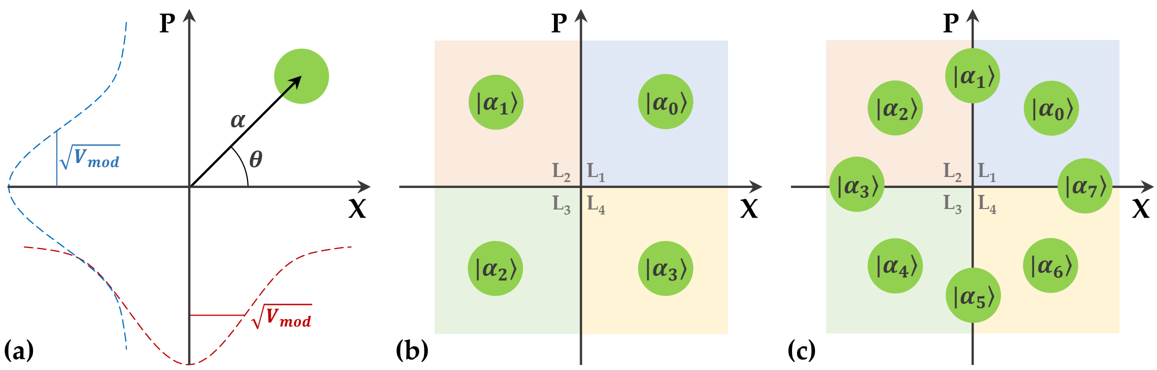

Figure 2.1: Coherent states encoded in electric field quadratures for (**a**) GMCS protocol showing α and θ, (**b**) DM quadrature PSK protocol, and (**c**) DM 8PSK protocol. The size of the circles represents the size of the vacuum noise.

CV-QKD protocols can, to a large extent, be separated into two main categories, GMCS CV-QKD and DM CV-QKD. GMCS CV-QKD involves randomly selecting quadrature values from independent Gaussian distributions with modulation variance V_{mod}, $X, P \in \mathcal{N}(0, V_{mod})$, which can be seen in Figure 2.1(a), where the Gaussian distribution for X is shown in red and the Gaussian distribution for P is shown in blue. Unlike

GMCS CV-QKD, DM CV-QKD involves encoding discrete values of the quadratures onto the signals. Encoding is usually achieved by modulating the quadratures through phase-shift keying (PSK) or quadrature amplitude modulation (QAM). In such encoding, a trade-off between increased information transfer and increased difficulty in measuring a larger pool of discretized states will be in effect. As example cases, two of the most common PSK protocols are the four-state quadrature PSK protocol, given in Figure 2.1(b), and the eight-state 8PSK protocol, given in Figure 2.1(c). It is important to note that GMCS CV-QKD security proofs are more mature than those of DM CV-QKD [14], [66]. However, DM CV-QKD has shown higher reconciliation efficiencies than GMCS CV-QKD at longer distances [67], [68] and at lower signal-to-noise ratios (SNRs) [69].

A generic CV-QKD protocol, which applies equally to GMCS and DM CV-QKD, involves the following steps:

1. Alice prepares and transmits states $|\alpha\rangle = |X_A + iP_A\rangle$ encoded on the signal across a channel (e.g. optical fiber or FSO).

2. Bob measures X_B or P_B of the signal using his homodyne detector.

3. Key sifting is performed, whereby Alice and Bob decide which variables are to be used for key generation, discarding any uncorrelated measurements.

4. Parameter estimation is undertaken to analyze the system parameters (transmissivity and excess noise), which can be determined from the amount of mutual information shared by Alice and Bob, as well as how much information Eve has access to.

5. Information reconciliation is carried out, in which, after the digitization of the symbols (using some pre-assigned scheme), an error correction code is used to correct differences in the keys held by Alice and Bob.

6. A confirmation protocol (usually via the use of hash functions) is used to bound the probability that the error correction has failed.

7. Finally, privacy amplification is performed on the keys, shortening their length, to reduce Eve's information on the key to a pre-assigned negligible level (again, usually via hash functions).

An overview of the generic CV-QKD protocol is given in Figure 2.2, where the current works on ML assistance for CV-QKD are highlighted at each step of the protocol.

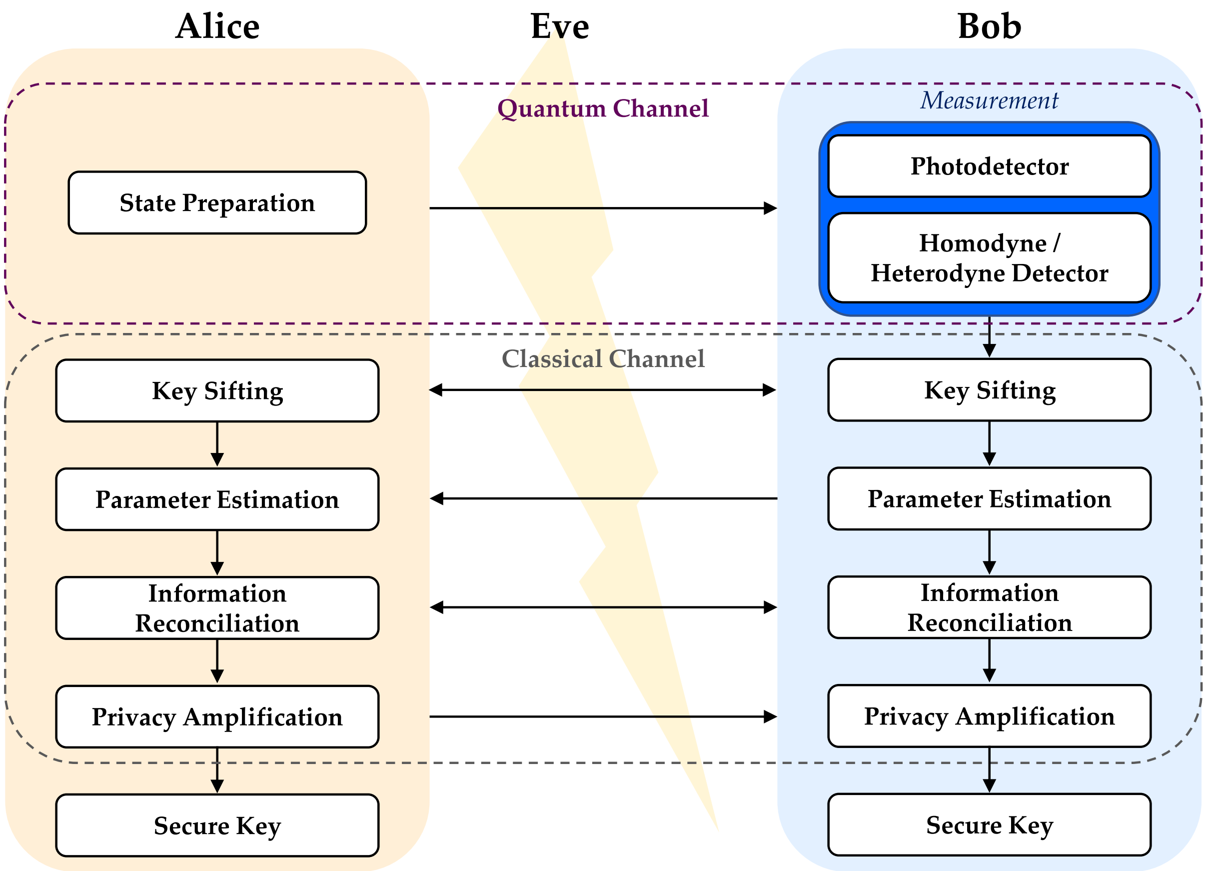

Figure 2.2: Generic CV-QKD protocol, where the works [11], [12], [20]–[37] applied ML to the measurement step, [42] applied ML to the key sifting step, [38]–[40], [70] applied ML to the parameter estimation and optimization step, and [43], [44] applied ML to the key rate estimation step.

Note in the above, a pre-shared secret key is used to authenticate all necessary classical communications. The presence of such an *a priori* shared key is why QKD is perhaps better described as a quantum key growing process.

2.2.1 Secure Key Rates

The aforementioned steps result in a final secure key rate R_{sec}, which is determined by two competing factors, the shared information I_{AB} between Alice and Bob, and the Holevo information χ_{BE} available to Eve. In the following analysis of secure key rates, we adopt a GMCS CV-QKD protocol, adapted from [10].

The shared information represents the amount of mutual information which can be transmitted between Alice and Bob, calculated as,

$$I_{AB} = \frac{\mu}{2} \log_2 \left(1 + \frac{\frac{\mu}{2} T V_{mod}}{1 + \frac{1}{\mu} \xi_{tot}} \right), \tag{2.2}$$

for the transmissivity T, total excess noise ξ_{tot}, and factor μ, which depends on whether the signal is measured using homodyne detection ($\mu = 1$) or heterodyne detection ($\mu = 2$).

The Holevo information represents the maximum amount of information Eve can ascertain about Bob's key. The Holevo information is essentially calculated as a function of the covariance matrix M_{AB}, which describes the quantum state ρ (a Bosonic multimode state). The diagonal entries of the covariance matrix represent the variances of the quadrature operators, while the off-diagonal terms represent the mutual covariance functions of the quadratures (see Appendix A for a derivation of the quadrature operators). For GMCS CV-QKD using homodyne detection, the covariance matrix is given by,

$$M_{AB} = \begin{pmatrix} (V_{mod}+1)\mathbb{1} & \sqrt{T(V_{mod}^2 + 2V_{mod})}\sigma_z \\ \sqrt{T(V_{mod}^2 + 2V_{mod})}\sigma_z & (TV_{mod} + 1 + \xi_{tot})\mathbb{1} \end{pmatrix}, \tag{2.3}$$

where $\mathbb{1}$ represents the identity matrix and σ_z represents the Pauli-z matrix, both of dimension 2×2. Using the shared information and Holevo information, the secure key rate can be calculated as,

$$R_{sec} = \beta I_{AB} - \chi_{BE}, \tag{2.4}$$

for the reverse reconciliation efficiency β. As can be seen, both the shared information and Holevo information are calculated as a function of the total excess noise, where the reduction of excess noise will increase the shared information between Alice and Bob, while decreasing the information available to Eve. Excess noise can arise from both channel noise and detector noise. A major contribution to channel noise comes from phase noise, which is caused by phase errors, the sources of which we discuss below. Note that the derivation of secure key rates here is limited, where more in-depth derivations are provided in Chapters 3 and 5, and the reader is referred to [10] for a comprehensive discussion of GMCS CV-QKD.

2.2.2 Phase Errors

For the purpose of the homodyne measurements for Bob, a transmitted local oscillator (TLO) is usually multiplexed with the signal for coherent measurement at the receiver. However, TLOs can be attacked and manipulated by Eve [71]. This has led to the use of real local oscillators (RLOs), also known as local local oscillators, which are phase-corrected using reference pulse information multiplexed with the signals. This latter approach using RLOs can be considered state-of-the-art for CV-QKD [16], [72].

Figure 2.3 illustrates a schematic of the preparation, transmission, and measurement of coherent states, including phase compensation using an RLO-based scheme, described as follows. At Alice, a laser source, L_A, is passed through a beamsplitter to generate the light source for the reference pulses and signals (coherent states). The (quantum) signals are encoded with randomly selected quadrature values, X_A or P_A, while the much stronger (classical) reference pulses are encoded with pre-assigned quadrature values (known publicly). The reference pulses and signals are then polarization-multiplexed using a beam combiner and transmitted across the channel to Bob, who de-multiplexes them. A laser source, L_B, at the receiver is split into two RLOs: RLO_1 and RLO_2. The first of these, RLO_1, is used to measure the reference pulses via heterodyne detection. The reference pulse quadrature measurements give us a first estimate $\Delta\phi_R$ of the signal phase error $\Delta\phi_S$, where $\Delta\phi_R$ is the phase difference between the transmitted and received reference pulse [15], [72]–[74].

This $\Delta\phi_R$ is then used to phase-correct the second RLO, RLO_2. Then, by combining the signal and the phase-corrected RLO_2 using a homodyne detector, one of the signal's quadratures, X_B or P_B, is measured (the signal and RLO_2 are delayed in this scheme). Note, in some literature, this method of phase-adjustment to the RLO_2 is termed "phase compensation".

The phase compensation technique described above represents but one type of implementation; however, it is important to realize that in essence all implementations have the same aim: the determination of the signal's quadratures corrected for errors incurred in the preparation, transmission, and measurement of the signal. In the context of CV-QKD, these errors in the signal's quadratures translate into phase noise, a contribution to total excess noise, the noise beyond vacuum fluctuations that impacts key rates negatively [10].

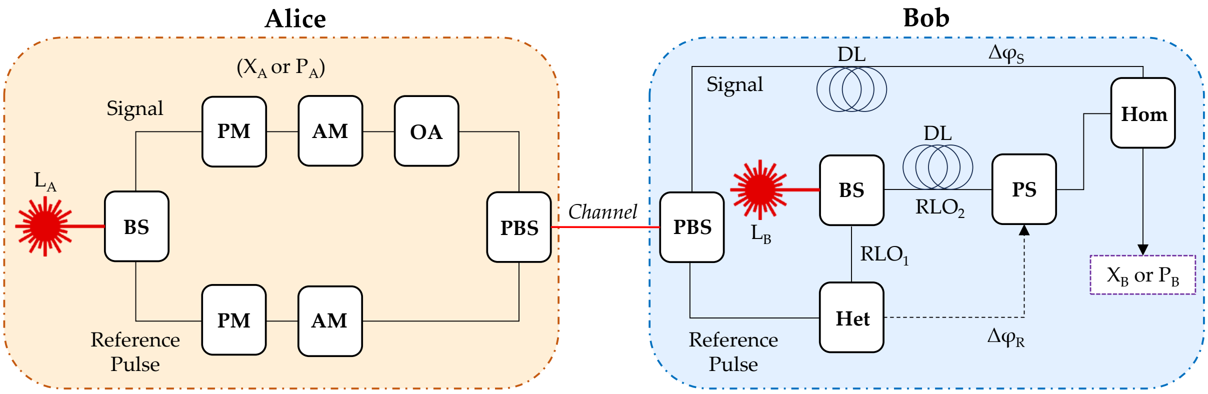

Figure 2.3: A schematic outlining the preparation and measurement of coherent states with phase compensation using reference pulses. L is a laser source, BS is a beamsplitter, AM is an amplitude modulator, PM is a phase modulator, OA is an optical attenuator, PBS is a polarized beamsplitter, Het is a heterodyne detector, Hom is a homodyne detector, PS is a phase shifter, and DL is delay line.

However, the measured $\Delta\phi_R$ does not exactly predict the phase error between the signal and RLO_1 — a relative phase error $\Delta\phi'$ between the signal and the RLO_2 exists, where $\Delta\phi' = \Delta\phi_R - \Delta\phi_S$. This relative phase error is due to several real-world limitations of the phase compensation method outlined above. Firstly, both the reference pulses and signals possess two-dimensional phase wavefronts, where the distortion of the wavefronts is not uniformly distributed. The resulting $\Delta\phi_R$ value is obtained after integrating across the reference pulse wavefront $\Delta\Phi_R(x, y)$. As such, phase correction of the RLO_2 wavefront $\Phi_{RLO_2}(x, y)$ using $\Delta\phi_R$ leads to inconsistencies across the wavefronts: $\Delta\Phi_{RLO_2}(x, y) \neq \Delta\Phi_R(x, y)$, which contributes to $\Delta\phi'$ [72], [74]. Other contributions to $\Delta\phi'$ are introduced by the independent laser sources L_A and L_B, through such factors as inaccurate clock synchronization and unequal spectral linewidths of the lasers [15], [73]. Furthermore, the reference pulses and signals could interact differently with the channel due to their different polarizations (such as via weak birefringence), adding to $\Delta\phi'$, as can contributions associated with the signal and RLO_2 propagating through delay lines [15], [73]. We note that the sources of $\Delta\phi'$ are modeled as independent from the secure information encoded on the signal quadratures (e.g. [15], [73]), thus estimation of $\Delta\phi'$ itself is assumed to not impact existing security proofs.

In the context of the literature reviewed in this chapter, we note that ML has been used for $\Delta\phi'$ correction at different stages of the signal quadrature measurement process, including being embedded in the system such that the $\Delta\phi'$ estimate is incorporated into the relative phase correction of the RLO [27], and the correction of the measured signal quadratures in post-processing using the estimated $\Delta\phi'$ [11], [12], [20]. In the rest of this chapter, we refer to these algorithms as *relative phase error correction* algorithms. Moreover, ML offers potential improvements in computational efficiency for key sifting and key rate estimation procedures, improving the feasibility of real-time CV-QKD.

We have not attempted to provide a complete methodology of CV-QKD here. A more in-depth explanation of relevant CV-QKD protocols and procedures can be found in the recent survey on CV-QKD [75], while the practical implementation of GMCS CV-QKD is outlined in [10], and a security analysis of DM CV-QKD is given in [66]. Next, we discuss the relevant ML algorithms in detail.

2.3 Machine Learning Methods

ML is a subclass of artificial intelligence, where we define ML as an algorithm designed to learn patterns from a data set, without being specifically programmed with a set of rules on how to do so.

ML encompasses a very wide variety of algorithm types, which have been categorized in several ways, sometimes with fuzzy distinctions between categories. This thesis does not present an in-depth review of different ML methodologies; we would recommend [76] for a survey on ML algorithms, their applications, and their future directions. Instead, we outline some key distinctions between ML algorithm types and then introduce some example algorithms to give readers with limited ML experience some perspective on the works presented in the background literature in this chapter.

The first distinction between ML algorithms is that they can either be supervised or unsupervised when learning from data. In supervised ML, data are split into training and testing data sets, whereby a model is constructed to map the relationship between known inputs and known outputs using the training data. The test data are then used as input to the trained model to test its output estimation performance. Unsupervised ML models are designed to analyze unlabeled data, discovering patterns in the data without being given example outputs, generally to cluster data, form associations, or perform dimensionality reduction.

Secondly, the output of an ML algorithm can either be continuous, forming regression algorithms, or discrete, forming classification algorithms. Further, both regression and classification ML algorithms can either incorporate the time evolution of a system or not when constructing relationships from the data, representing another distinction between ML algorithm types.

Neural networks (NNs) are a popular subclass of ML, themselves encompassing a wide range of algorithm types, which can be used for supervised or unsupervised learning, classification, regression, and time-series analyses. At their core, NNs are comprised of a series of layers, where weighted connections are formed between features in consecutive layers. Data are fed into an input layer, then a relationship is mapped to an output layer via one or more hidden layers. Deep learning is a subclass of NN, where the *depth* of the NN represents the number of hidden layers within it. The consensus on the exact number of hidden layers required for an algorithm to be considered *deep* is fuzzy, though we will define any NN with more than three hidden layers as a deep learning algorithm.

Figure 2.4 outlines a taxonomy of the ML algorithms included in the background literature on ML-assisted CV-QKD. As can be seen, all of the algorithm types fall under the title of ML, with regression and classification algorithms forming two distinct categories. The distinction between supervised and unsupervised ML algorithms is identified by the unsupervised algorithms written in *italics* (with supervised algorithms written in normal text). As mentioned, while NN and time-evolution algorithms are only presented within the regression category, their disconnect from classification applies only to the work contained within this survey. NNs can be used for both classification and time-evolution analyses.

A multi-layer perceptron (MLP) is a simple feedforward NN, which takes a vector as input. Perceptrons (nodes) in each layer of the MLP receive inputs x_i from the perceptrons i in the previous layer, manipulating the input data using an activation function $\psi(\cdot)$ to calculate the output of each perceptron σ_j as,

$$\sigma_j = \psi(\sum_{i=1}^{n} x_i w_{ij} + b_j), \tag{2.5}$$

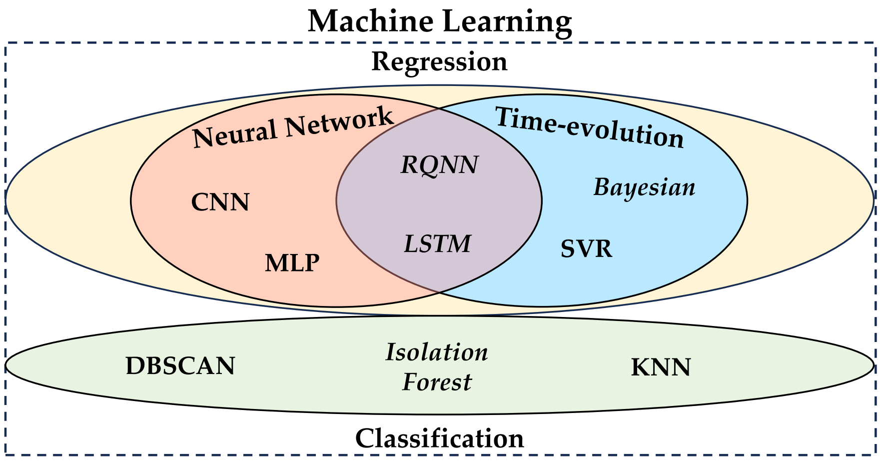

Figure 2.4: Taxonomy of ML algorithms used in the works in this survey. Regression and classification algorithms form distinct categories, while subclasses of NNs and time-evolution algorithms can fall within either category and overlap. Unsupervised ML algorithms are written in *italics*. Acronyms are defined in the following sections, while a summary of the ML algorithms applied in each of the surveyed works can be found in Table 2.1.

using the weight w_{ij} on the ij-th path, and a bias term b_j of the relative importance of the activation function. Backpropagation then passes error information back through the layers to update the network weights and biases during training. The error is calculated layer-by-layer, where a type of optimization (often gradient descent) is used to minimize a cost function (often mean squared error), improving the estimation results. Different types of activation functions are used for different purposes. For example, the rectified linear unit function is one of the most commonly used activation functions for hidden layers, where $\psi(x) = 0$ if $x < 0$, else $\psi(x) = x$ if $x \geq 0$. A linear activation function is commonly used in the output layer of NNs designed for regression, such that the output can take any continuous value, while sigmoid activation functions are commonly used for classification NNs to return discrete outputs. In the background literature, MLPs were applied to excess noise filtering in [26], parameter optimization in [39], [40], and key rate estimation in [43].

Figure 2.5(a) gives an example of a generic fully connected MLP, where the connections between the input layer, hidden layer, and output layer are shown, as well as an example perceptron in Figure 2.5(b), indicating how the inputs and weights from the previous layer are transferred into an output via the activation function (following the path outlined in red).

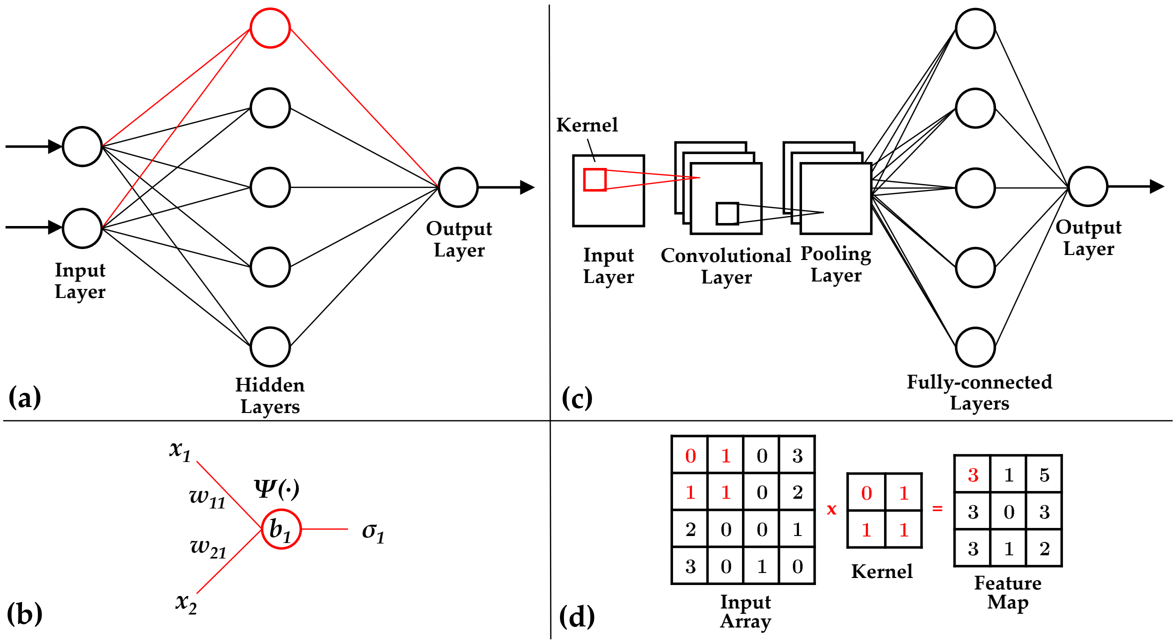

Figure 2.5: (**a**) Fully connected MLP schematic with (**b**) example perceptron outlining the inputs x_i, weights w_{ij}, biases b_j, activation function $\psi(\cdot)$, and outputs σ_j, isolating the path marked in red. (**c**) CNN schematic with (**d**) example convolution, isolating the kernel operation marked in red.

Convolutional neural networks (CNNs) are another common architecture type of NN, which take an n-dimensional array as input (usually two- or three-dimensional), such that spatial relationships between array elements are exploited. CNNs are generally constructed using three layer types: convolutional layers, pooling layers, and fully connected layers. In convolutional layers, a kernal (filter) is passed over a sub-array of the input data, sifting out a feature map of the relationships between elements. The pooling layers then decrease the size of the convolved feature map to reduce the network's computational complexity. Often, pooling layers are flattened into a vector to use as input for fully connected layers, which construct the model output. Fully connected layers are the same as hidden layers in an MLP, where each perceptron in each layer is connected to every perceptron in the next. Figure 2.5(c) gives an example of CNN architecture, outlining common layer types and the mapping of a two-dimensional array as input to a vector output. Figure 2.5(d) then gives an example convolution of an input array to a feature map. ML-assisted CV-QKD literature include CNNs developed for relative phase error correction in [27].

Note that the subsequent background literature on ML assistance for CV-QKD includes works using Bayesian inference methods combined with a type of Kalman filter (KF), defined as ML in [11], [20]–[25], as well as combined with particle smoothing, defined as ML in [30]–[32], though it does not include literature where Bayesian inference alone has assisted CV-QKD (as in [77]).

2.3.1 Regression

Encoder–decoder NNs are a type of supervised NN commonly used for regression. The encoder maps relationships from input data points to a reduced feature space; then, the decoder uses the reduced feature space to reconstruct a prediction of the output space. For example, image-to-image translation can be undertaken using encoder–decoder NNs, such as transforming a picture from a summer scene to a winter scene. Using this example, the common structure between an image in summer and winter would be encoded into the reduced feature space, while the adaptation to winter would be constructed in the decoder. CNNs are often used as encoder–decoder NNs, for which the architecture in Figure 2.5(c) could be used as the encoder; then, an inverse architecture could be used as the decoder. Note that encoder–decoder NNs can also be used for classification. ML-assisted CV-QKD literature used encoder–decoder NNs for excess noise reduction in [28].

2.3.2 Classification

One of the most common supervised ML classification algorithms is the k-nearest neighbor (KNN) algorithm. Test data points are connected to their k-nearest matching neighbors from the labeled training data by measuring their distances, then classified as being in the group containing the most neighbors. Distance-weighted KNN algorithms extend the original KNN methodology by including a weighting to each of the k-nearest neighbors, such that closer neighbors are prioritized. The value of k should be tuned to minimize misclassification. Turbulence strength classification using a KNN algorithm was undertaken in [26], while DM CV-QKD state classification using KNN algorithms was undertaken by [34], [35], [37]. Note that no works on ML-assisted CV-QKD utilized an NN for classification.

2.3.3 Time Evolution

Recurrent NNs represent a subclass of NN designed to store time-series data in a form of memory, such that dependencies can be mapped between the previous data inputs and the next output instance in the series. Essentially, a feedback loop is created, where the network weights and biases of previous time instances are fed into the activation function of the consecutive time instance, forming *recurrent* layers. A long short-term memory (LSTM) NN is a type of recurrent NN that incorporates a memory cell, which controls the information fed back into the network at consecutive time instances, retaining or discarding certain information to form long-term dependencies. While LSTMs are considered as unsupervised NNs, they have been described as *self-supervised* given that they learn relationships from the error of previous time instances, and they can be designed for both regression and classification. The works [28], [33] used an LSTM and a type of recurrent NN for excess noise filtering, respectively, while [12], [29] used an LSTM for relative phase error correction.

2.3.4 Unsupervised Learning

Isolation forest algorithms are a type of unsupervised learning ML algorithm used for anomaly detection (a type of classification). Isolation forest algorithms work by forming isolated trees, where the tree randomly selects a dimension from a dataset and then randomly splits the data along that dimension. Each of the new subspaces forms its own *sub tree*. This process is repeated until every data point has been isolated as its own *leaf* node; then, the whole process is repeated for multiple isolated trees, which form a *forest*. The algorithm traverses each tree, assigning a score to each leaf as a function of its depth in the tree, where it is assumed that anomalous data will be isolated along a shorter path length. The data point with the lowest composite score across all trees in the forest is output as the anomalous data point. The removal of abnormal data points for key sifting using an isolation forest algorithm was undertaken by [42].

2.4 Gaussian Modulated Coherent State CV-QKD

The primary application of ML for GMCS CV-QKD has been recovering the transmitted quadrature measurements X or P at the receiver. This has been achieved using three different approaches. The first was relative phase error correction, where an estimate $\Delta\phi'$ was obtained from the reference pulses and then used to phase-correct the RLO for the coherent measurement of the signal [27], [29]. The second was the correction of the signal quadratures in post-processing using $\Delta\phi'$ estimates [11], [12], [20]. The third was, more broadly, the excess noise filtering of the signal quadratures in post-processing [26], [28], not focusing specifically on the phase noise. Note that [12], [26], [27], [29] used regression-type ML algorithms for signal recovery for GMCS CV-QKD.

A Bayesian inference-unscented KF relative phase error correction technique was developed in [11]. A state space model of reference pulse evolution over time was implemented according to discrete Markovian dynamics, with an included $\Delta\phi'$ term. The Bayesian inference-unscented KF model was used to evaluate a probability distribution of $\Delta\phi_S$ at each time step, using the current and previous reference pulse heterodyne measurements and $\Delta\phi_S$ estimations, such that the distribution represented the $\Delta\phi'$ distribution, and the mean represented the optimal estimation of $\Delta\phi_S$. This process allowed the algorithm to update itself based on information from the reference pulses in real-time. The Bayesian inference-unscented KF method was compared to a *standard reference method* and an extended KF, where it was found to outperform both of them in terms of excess noise filtering (by reducing the phase noise term), particularly for low SNRs, which resulted in higher key rates. An extension of [11] was published in [20], where a joint polarization and relative phase error correction model was developed based on the same Bayesian inference-unscented KF approach. Higher key rates were achieved using the proposed ML algorithm than the presented traditional constant modulus algorithm. The relative phase error correction methodology of [11] was then applied to several works, including [21] for their work on CV-QKD across a 60 km optical fiber channel, then again across a 100 km optical fiber channel [22], as well as in another work on modulation leakage-free CV-QKD [23]. Further, the authors of [24] implemented the same methodology in their research on a practical CV-QKD approach with composable keys, and again in another work on modulator vulnerabilities in CV-QKD [25].

Another approach to relative phase error correction was developed in [27] using a CNN architecture. The input to the CNN was discussed as being the reference pulse quadratures, defined by an array of dimension 5×2, while the output was an estimate of $\Delta\phi_S$ to apply to an RLO, incorporating $\Delta\phi'$. Figure 2.6 gives a schematic of the measurement methodology of [27], showing how the CNN was incorporated into the measurement process. Of note, the authors incorporated the frequency offsets and phase jitter of L_A and L_B into their model of $\Delta\phi'$. The results from the CNN were compared with those attained using a KF, showing that the CNN estimated $\Delta\phi_S$ more accurately than the KF across a range of SNRs.

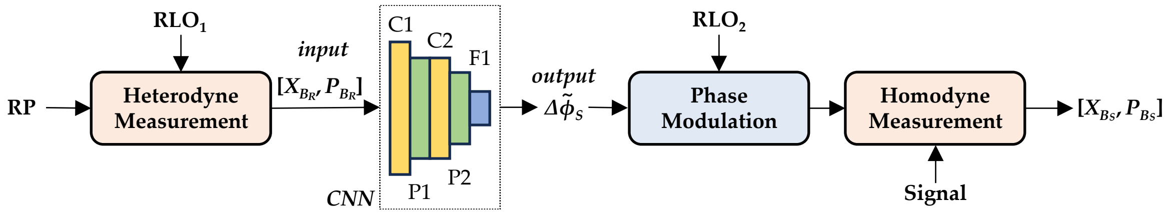

Figure 2.6: Relative phase error correction methodology from [27] using a CNN, where RP represents the reference pulse, $[X_{B_R},P_{B_R}]$ represent the reference pulse quadratures, $\Delta\tilde{\phi}_S$ is the estimated signal phase error, and $[X_{B_S},P_{B_S}]$ represent the signal quadratures. The CNN is comprised of convolutional layers C1 and C2, pooling layers P1 and P2, and the fully connected layer F1.

In order to compensate for signal distortion across an optical fiber channel, the authors of [12] developed an LSTM NN to learn the long-term dependencies of $\Delta\phi'$ over time, predicting $\Delta\phi'$ at current and future time steps. Unlike previous approaches, a TLO was used, where Bob measures the signal quadratures and then trains the NN using the quadrature measurements and historic $\Delta\phi'$ information. Then $\Delta\phi'$ estimations were sent to Alice, such that she could reconstruct her quadrature values. It was stated that the advantage of their proposed signal recovery methodology was maintaining the security of the CV-QKD system without requiring additional hardware or quantum resources. Key rates were calculated for the case of perfect $\Delta\phi'$ estimation, no $\Delta\phi'$ estimation, and using the LSTM $\Delta\phi'$ estimation method, where results showed the LSTM-based case approached the perfect $\Delta\phi'$ estimation case across a range of channel lengths, while the case of no $\Delta\phi'$ estimation performed significantly worse.

A relative phase error correction algorithm, based on a LSTM NN, was implemented experimentally for both a 20 km fiber-optic channel and a 1 km FSO channel in [29], using an RLO. The LSTM took previous $\Delta\phi_R$ measurements as input, then output an estimate of $\Delta\phi_S$ for the following time step. Three reference pulse powers were tested across the fiber-optic channel, -40, -45, and -50 dBm, where lower powers resulted in worse LSTM $\Delta\phi_S$ estimation performance. Further, results using the -45 dBm reference pulses showed information destruction in quadrature PSK before relative phase error correction, while four distinct PSK states are identifiable after relative phase error correction. A candle and fan were used to produce turbulence in the FSO experiments, where three different data sets were collected, with varying transmissivity strengths. Results showed poorer relative phase error correction performance when transmissivity values were lower, likely due to greater SNRs. Relative phase error variances for the LSTM were found to be approximately halved, when compared with relative phase error correction using an unscented KF.

A noise filtering methodology was developed in [28] using an LSTM and an autoencoder to estimate corrected quadrature measurements, where the primary emphasis was on noise introduced by the nonlinear imperfections of the balanced homodyne detector and consequent analogue-to-digital converter (termed inter-symbol interference reduction). When the received signal pulse repetition rates were lower than the homodyne detector's, the amplitude of the signal was maintained; however, as repetition rates increased, the lag of sequential signal pulses overlapped. Further, as the sampling frequency of the analogue-to-digital converter should match that of the signal pulse duration, the mismatch of the system bandwidth and sampling frequency led to nonlinear distortions of the quadrature measurements. Increased repetition rates greatly improved the secure key bit rate. The autoencoder was used as a noise filtering NN, where its encoder and decoder architectures were constructed using LSTM layers, such that the LSTM memory component could be used to harness the time-based relationships between signal pulses. The NN was trained on three different signal pulse repetition rates: 250 MHz, 500 MHz, and 1 GHz. While positive key rates were possible without the noise filtering algorithm for the 250 MHz case, the 500 MHz and 1 GHz cases resulted in null key rates. However, once the noise filtering NN algorithm was applied, excess noise was reduced below the null key rate threshold. The results indicated how ML can assist in increasing the secure key bit rate across a channel by compensating for noise introduced by CV-QKD hardware.

An equalization method, assisted by an MLP, was developed in [26] to reduce excess noise in the signal quadrature measurements across an optical fiber channel. The authors then applied their equalization methodology for an FSO channel, such that a KNN algorithm was used to classify turbulence strength; then, a different equalization MLP was trained for each turbulence class, where each NN had a different set of connection weights and perceptron biases. The results showed that signal quadrature measurement corrections were improved when the classification scheme was implemented, particularly for channels with medium-to-strong turbulence.

Relative phase error correction can be undertaken during the measurement process using digital signal processing and RLO modulation [27], [29], or using digital signal processing alone [11], [12], [20]. The use of hardware to modulate the RLO requires additional time for actuation, which can introduce its own errors, but leads to the measurement of signals with greater coherence. Digital signal processing can be applied in post-processing, where noise sources beyond relative phase errors can be filtered, and the time required for hardware actuation in the loop is removed. Thus, using digital signal processing alone could increase the real-time feasibility of a CV-QKD system. However, we note that errors in the measurement will be greater, and this could lead to less accurate relative phase error correction overall. When modulating the RLO, channel conditions should be taken into consideration when determining how to measure reference pulses for relative phase error correction. Signals that pass through atmospheric turbulence and experience uneven phase distortion across their wavefront should use a wavefront sensor to measure the reference pulses, such that the RLO wavefront can be heterogeneously modulated. However, if the assumption of even relative phase errors across the signal wavefront is applicable, then heterodyne detection can be used to measure the reference pulses. The aforementioned assumption could be applicable to FSO channels with weak turbulence.

2.5 Discretely Modulated CV-QKD

Similarly to GMCS CV-QKD, ML algorithms designed for excess noise filtering [33] and relative phase error correction [30] have been applied to DM CV-QKD, where the recovered signal was then classified as one of the discrete states used in the DM protocol (i.e., PSK or QAM states). However, unlike GMCS CV-QKD, classification ML algorithms have also been applied to state discrimination in DM CV-QKD, such that the output of the algorithms is the discrete state classification itself [34], [35], [37], as well as modulation format identification [36].

A distance-weighted KNN was used to classify states measured by Bob for a four-state quadrature PSK DM CV-QKD protocol in [34], as shown in Figure 2.1(b). Errors rates were compared against SNRs for the distance-weighted KNN approach and a traditional maximum likelihood classification scheme, where the KNN classification resulted in lower error rates (misclassifications) across the entire SNR range, with greater improvements at higher SNRs. Secure key rates were also calculated for varying channel lengths and modulation variances, with the ML-assisted classification model outperforming the traditional approach in each case. The improvement in secure key rates using the distance-weighted KNN was more pronounced for longer channels and greater modulation variance values.

A multi-label classification algorithm, based on KNN, was developed in [35] to classify 8PSK states. Each quadrant was given a label $L_i (i = 1, 2, 3, 4)$, where each discrete state uses a single label in quadrant PSK, as depicted in Figure 2.1(b). However, for 8PSK, states $|\alpha_1\rangle$, $|\alpha_3\rangle$, $|\alpha_5\rangle$, and $|\alpha_7\rangle$ simultaneously belong to two labels each, as shown in Figure 2.1(c). A feature vector of Euclidean distances to each of the 8PSK state quadrature locations was constructed for each received state to use as input to the classification algorithm. KNN was performed, and then posteriori probabilities were calculated for the condition of the received state belonging to a label, applying the label when a threshold probability was reached. The key rates calculated for a range of channel lengths and modulation variances were compared for the proposed 8PSK classification algorithm, as well as quadrature PSK, 8PSK, and GMCS CV-QKD, all without the classification algorithm. It was found that the proposed approach achieved higher key rates than each of the protocols without the classification algorithm, except across short channels (approximately

<10 km).

A quantum KNN algorithm was designed for state classification for 8PSK DM CV-QKD in [37]. This quantum KNN algorithm differs from a classical KNN by using quantum computation to simultaneously calculate the distances between an unclassified state and its KNNs. The algorithm then sorts the distances using an optimization technique based on the Grover's algorithm to find the closest neighbors. The authors outline a theoretical security proof of this specific quantum KNN algorithm using a semi-definite program method, where they show that less Holevo information was accessible to Eve than in a classical KNN algorithm. Secure key rates were compared for traditional quadrature PSK, 8PSK, and the proposed quantum KNN algorithm, with the quantum KNN algorithm having higher key rates across a range of modulation variances and channel-noise conditions. In [37], key rates were not calculated for a classical KNN algorithm applied to state classification in DM CV-QKD, though a complexity analysis demonstrated the increased computational efficiency attained by the quantum KNN algorithm when compared to a classical KNN. The work of [37] provides an in-depth definition of their methodology; however, it should be noted that the work is purely theoretical as quantum computation has not reached the level of maturity required to execute the algorithms.

One work [30] investigated a DM CV-QKD protocol without sending a reference pulse or TLO, only using an RLO and a noise filtering particle smoother model based on Bayesian inference, which filtered the signal quadrature measurements for $\Delta\phi_S$ estimation. The Bayesian particle smoother was trained using a Monte Carlo Markov chain method, which optimized the tracking of dynamic signal distortion using the set of keys shared between Alice and Bob for parameter estimation. Experimental and simulated data were used to test their noise filtering methodology for both binary PSK and quadrature PSK, with the cases of 0%, 5%, and 100% of the keys being used to train the Bayesian model. Using simulated data, excess noise was reduced by approximately an order of magnitude at low SNRs when 100% of keys were shared, compared to 5%, before converging on similar noise levels at higher SNRs. The results were volatile for 0% shared measurements at low SNRs. The experimental results showed that for low SNRs, the cases of 5% and 100% of shared measurements had similar noise filtering efficiencies, both improving upon the 0% case. The absence of a TLO or reference pulses in the protocol presented in [30] could help to mitigate security vulnerabilities introduced by their transmission, as well as reduce system complexity. Two works that used the same methodology were also published [31], [32].

While [30]–[32], [34], [35] applied ML to aid in state discrimination for fixed PSK state modulation schemes, the authors of [36] investigated a variable modulation scheme for increased network adaptability and receiver flexibility. A density-based spatial clustering of applications with the noise (DBSCAN) algorithm was implemented for modulation format identification, which could identify M-QAM protocols, assuming that Alice had not informed Bob of which modulation scheme was being used. DBSCAN classified the received states as being modulated using a 4-QAM, 16-QAM, 64-QAM, or 256-QAM scheme, where it was shown that higher M-QAM values resulted in higher key rates across a range of channel lengths. When comparing SNRs with the success rates of correct modulation format identification, lower M-QAM values could achieve higher success rates at lower SNRs. For example, 100% success rates were attained for 4-QAM at an SNR of ~10 dB, while 256-QAM required an SNR of ~23 dB. The accuracy of DBSCAN was found to increase as the number of training samples increased. One advantage of DBSCAN over other clustering techniques (such as KNN) was that the number of clusters was determined by the algorithm itself rather than being defined by the user.

Lu et al. [33] implemented a recurrent quantum NN (RQNN) to assist in coherent state $\psi(x,t)$ recovery for a two-state coherent polarization CV-QKD protocol, where the input to the RQNN was the signal quadrature measurements and the output was the filtered signal. It was assumed that the probability density function (PDF) of the Schrödinger equation, calculated as,

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi(x,t) + V(x,t)\psi(x,t), \tag{2.6}$$

could act as a stochastic filter of noisy signals, where $\hbar$ is the reduced Planck constant, m is mass, and (x,t) represent the spatial and temporal coordinates of the quantum state. The spatial potential $V(x,t)$ of the Schrödinger equation, given by,

$$V_j(x,t) = \phi_j(\nu(t))\left(\sum_{i=1}^{n} w_{ij}(x,t)\right), \tag{2.7}$$

was implemented as the activation function of each perceptron. The recurrent input of the RQNN was an expected error term $\nu(t)$, which was used to construct $V(x,t)$ via a Gaussian kernel function $\phi(\nu(t))$, where the connection weights between perceptrons were modified to update $V_j(x,t)$ for each perceptron j. The summed $V(x,t)$ from the output of the perceptrons in the second layer was used to update the Schrödinger equation, which was then used to calculate the PDF as $|\psi(x,t)|^2$. Noise parameters were calibrated at each time step for each noise source, constructing $\phi(\nu(t))$, which was stated as the core process of the RQNN. By adjusting the PDF, the RQNN decreased the expectation error, improving the accuracy of the filtered signal. The noise filtering performance of the RQNN was compared with a KF and an MLP. Similar noise filtering performance was found for all approaches, given unlimited computational time; however, the RQNN was able to achieve the same performance in far less time. Further, when compared with the MLP, the RQNN is an unsupervised ML algorithm and thus does not need to be trained on existing data beforehand and is able to adapt to unknown fluctuations in the channel. The ML classification of coherent polarization states has not been investigated, while more research should be undertaken to understand the advantages and disadvantages of encoding information using the polarization of coherent states, when compared to quadrature encoding.

2.6 Parameter Estimation and Optimization

Beyond the signal recovery and state discrimination methods given in Sections 2.4 and 2.5, effective parameter estimation and optimization can further improve secure key rates for CV-QKD, where ML has been used to assist in predicting the time evolution of intensity distributions [38], modulation optimization [39], and key estimation for parameter estimation [70].

A support vector regression (SVR) model was adopted in [38] to predict the time evolution of the intensity of the laser source at the transmitter and TLO at the receiver in their CV-QKD setup. Experimental intensity distributions were used to train and test the SVR model, with a radial basis function kernel used to fit the hyperparameters of the predictive function. The distorted TLO intensity predictions were then used to amplify or attenuate the TLO at the receiver to compensate for the predicted intensity fluctuations. Secure key rates were analyzed across a range of channel lengths for the proposed SVR-based dynamic TLO intensity modulation approach and a traditional stabilization technique. The SVR-based approach achieved higher key rates across all channel lengths, as well as increasing the maximum channel length where non-zero key rates are attained (up to approximately 50 km, compared to 45 km using the traditional method). The fluctuations in TLO intensity can be exploited by Eve to conceal attacks on the system, highlighting the need for accurate TLO intensity predictions to identify whether signal distortions are channel or hardware-based versus introduced by Eve.

A signal modulation technique in [39] utilized an MLP to optimize modulation variance for a DM CV-QKD protocol as a function of FSO channel conditions. Specifically, the Rytov variance (as a function of refractive index structure parameter C_n^2), transmissivity, and channel length were used as inputs to an MLP to map a relationship to the optimal modulation variance output for the varying channels, as outlined in Figure 2.7. A horizontal FSO channel (fixed C_n^2) was simulated, of length 0 to 11 km, with beam wandering, broadening, and deformation all accounted for in the calculation of transmissivity. Higher key rates were achieved when the MLP-based signal modulation optimization was applied, compared to a traditional local search algorithm, though it was not stated which local search algorithm was tested.

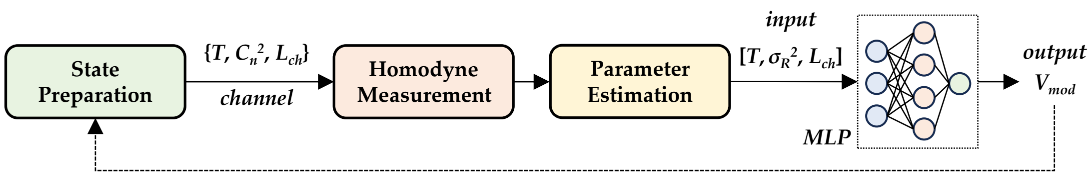

Figure 2.7: Modulation variance V_{mod} optimization from [39] using an MLP, where a feedback loop is formed as a function of channel conditions. C_n^2 represents the refractive index structure parameter, T represents transmissivity, σ_R^2 represents Rytov variance, and L_{ch} represents channel length.

The work [40] developed an adaptation of the ResNet-18 NN, termed the ResFCN-18, to estimate optimal modulation variances when undertaking satellite-to-Earth FSO GMCS CV-QKD. The convolutional layers in the ResNet-18 were replaced by fully-connected layers, termed the ResFCN-18. The ResFCN-18 took satellite altitude, channel length, satellite zenith angle, and signal excess noise as inputs, then output an estimation of the optimal modulation variance. An average estimation accuracy of 99.54% was reported, when compared with a local search algorithm, while the computational times were also reduced by 3-4 orders of magnitude.

The work [70] used an MLP to estimate Alice's key string by mapping the relationship between a proportion of Alice and Bob's normalized keys during training, then using the trained MLP to estimate the rest of the key. The objective was to use the estimated key for parameter estimation. Their results indicated that higher key rates could be achieved using their methodology than a claimed conventional scheme.

In general, ML for parameter estimation for FSO channels, where turbulence can lead to greater fluctuations of the channel parameters than in fiber optics, could be well-suited when channel conditions are highly volatile. Further, a combined approach to signal intensity and variance modulation could lead to improved key rates, rather than single-variable optimizations (as in [38]–[40]).

2.7 Key Sifting and Key Rate Estimation

ML algorithms have also been applied to key sifting [42] and key rate estimation [43], [44], with [43], [44] providing examples of improved computational efficiency of NNs over traditional approaches, while [42] gives an example of ML used for anomaly detection.

One method for key sifting in [42] used an isolation forest ML algorithm to sift out abnormal bits, affected by noise, from the raw key. After uncorrelated keys were discarded, the remaining bits were divided into equal-sized miniblocks to input to the isolation forest algorithm, which recursively split the data until each bit was isolated. An anomaly score was calculated as a function of each bit's height in the miniblock tree to calculate the probability of them being anomalous. Anomalies were introduced to 0.3% of miniblocks of size 10^3 bits, where noise was introduced to 20% of the anomalous miniblocks. Excess noise filtering was compared between the isolation forest method and a traditional Wiener filter for anomalous bit detection, where noise filtering was found to be similar for both methods across a range of channel distances up to 100 km. Key rates showed a similar trend, though the Wiener filter approach resulted in slightly higher key rates for most channel conditions. This work presented an example of a traditional algorithm outperforming ML in terms of anomaly detection, highlighting the need not to blindly apply ML algorithms to all problems by assuming that they will achieve the highest performance.

Calculating secure key rates traditionally involves numerically minimizing a convex function over all possible eavesdropping attacks. The authors of [43] developed an ML-assisted key rate estimation methodology for the purpose of reducing computational time and power requirements, improving the real-time practicality of CV-QKD. A minimization function, designed to optimize the bipartite density matrix shared between Alice and Bob, was encoded as the loss function of an MLP, where the objective of the loss function was to maximize key rate security. The MLP was applied to a quadrature PSK DM CV-QKD protocol, with the input vector containing 29 variables corresponding to density matrix elements, optimization restrictions, and an excess noise term; then, a secure key rate was output. The input vectors were generated for a range of different experimental parameters: channel length, signal intensity, excess noise, and state probabilities. It was shown that calculation time was reduced by up to six orders of magnitude when comparing the MLP to the traditional convex function minimization. Key rates were found to be lower for the MLP-based key rate calculation than for the traditional numerically minimized convex function, though secure probabilities of up to 99.2% were reported, being too insecure for practical CV-QKD. However, the authors stated that, given enough training data, the secure probability could approach traditional key rate estimation methods (in much less time). Taking the methodology of [43] a step further, another work [44] used a Bayesian optimization technique - a tree-structured Parzen estimator — to optimize the structure and hyperparameters of the architecture of a key rate estimation MLP. The output key rates had secure probabilities of 99.15% for quadrature PSK using heterodyne detection and 99.59% using homodyne detection. The optimized MLP architecture was able to reduce key rate calculation times by up to eight orders of magnitude when compared with the traditional convex function minimization technique, showcasing the primary benefit of the included Parzen estimator. Higher security probabilities should be validated before ML is implemented for key rate estimation.

2.8 Discussion

Overall, the primary objective of assisting CV-QKD with ML was to increase secure key rates, improving the practical implementation of CV-QKD. In general, the results from the literature presented in this chapter found that, when compared with their traditional counterparts, ML was better able to compensate for unknown conditions when recovering transmitted signal information by performing relative phase error correction, excess noise filtering, discriminating states, estimating and optimizing CV-QKD system parameters, and compensating for key errors.

However, ML should not be considered as an optimal solution to all problems, as shown by the example of a traditional algorithm showing improved results over the ML-based approach [42]. Some works also did not compare their ML methodologies with traditional approaches [12], [26], [28], [30], [35], [36], [38], instead focusing on improvements compared to the use of alternative ML algorithms or no algorithm. It should also be noted that the most state-of-the-art traditional methods may not have been tested, though equally, the best ML models may also not have been applied.

An overview of the literature on ML assistance for CV-QKD is given in Table 2.1, outlining each work's objective, the ML algorithm(s) applied, the algorithm comparisons, the channel type, and types of the analyses (further discussed below).

Consideration should also be given to the applied assumptions in the presented works analyzing the benefits of ML assistance for CV-QKD, where the performance of the ML algorithms could vary as system fidelity is improved. Moreover, the architecture of the ML algorithms should be constructed deliberately, where an approach of *bigger is better* does not automatically apply.

Table 2.1: Literature summary.

Work	Objective	ML Alg.	Algorithm Comp.	Channel Type	Analyses
[11]	$\Delta\phi'$ correction	Bayesian inference & unscented KF	Standard reference method, extended KF	Optical fiber	Asymptotic key rate, time-domain, experimental and simulation
[12]	$\Delta\phi'$ correction	LSTM	-	Optical fiber	Finite key rate, time-domain, experimental
[20]	$\Delta\phi'$ correction	Bayesian inference & unscented KF	Constant modulus algorithm	Optical fiber	Asymptotic key rate, time-domain, experimental
[21]	$\Delta\phi'$ correction	Bayesian inference & unscented KF	-	Optical fiber	Asymptotic key rate, time-domain, experimental
[22]	$\Delta\phi'$ correction	Bayesian inference & unscented KF	-	Optical fiber	Asymptotic key rate, time-domain, experimental
[23]	$\Delta\phi'$ correction	Bayesian inference & unscented KF	-	Optical fiber	Asymptotic key rate, time-domain, experimental

[24]	$\Delta\phi'$ correction	Bayesian inference & unscented KF	-	Optical fiber	Asymptotic and composable key rate, time-domain, experimental
[25]	$\Delta\phi'$ correction	Bayesian inference & unscented KF	-	Optical fiber	Asymptotic key rate, time-domain, experimental
[27]	$\Delta\phi'$ correction	CNN	KF	Optical fiber	No time-domain, simulation
[29]	$\Delta\phi'$ correction	LSTM	KF	Optical fiber, FSO	Asymptotic and finite key rate, time-domain, experimental
[28]	Noise filtering	LSTM & autoencoder	-	Optical fiber	Finite key rate, time-domain, simulation
[26]	Noise filtering	KNN & MLP	-	Optical fiber, FSO	Asymptotic key rate, time-domain, experimental and simulation
[34]	State classification	Distance-weighted KNN	-	Optical fiber	Finite key rate, no time-domain, simulation

[35]	State classification	Multi-label classification algorithm (KNN)	-	Optical fiber	Asymptotic key rate, no time-domain, simulation
[37]	State classification	Quantum KNN	Quadrature PSK, 8PSK	Optical fiber	Asymptotic key rate, no time-domain, simulation
[30]	Noise filtering	Bayesian inference + particle smoother	-	Optical fiber	Asymptotic key rate, time-domain, experimental and simulation
[31]	Noise filtering	Bayesian inference + particle smoother	-	Optical fiber	Asymptotic key rate, time-domain, experimental
[32]	Noise filtering	Bayesian inference + particle smoother	-	Optical fiber	Asymptotic key rate, time-domain, experimental
[36]	Modulation format identification	DBSCAN	KNN, BIRCH, and CLARANS	Optical fiber	No time-domain, simulation
[33]	Noise filtering	RQNN	KF, MLP	FSO	Time-domain, experimental

[38]	Parameter estimation	SVR	-	Optical fiber	Finite key rate, time-domain, and experimental
[39]	Parameter optimization	MLP	-	FSO	No time-domain, simulation
[40]	Parameter optimization	MLP	Local search	FSO	No time-domain, simulation
[70]	Parameter estimation	MLP	Conventional scheme	Optical fiber	Asymptotic key rate, no time-domain
[42]	Key sifting	Isolation forest	Wiener filter, COPOD, HBOS, LOF, KNN, MCD, ABOD, and PCA	Optical fiber	Finite key rate, no time-domain, simulation
[43]	Key rate estimation	MLP	-	Optical fiber	Asymptotic key rate, no time-domain, simulation
[44]	Key rate estimation	MLP + Parzen estimator	-	Optical fiber	Asymptotic key rate, no time-domain, simulation

2.8.1 Assumptions

The consideration of the time-domain changes the type of ML model used to assist in CV-QKD. While some literature investigated independent instances of the signal propagation and detection [27], [34]–[36], in reality, there would be time-dependent variations in channel and detection parameters, as well as varying signal modulation. The inclusion of the time-domain provides additional information to learn from, likely resulting in higher accuracy model predictions. As such, several of the reviewed works implemented ML models that learned from the time evolution of the system [11], [12], [20]–[26], [28]–[33], [38].

The channel type has a big impact on the noise affecting the signals, where the two main types are atmospheric FSO and optical fiber-based. The majority of literature in this survey involved signal transmission across simulated [27], [28], [34]–[36], [42]–[44], [70] or experimental [11], [12], [20]–[26], [29]–[32], [38] optical fiber channels, with simulated FSO channels explored in [39], [40] and experimentally in [26], [29], [33]. For example, turbulence fluctuations are highly volatile in atmospheric FSO CV-QKD, where ML-based algorithms designed to compensate for noise could prove more adaptable to unknown channel conditions than traditional model-based approaches.

Another commonly applied assumption was whether key rates were calculated at the asymptotic limit [11], [20]–[26], [29]–[32], [35], [43], [44], [70], or if finite [12], [28], [29], [34], [38], [42] or composable effects [24] were taken into account, though this factor may not directly influence the choice of ML algorithm used.

2.8.2 ML Architecture

The selection and design of ML algorithms requires the careful consideration of their application, input and output variables, and modeled system complexity. While this applies to all ML algorithms, we will focus on the use of NNs in this discussion, being the most common category of ML algorithm used in the presented works.

When the input of an NN model is a single value or short vector, such as the measured quadrature X or P values, and the output is a single value, such as the filtered X or P value, then a simple NN structure should suffice. For example, [26], [39] only had one hidden layer in their MLP architectures, with a single value output. Models considering the time evolution of a system should inevitably be more complex, given that they require historic information to be utilized, such as using LSTM layers in [28] as the encoder and decoder, or the two recurrent layers used in [33]. Increasing the channel complexity, such as for atmospheric channels when compared to optical fiber channels, could also warrant increasing the complexity of the NN architecture, as the variation in signal distortions is greater and caused by a more complex process. An example could be for wavefront correction, where a relationship between an input spatial distribution and an output spatial distribution should be found, requiring a more complex NN model architecture. Further, signals propagated across vertical atmospheric channels, which vary as a function of altitude, could increase the complexity of the relationship between the input and output distributions, such as for satellite-Earth channels.

The NN architecture in [33] offers an example of an ML algorithm designed specifically for its application, where the NN architecture was designed to update the Schrödinger equation by encoding the spatial potential as the activation function, which could potentially improve the modeling of a quantum system's evolution. On the contrary, Xing et al. [27] did not explicitly define their input variables for their CNN, other than its array shape of 5×2, making it difficult to replicate their results and understand their choice of ML algorithm.

Given that the feasibility of widespread CV-QKD relies on its real-time implementation, the complexity of the trained NN model should also be considered. Increasing the complexity of the NN increases the computational requirements to output results, reducing the real-time capabilities of the ML-based approaches.

A final note on the ML algorithms utilized for GMCS CV-QKD versus DM CV-QKD: although continuous quadrature values are encoded in GMCS CV-QKD and discrete quadrature values are encoded for DM CV-QKD, the distortion of the signals is continuous. For this reason, regression ML algorithms were applied to GMCS CV-QKD, while both classification ML algorithms [34]–[36] and regression ML algorithms [30]–[33] were applied to DM CV-QKD. Classification was used for state discrimination, while regression was used in the same manner as GMCS CV-QKD, correcting the distorted signals, before classifying them using traditional techniques. As such, the regression ML algorithms applied to DM CV-QKD could equally be applied to GMCS CV-QKD.

2.8.3 Key Research Gaps

In general, there are few works on ML-assisted CV-QKD for FSO channels [26], [29], [33], [39], [40], even fewer with experimental results [26], [29], [33], and only one satellite-based work [40]. In order to implement a global QKD network, satellite-based FSO CV-QKD could prove to be the most efficient approach. As such, further investigation of ML to assist in signal recovery and parameter estimation across complex atmospheric FSO channels should be undertaken to better understand its feasibility and limitations.

Although valid methodologies for achieving CV-QKD can be defined without analyzing the time evolution of the signal or channel and hardware parameters, which would be ideal for situations where system parameter variation is time-independent, in reality, there would be some time-dependence. Further work should be undertaken to assess the advantages of applying ML algorithms, which incorporate time evolution.

Given that one of the speculated advantages of ML algorithms is the reduction of computational time and power, when compared to their traditional counterparts, we recommend that more work should be done to analyze the computational requirements of the ML algorithms, as well as linking those requirements to the practical requirements of CV-QKD hardware.

Finally, ML should be investigated in the context of two-dimensional spatial phase wavefront correction, where all of the current literature focuses on ML for one-dimensional $\Delta\phi'$ correction, post-heterodyne/homodyne detection. This could be particularly pertinent in the context of CV-QKD across FSO channels, where atmospheric turbulence is known to distort optical wavefronts propagating through it (e.g. [78]).

2.9 Summary

A comprehensive survey of machine learning algorithms designed to assist in continuous-variable quantum key distribution was presented in this chapter. The literature covers a range of applications, including phase error estimation, excess noise estimation, state discrimination, parameter estimation and optimization, key sifting, and key rate estimation. Results from the literature generally indicated that machine learning algorithms were able to outperform their traditional counterparts, though instances of traditional algorithms outperforming or equaling the performance of machine learning were also found. An analysis of machine learning algorithm applicability to different protocols, channel types, and assumptions was undertaken. Overall, from the current field of literature, it can be concluded that machine learning has been shown to be a powerful tool in assisting to make real-time continuous-variable quantum key distribution a practical reality, thereby suggesting that further research on machine learning assisted continuous-variable quantum key distribution should be undertaken.

Chapters 3 to 6 in this thesis aim to address some of the key research gaps identified in the literature, namely: (i) to expand the research on free-space optical continuous-variable quantum key distribution, (ii) to assess how machine learning can assist in reducing both hardware and software complexity requirements for practical phase error correction in free-space optical continuous-variable quantum key distribution, (iii) to expand upon the literature on machine learning assistance for temporal phase error correction (iv) to develop machine learning models for spatial relative wavefront error correction, and (v) to take preliminary steps towards experimental identification, characterization, and machine learning assisted correction of spatial relative wavefront errors, in the context of free-space optical continuous-variable quantum key distribution across turbulent atmospheric channels.

Chapter 3

Intensity-Only Wavefront Correction Using Machine Learning for Satellite-to-Earth CV-QKD

This chapter expands the research on machine learning assisted free-space optical continuous-variable quantum key distribution by developing an intensity-only real local oscillator phase wavefront correction algorithm, based on a convolutional neural network architecture. The convolutional neural network is designed to reduce hardware complexity requirements in continuous-variable quantum key distribution by taking intensity-only measurements as input, thereby removing the need for complex wavefront sensing instrumentation. Reducing hardware complexity requirements could accelerate the implementation of a global continuous-variable quantum key distribution network, as well as expand the variety of platforms capable of accommodating quantum communications infrastructure. Note that the phase screen model adopted in this chapter is validated in Appendix B, and that the nomenclature used in this chapter applies solely to the content within this chapter.

3.1 Introduction

A global continuous-variable quantum key distribution (CV-QKD) network could be established by connecting a network of satellite and ground stations via atmospheric free-space optical (FSO) channels [13], [79]–[83]. FSO transmission offers several advantages over optical fiber, including lower attenuation and higher bit-rate communications [84], [85]. However, FSO communications are susceptible to several issues, the most important of which is the distortion of the quantum signal, caused by atmospheric turbulence in the channel [86]. Recovery of the information encoded on the quantum signals (hereafter, referred to as “signals”) is of critical importance to the implementation of CV-QKD between satellites and ground stations.

In state-of-the-art CV-QKD protocols, reference pulses are often multiplexed (e.g. using a different polarization) with the transmitted signals, such that phase wavefront distortion of the reference pulses can be measured, then encoded onto a real local oscillator (RLO) at the receiver, which is used for coherent measurement of the signals. While it has been proposed that wavefront sensors be used to directly measure wavefront distortions across atmospheric channels, real-time measurement is restricted by the computational requirements of wavefront reconstruction algorithms [87]. This chapter instead proposes the development of machine learning (ML)-based algorithms to estimate wavefront corrections using only reference pulse intensity distributions as input. Estimation of wavefront corrections using intensity information on the ground could lead to faster and simpler signal measurements, increasing the practicality of satellite-based CV-QKD.

One motivation for reducing hardware and software requirements is the development of a QKD network of satellite, aerial vehicle, and terrestrial links, forming highly resilient and adaptable secure connections across atmospheric FSO channels (e.g. see [81], [88]). Complexity reduction is vital for aerial vehicles, where fuel consumption is a major limitation to long range and high endurance flight (e.g. [89]). Thus, innovative software-based solutions to replace hardware and computationally heavy resources are imperative to weight reduction, highlighting an advantage of our wavefront correction algorithm.

The open question we investigate here is the following: Can simple intensity measurements of reference pulses, transmitted through a satellite-to-Earth channel, provide accurate assessments on the wavefront distortions of reference pulses — *sufficient to enable non-zero secure key rates for CV-QKD?* Surprisingly, we find the answer in the affirmative. Moreover, we find reasonable secure key rates for a range of channel settings. We believe our results to be the first to show how ML algorithms can replace complex measuring equipment in the context of CV-QKD delivered from space, thereby opening up a new avenue for real-world deployments of this emerging quantum-communication technology.

The remainder of this chapter is as follows: A description of turbulent atmospheric satellite-to-Earth channel modeling is presented in Section 3.2. A methodology for estimating phase wavefront corrections across satellite-to-Earth channels using only intensity distributions as input to an ML model is provided in Section 3.3. A secure key rate analysis for CV-QKD is provided in Section 3.4, and our concluding remarks are given in Section 3.5.

3.2 The Satellite-to-Earth Channel

To model the satellite-to-Earth channels, reference pulse beam propagation is simulated through varying turbulent atmospheres using a phase screen model based on WavePy [90] (validated against experimental data in Appendix B), which uses a Fresnel split-step propagator. Phase screens represent a turbulent atmospheric volume by assuming that optical beams travel as if in a vacuum until interaction with the screen. The screens then perturb the phase of the beam when it passes through them, replicating how the beam would be perturbed if it had passed through an equivalent atmospheric volume.

The WavePy model in [90] is modified to account for the vertical variation in atmospheric turbulence across various satellite-to-Earth channels. The atmosphere is separated into N layers (as described in [91]), with varying propagation distances to account for the increase in atmospheric density at lower altitudes. Values for the refractive index structure parameter of turbulence C_n^2 were calculated as a function of altitude h in m, root mean square wind speed ν_{rms} in m/s, and ground-level C_n^2 in m$^{-2/3}$, $C_n^2(0)$, given by,

$$\begin{aligned} C_n^2(h) = 0.00594(\nu_{rms}/27)^2 \; (h \times 10^{-5})^{10} \; \exp(-h/1000) \\ + 2.7 \times 10^{-16} \; \exp(-h/1500) + C_n^2(0) \exp(-h/100). \end{aligned} \tag{3.1}$$

Atmospheric stratification into N slabs in the phase screen model involves an optimization based on the scintillation index σ_I^2, calculated as,

$$\sigma_I^2 = \exp\left[\frac{0.49\sigma_R^2}{(1 + 1.11\sigma_R^{12/5})^{7/6}} + \frac{0.51\sigma_R^2}{(1 + 0.69\sigma_R^{12/5})^{5/6}}\right] - 1, \tag{3.2}$$

and as a function of Rytov variance σ_R^2, given by,

$$\sigma_R^2 = 2.25k^{7/6} \sec^{11/6}(\theta_z) \int_{h_0}^{H} C_n^2(h)(h - h_0)^{5/6} dh. \tag{3.3}$$

Here, k represents the wavenumber, $k = 2\pi/\lambda$. For simplicity in the modeling, the satellite zenith angle θ_z is assumed to be 0° (as such, our results refer to the time when the satellite is directly over the ground station). The atmospheric channel parameters used are outlined in Table 3.1.

Table 3.1: Simulation parameters.

Parameter	**Value**
Satellite altitude (H)	300 - 500 km
Receiver altitude (h_0)	0 - 2 km
Ground-level C_n^2 ($C_n^2(0)$)	1.70×10^{-15} - 10^{-14} m$^{-2/3}$
Outer scale of turbulence (L_0)	5 m
Inner scale of turbulence (l_0)	0.025 m
Root mean square wind speed (ν_{rms})	21 m/s
Satellite zenith angle (θ_z)	0°
Laser wavelength (λ)	1550 nm
Beam waist radius (w_0)	0.15 m
Receiver radius (R_r)	0.750 - 1.25 m
Reference pulse photon number (n_{ph})	55,000

Several satellite-to-Earth channels within the ranges outlined in Table 3.1 are simulated to assess the robustness of the phase correction methodology. However, a demonstrative channel (termed *channel one*), with $H = 300$ km, $h_0 = 2$ km, $C_n^2(0) = 1.70 \times 10^{-14}$ m$^{-2/3}$, and $R_r = 0.75$ m, are used, in the first instance, as our key illustrative example (additional channels discussed later).

A beam transmitted across a vacuum (no atmospheric disturbance) diffracts as a function of distance traveled, with a beam traveling through an atmosphere of equivalent distance experiencing equivalent diffraction. Therefore, in order to quantify the phase wavefront distortion caused by the atmosphere, phase distributions at the ground station are compared to the phase wavefront distribution of a beam having traveled across an equivalent distance through a vacuum. The intensity $I_R(x, y)$ and wavefront $\Delta\Phi_R(x, y)$ of a reference pulse after transmission through a vacuum of distance 298 km are shown in Figure 3.1(a), while example intensity and wavefront distributions of a random reference pulse after passing through channel one are shown in Figure 3.1(b).

64,000 instances of reference pulse propagation are simulated for several satellite-to-Earth channels within the ranges outlined in Table 3.1. Intensity and phase wavefront

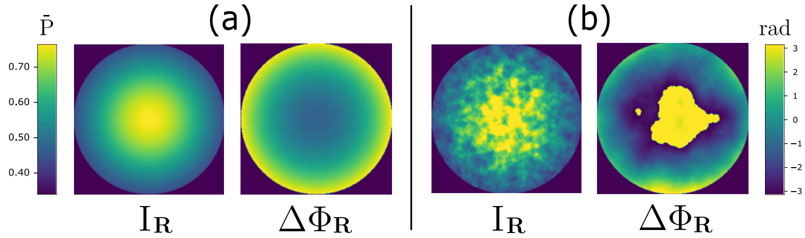

Figure 3.1: Intensity $I_R(x, y)$ and phase $\Delta\Phi_R(x, y)$ distributions after reference pulse transmission through (**a**) a 298 km vacuum channel and (**b**) an example instance of channel one. Note that the scales for intensity on the left (where $\bar{P}$ is the power normalized with respect to the transmit power), and for phase on the right, apply to both (**a**) and (**b**).

information is recorded at the receiver for each channel type, then used to train an ML model to estimate an RLO phase wavefront correction.

3.3 Phase Correction Machine Learning Model

We analyze an alternative approach to modulating the RLO phase wavefront in this chapter, whereby only the intensity distributions of the reference pulses need to be measured at the receiver. Using these measurements, a convolutional neural network (CNN) estimates the phase wavefront correction $\Delta\Phi(x, y)$ to be applied to the RLO. Encoder-decoder CNNs are designed for image-to-image translation. The encoder maps relationships between input data points to a reduced feature space, then the decoder uses the reduced feature space to reconstruct a prediction of the output data points. As such, we develop an encoder-decoder CNN model, where the encoder creates a feature space from the received reference pulse intensity distribution, then the decoder constructs an estimated RLO wavefront correction $\Delta\tilde{\Phi}(x, y)$ for coherent measurement of the signals.

Figure 3.2 outlines the SegNet CNN architecture used to estimate the wavefront correction (hereafter, referred to as "the network"), based on the SegNet model described in [92], which is comprised of eleven convolutional layers, three maxpool layers, and three deconvolutional layers. The encoder architecture is based on the VGG16 model [93], while the decoder inverts the layer sequence of the encoder.

While many image translation networks have been designed to only modify the surface appearance of the same underlying structure (e.g. converting a satellite photo to a map [94]), a common underlying structure is assumed not to exist between intensity dis-

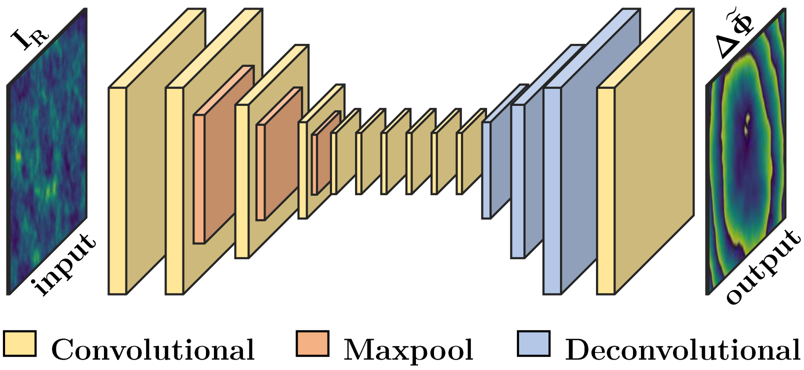

Figure 3.2: SegNet network layer architecture (further defined in [92]). Reference pulse intensity distributions I_R at the receiver were used as input to the network, which were then mapped to output estimated wavefront corrections $\Delta\tilde{\Phi}$.

tributions and phase wavefront distributions. Therefore, linear regression is implemented to map the final network layer to the predicted phase correction, without pixel-wise classification or skip connections between the encoder and decoder layers.

The first convolutional layer has a 5×5 kernel, while the rest of the layers have a 3×3 kernel. Batch normalization is performed on all of the layers, with each using the rectified linear unit activation function. A mean squared error (MSE) loss function is used to optimize the network weights, where the MSE is taken for each pixel and then averaged, via,

$$\mathrm{MSE} = \frac{\sum_{i=1}^{N_r}\sum_{j=1}^{N_r}(\Delta\tilde{\phi}_{ij} - \Delta\phi_{ij})^2}{N_r^2}. \tag{3.4}$$

Here, N_r is the number of pixels of the measured reference pulse phase at the receiver, $\Delta\phi_{ij}$ is the true phase correction, and $\Delta\tilde{\phi}_{ij}$ is the estimated phase correction, at each pixel.

The 64,000 reference pulse propagation simulations for each channel are split into 57,600 training instances and 6,400 testing instances. Three examples of the CNN phase wavefront correction output from the test data, for channel one, are shown in Figure 3.3(a)-(c), where the true RLO phase corrections $\Delta\Phi(x, y)$ at the receiver (top) are compared with RLO wavefront correction estimations $\Delta\tilde{\Phi}(x, y)$ (bottom).

Considering channel one, the network is generally able to estimate phase gradient trends across the wavefront correction distributions. For example, higher phase correction values at the edges of the wavefront distributions are estimated as decreasing towards the center of the distributions, as is present for the true wavefront corrections in Figure 3.3.

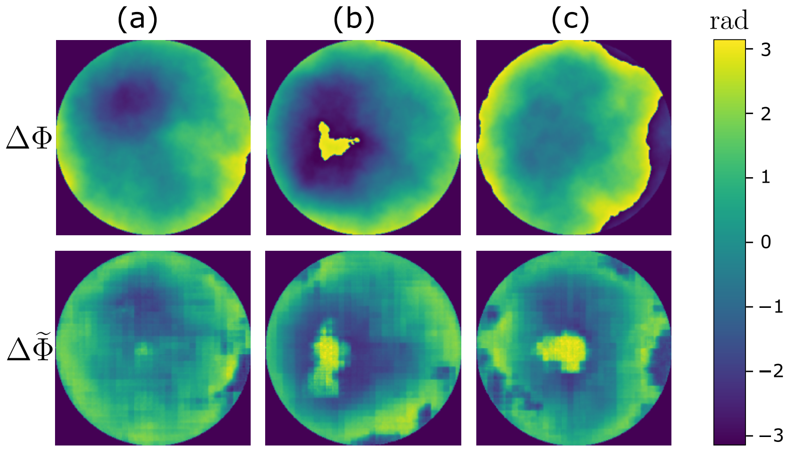

Figure 3.3: Examples (**a**)-(**c**) give RLO wavefront corrections at the receiver for three independent instances of reference pulse propagation across channel one, with true phase corrections $\Delta\Phi$ compared to estimated phase corrections $\Delta\tilde{\Phi}$.

Another example of accurate phase gradient estimation can be seen in Figure 3.3(b), where a phase correction peak is estimated in the center of the distribution, as in the true wavefront correction.

The wavefront gradients in the estimations are found to be more segmented than the true wavefront gradients, which could be explained as the optimization function segmenting the output into regional averages to reduce the total MSE between pixels (defining sharp gradients incorrectly would lead to greater MSE values). Further, some output wavefront corrections estimate false peaks, as in Figure 3.3(c), while others fail to estimate any peaks. However, overall the estimated phase corrections improve coherence between the RLO and signals (defined below). Note, although not explicitly shown, similar trends are found from the estimation results of the other channel types used in the analyses of CV-QKD provided next.

We note the prevailing method for recovering reference pulse wavefront distributions at the receiver is by the use of Zernike polynomials, e.g. [72], [74], which require the wavefront of the reference pulses to be measured in order to parameterize the Zernike coefficients. However, wavefront sensor performance deteriorates when the reference pulses encounter strong turbulence, while also requiring computationally intensive algorithms for wavefront reconstruction [87], reducing the practicality of real-time implementation. We also note that [95] uses an encoder-decoder CNN architecture to estimate wavefront distortion across an atmospheric channel with intensity distributions as input. However, the calculations in [95] focus on single-photon OAM mode recovery for high-dimensional discrete variable QKD across a horizontal atmospheric channel, rather than the wavefront correction methodology for satellite-to-Earth CV-QKD that is presented here. OAM CV-QKD, although in principle possessing higher throughput advantages, is very difficult to implement in practice, whereas the standard CV-QKD we investigate here is considered mainstream and easier to implement. Further, while [95] uses both distorted and undistorted intensity distributions as inputs to a CNN for wavefront correction in free-space OAM-QKD over horizontal links, our work uses only the distorted intensity distribution and applies the method to satellite-to-Earth links.

3.4 Connections to CV-QKD

In connecting to CV-QKD, we must apply our wavefront corrections to the RLO used to measure the incoming signal.

3.4.1 Coherent Efficiency

The coherent efficiency γ is used as a singular metric to describe the relative difference between two wavefronts. The coherent efficiency is derived by comparing the difference between the electric fields of the RLO $E_{RLO}(x, y)$ and the signals $E_S(x, y)$ via the following relation [78],

$$\gamma = \frac{|\frac{1}{2}\iint_{\mathcal{D}_R} \left(E^*_{RLO}(x,y)E_S(x,y) + E_{RLO}(x,y)E^*_S(x,y)\right)\ dr|^2}{\iint_{\mathcal{D}_R} |E_{RLO}(x,y)|^2\ dr \iint_{\mathcal{D}_R} |E_S(x,y)|^2\ dr}, \tag{3.5}$$

for the receiver aperture diameter $\mathcal{D}_R$ and the radial integral dr. Correcting the RLO phase wavefront increases the coherent efficiency, where a value of $\gamma = 1$ would indicate a perfect correction of the wavefront. In this chapter, it is assumed that the wavefront distortion is equivalent for the reference pulses and signals.

3.4.2 CV-QKD

Beyond atmospheric attenuation, absorption, and scintillation, optical signals are affected by channel excess noise ξ_{ch} and detector noise ξ_{det}, both contributing to secure key rate reduction in CV-QKD. Detector noise is calculated as a function of the coherent efficiency, electronic noise ξ_{el}, and detector efficiency η_{det} as,

$$\xi_{det} = \frac{((1-\gamma) + \xi_{el})\eta_{det}}{\gamma}. \tag{3.6}$$

The transmissivity T of a channel further affects secure key rates for satellite-to-Earth CV-QKD [13]. This parameter is defined as $T = (P_R/P_T)\eta_{det}$, where P_T and P_R represent the power at the transmitter and receiver, respectively. Rather than using ensemble-average quantum state information to analyze secure key rates for satellite-to-Earth CV-QKD, a combined *effective transmissivity* and *effective excess noise* term $T_f\xi_f$ can be used for a non-fluctuating channel (described in [74]), calculated as,

$$T_f\xi_f = \xi_{ch}\langle T\rangle + \langle \xi_{det}\rangle + \left(\langle T\rangle - \langle\sqrt{T}\rangle^2\right) V_{mod}. \tag{3.7}$$

Average transmissivity $\langle T\rangle$ and average detector noise $\langle \xi_{det}\rangle$ values are determined for the reference pulses for each channel type simulated. We note that this non-fluctuating effective-channel approximation could underestimate the detector noise contribution from low-coherence channel realizations, and should therefore be accounted for during channel estimation. Using these parameters, the shared information I_{AB} between Alice (at the transmitter) and Bob (at the receiver) can be calculated as,

$$I_{AB} = \frac{1}{2}\log_2(1 + \frac{T_f V_{mod}}{1 + T_f\xi_f}). \tag{3.8}$$

Table 3.2 outlines the noise and efficiency values used to analyze secure key rates in this chapter.

Table 3.2: Noise parameters (where SNU represents shot noise units).

Parameter	**Value**
Detector efficiency (η_{det})	95% [72]
Reverse reconciliation efficiency (β)	95% [72]
Detector electronic noise (ξ_{el})	0.010 SNU [96]
Channel noise (ξ_{ch})	0.0172 SNU [72]

We adopt the entanglement-based GG02 protocol [8] to calculate secure key rates before and after the wavefront correction is applied. Homodyne detection is used, as this is known to achieve higher key rates than heterodyne detection for low transmissivity values [96]. As such, the covariance matrix M_{AB} describing the quantum state ρ at an untrusted ground station is given by,

$$M_{AB} = \begin{pmatrix} a\mathbb{1} & c\sigma_z \\ c\sigma_z & b\mathbb{1} \end{pmatrix} = \begin{pmatrix} (V_{mod}+1)\mathbb{1} & \sqrt{T_f(V_{mod}^2+2V_{mod})}\sigma_z \\ \sqrt{T_f(V_{mod}^2+2V_{mod})}\sigma_z & (T_f V_{mod}+1+T_f\xi_f)\mathbb{1} \end{pmatrix}, \tag{3.9}$$

where $\mathbb{1}$ represents the identity matrix and σ_z represents the Pauli-z matrix (of dimension 2×2).

As we shall see, given atmospheric turbulence, the effects of trust in the detector noise become important. A detector is considered trusted if an eavesdropper (Eve) cannot control the imperfection of Bob's detector [97]. For the GG02 CV-QKD protocol with reverse reconciliation, trusted detector noise only affects the shared information between Alice and Bob, not the Holevo information between Bob and Eve, whereas effective transmissivity and excess noise affect the Holevo information when detector noise is untrusted [10]. Therefore, for a trusted detector, $b = T_f V_{mod} + 1 + \xi_{ch}$ (note the difference from that explicitly given in Eq. (3.9) for the untrusted model).

The upper bound on the amount of information that Eve can determine about a quantum state is known as Holevo information [10]. Holevo information is calculated as $\chi_{BE} = S_{AB} - S_{A|B}$, where S_{AB} is the von Neumann entropy of the state accessible to Eve and $S_{A|B}$ is the entropy after homodyne measurement by Bob. Here, the subscripts A, B, and E represent Alice, Bob, and Eve, respectively. Assuming that Eve has a purification of Alice and Bob's shared quantum state ρ_{AB}, the von Neumman entropy S is $S(\rho_{AB}) = \sum_i g(\nu_i)$, using the function $g(x)$ given by,

$$g(x) = \frac{x+1}{2}\log_2\left(\frac{x+1}{2}\right) - \frac{x-1}{2}\log_2\left(\frac{x-1}{2}\right). \tag{3.10}$$

Here, ν_i are three symplectic eigenvalues (defined as the modulus of any eigenvalue) of the following matrices. The first two, $\nu_{1,2}$, are the symplectic eigenvalues of M_{AB}, and are derived as $\nu_{1,2} = \frac{1}{2}(z \pm [b-a])$, where $z = ([a+b]^2 - 4c^2)^{1/2}$ [98]. The third, ν_3, is the symplectic eigenvalue of the conditional matrix $M_{A|B}$, where

$$\tilde{M}_{A|B} = i\Omega M_{A|B} = i \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a - \frac{c^2}{b} & 0 \\ 0 & a \end{pmatrix} = i \begin{pmatrix} 0 & a \\ -a + \frac{c^2}{b} & 0 \end{pmatrix}, \tag{3.11}$$

which results in $\nu_3 = (a[a - \frac{c^2}{b}])^{1/2}$ [10]. The Holevo information is then calculated as $\chi_{BE} = g(\nu_1) + g(\nu_2) - g(\nu_3)$. Finally, the secure key rates R_{sec}, in bits/pulse, can be calculated for the simulated satellite-to-Earth channels, where $R_{sec} = \beta I_{AB} - \chi_{BE}$, with reverse reconciliation efficiency β. We note this rate derivation assumes the asymptotic limit for the signal under the assumption of a collective attack.

3.4.3 Results

The probability density functions (PDFs) for the coherent efficiency, both before and after RLO wavefront correction, are given in Figure 3.4 for the 6,400 reference pulse simulations used as test data for the CNN across channel one.

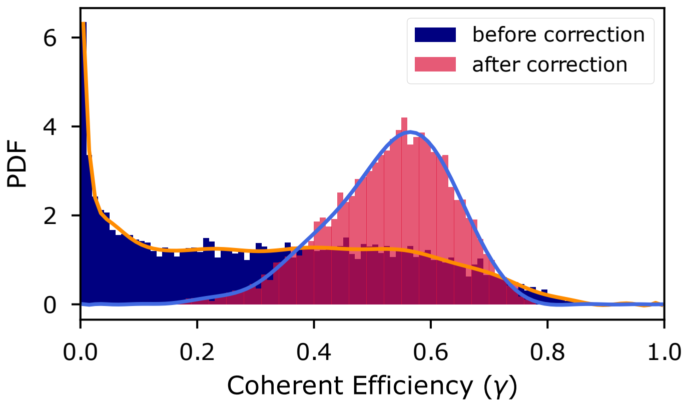

Figure 3.4: Coherent efficiency probability density (PDF) distributions, before wavefront correction are applied (purple) and after wavefront correction are applied (red), for channel one.

Our results in Figure 3.4 show that before the wavefront corrections are applied, coherent efficiencies are skewed towards zero, where a value of zero indicates no coherence between the signal wavefronts and RLO wavefronts. However, once the estimated wavefront correction is applied to the RLO, a mean coherent efficiency of approximately 0.53 is achieved. For comparison, Shack-Hartmann wavefront sensors, which are commonly used in adaptive optics, can achieve a wide variety of coherent efficiencies depending on their system architecture, with potential wavefront corrections leading to coherent efficiencies upward of 0.90, at the cost of increased system complexity [99].

Our wavefront corrections increase the shared information between Alice and Bob from $I_{AB} = 1.910 \times 10^{-5}$ bits/pulse to $I_{AB} = 0.284$ bits/pulse. As such, after the wavefront corrections are applied, the signals are able to be measured with greater coherence at the receiver, leading to an increase in shared information between Alice and Bob.

As mentioned earlier, a range of channel conditions are tested in order to assess the flexibility of the wavefront correction methodology across different satellite-to-Earth channels (as per Table 3.1). We illustrate these results using three channels, as defined in Table 3.3. Compared to channel one, the refractive index structure parameter is weakened across *channel two* by lowering $C_n^2(0)$, while the channel distance is lengthened in *channel three* by increasing the altitude of the satellite. A new network is trained for each channel using 64,000 independent simulation instances, using the same architecture outlined in Section 3.3. Channels one and two have an average transmissivity of $\langle T \rangle \approx 0.71$, whereas the receiver radius is increased for channel three to achieve an average transmissivity of $\langle T \rangle \approx 0.83$ to compensate for the lower coherent efficiencies attained from the wavefront corrections for the longer channel.

Table 3.3: Channel variables.

Parameter	Channel One	Channel Two	Channel Three
H [km]	300	300	500
$C_n^2(0)$ [m$^{-2/3}$]	1.7×10^{-14}	1.7×10^{-15}	1.7×10^{-14}
R_r [m]	0.75	0.75	1.25

Secure key rates, after RLO wavefront correction across the three channels outlined in Table 3.3, are given as a function of V_{mod} and average coherent efficiency in Figure 3.5. Wavefront corrections for channel one result in the highest secure key rate of the simulated channels, with an average coherent efficiency of $\gamma_1 \approx 0.53$ resulting in a maximum secure key rate of 0.112 bits/pulse. The corrections for channel two produce a maximum secure key rate of 0.007 bits/pulse at $V_{mod} = 3.2$ SNU, with an average coherent efficiency of $\gamma_2 \approx 0.42$, while the wavefront corrections for channel three leads to a maximum secure key rate of 0.004 bits/pulse, with an average coherent efficiency of $\gamma_3 \approx 0.31$. The resulting secure key rates indicate that channels with weaker turbulence actually result in worse wavefront correction performance when using the methodology outlined in this chapter, as does increasing the satellite altitude. Improved wavefront correction performance in stronger turbulence regimes could be due to larger fluctuations in the intensity and wavefront distributions providing more information for the CNN to learn from.

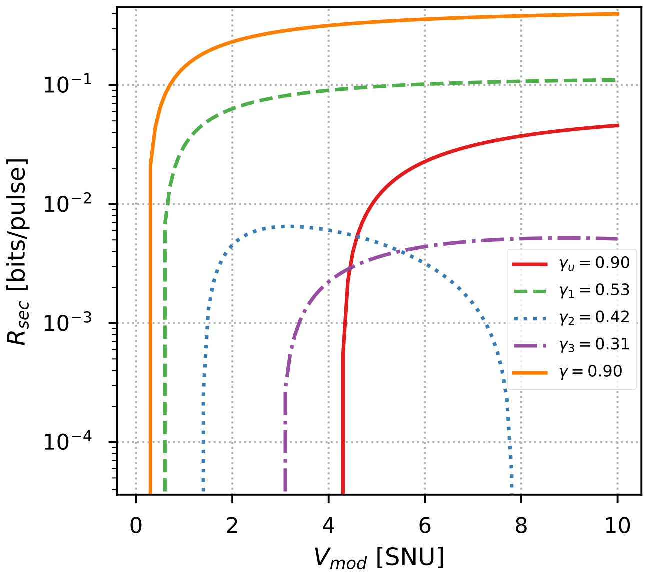

Figure 3.5: Secure key rate versus V_{mod}, for different average coherent efficiency values. Results after wavefront correction are shown for channel one in the dashed green line ($\gamma_1 \approx 0.53$), channel two in the dotted line ($\gamma_2 \approx 0.42$), and channel three in the dash-dotted purple line ($\gamma_3 \approx 0.31$), while results for untrusted and trusted detector noise with $\gamma = 0.90$ are plotted in the red and orange lines, respectively.

Further, secure key rate results in Figure 3.5, for a hypothetical average coherent efficiency of $\gamma = 0.90$, are shown for channels with both untrusted (with a coherent efficiency γ_u) and trusted (with a coherent efficiency γ) detector noise, given $\langle T \rangle \approx 0.71$. Secure key rates are found to vary greatly depending on whether the detector noise is trusted or untrusted, primarily due to the influence of coherent efficiency on detector noise (see Eq. 3.6). Shown in Figure 3.5, an untrusted detector requires a coherent efficiency of $\gamma_u \approx 0.90$ to achieve positive key rates, as a result of the trade-off between shared information and Holevo information. However, positive key rates are attained for coherent efficiencies above $\gamma \approx 0.42$ for a trusted detector (for $\langle T \rangle \approx 0.71$), when Eve cannot manipulate the detector noise, thereby reducing the Holevo information.

It is found that increasing V_{mod} up to 10 SNU generally leads to higher secure key rates for each of the channels tested, except for channel two, which is approaching the limit of achieving non-zero key rates (at $\gamma_2 \approx 0.42$). However, a similar trend is found for each of the other channels tested, where key rates would rapidly increase as V_{mod} increased before reaching an asymptote, reducing the advantage of continually encoding higher V_{mod} values on the quantum states.

A network is then trained to estimate wavefront corrections for channels of varying $C_n^2(0)$ (across the range in Table 3.1) with $R_r = 0.75$ m, achieving an average coherent efficiency of 0.49 and maximum key rate of 0.072 bits/pulse. As such, given enough training data, it is assumed that a network would be able to output wavefront corrections for a wide range of channel conditions.

We should be clear: not all satellite-to-Earth channel settings we investigate lead to a non-zero key rate. Example settings where $R_{sec} = 0$ bits/pulse include channels with a ground station altitude of $h_0 = 0$ km, where the refractive index structure parameter of turbulence is strongest, leading to greater wavefront distortions. Of course, secure key rates can be increased by increasing the receiver aperture diameter, but we have not explored that further, as large ground-based telescopes increase costs substantially. We also note that the assumptions of asymptotic key rates and idealized channel estimation likely lead to optimistic key rates. Future work could investigate finite-size effects and practical channel estimation algorithms to more accurately assess achievable secure key rates for a given channel. Finally, global CV-QKD has been our principal interest in this chapter; as such, we have not explored our setup in the context of terrestrial-only horizontal channels. However, we believe our method of intensity-only wavefront correction could play a substantial role in many terrestrial CV-QKD implementations.

3.5 Summary

A convolutional neural network was designed to estimate phase wavefront corrections to modulate the phase wavefront of a real local oscillator, using only the reference pulse intensity distributions at the ground receiver as input. We found that the estimated phase wavefront corrections increased coherent efficiencies, resulting in positive key rates for several satellite-to-Earth channels when the detector noise was trusted, without the need to measure reference pulse wavefronts directly. Successful implementation of the proposed method would reduce the complexity of wavefront sensing hardware requirements, and thus would increase the feasibility of a global satellite-based continuous-variable quantum key distribution network, highlighting an important application of machine learning in assisting free-space optical continuous-variable quantum key distribution.

After assessing an application of machine learning in reducing hardware complexity requirements in continuous-variable quantum key distribution in this chapter, we instead focus on reducing software complexity requirements in Chapter 4. We assess the complexity requirements of machine learning-based signal phase error estimation algorithms, with a focus on increasing the real-time feasibility of implementing relative phase error correction in a practical free-space optical continuous-variable quantum key distribution setup. As opposed to the spatial wavefront corrections investigated in this chapter, Chapter 4 investigates temporal phase errors post-homodyne/heterodyne detection, a separate, though related, challenge.

Chapter 4

Machine Learning for Temporal Phase Estimation in Satellite-to-Earth Quantum Communication

An application of machine learning algorithms designed for hardware complexity reduction in continuous-variable quantum key distribution was shown in Chapter 3, where spatial wavefront correction was achieved without the need for complex wavefront sensing instrumentation. This chapter instead assesses the software-based computational complexity required for temporal signal phase error estimation using machine learning. We focus on reducing the scale of long short-term memory neural networks, designed to correct for relative phase errors. Reducing the computational requirements in free-space optical continuous-variable quantum key distribution would increase the practical feasibility of implementing phase error correction in real-time. Note that the phase screen model adopted in this chapter is validated in Appendix B, and that the nomenclature used in this chapter applies solely to the content within this chapter.

4.1 Introduction

Global information security could be achieved via the use of a low Earth orbit satellite-based continuous-variable quantum key distribution (CV-QKD) network, used to distribute secure keys between parties. In the context of free-space optical CV-QKD using Gaussian modulated coherent states, secure key information is encoded on the quadratures of the electric field of very weak optical quantum signals (hereafter, referred to simply as "signals") [8]. Current state-of-the-art CV-QKD protocols consider using a real local oscillator (RLO) created locally at the receiver. The RLO is manipulated so as to correct for the phase errors accumulated across the channel, such errors being measured from reference pulses that are polarization-multiplexed with the signals. The signals are then measured with the RLO using balanced homodyne/heterodyne detection.

However, there remains a difference in the phase error measured using the reference pulses and the phase error of the signals, which we refer to as the *relative phase errors*, which are one-dimensional temporal phase errors (post-homodyne/heterodyne detection). The reduction of temporal relative phase errors, using signal phase error estimation algorithms, is an ongoing area of research (e.g. [11], [12], [17]). Given that operational speed remains a primary factor in designing CV-QKD systems to function in real-time, signal phase error estimation algorithms should be as low-complexity as possible (whilst achieving the required accuracy).

Machine learning (ML) is able to adapt to noisy data by learning patterns between input and output data, where no assumption-based physical models are required. Previous works have used a variety of ML algorithms for signal phase error estimation (as discussed in Chapter 2), including convolutional neural networks (NNs) [27], long short-term memory (LSTM) NNs [12], [29], and Bayesian inference with Kalman filters [11]. However, these relative phase error correction approaches are not explored for satellite-to-Earth CV-QKD links. Moreover, a question that remains unexplored within the context of ML applied to signal phase error estimation is: what is the lowest complexity (highest operational speed) NN that can achieve a required accuracy? This is the main question we answer in this chapter.

Our contributions can be summarized thus: (i) We analyze the relationship between LSTM NN model architecture and signal phase error estimation accuracy for satellite-to-Earth channels, with a focus on extracting an efficient model for the required signal phase error estimation, while limiting the number of free parameters within the NN (thereby offering lower complexity). (ii) We compare several architectures, their estimation performance, and how their estimations relate to formal uncertainty bounds, calculated using Fisher information. (iii) We show how our results lead to a feasible ML-based CV-QKD system for satellite-to-Earth channels that have improved outcomes relative to standard, non-ML-based systems.

The remainder of this chapter is as follows: The system model is presented in Section 4.2, the NN architectures are outlined in Section 4.3, the NNs are analyzed in Section 4.4, and concluding remarks are made in Section 4.5.

4.2 System Model

A Gaussian modulated coherent state, RLO-based, CV-QKD protocol is implemented in this chapter. In the protocol, Alice (subscript A) at the satellite encodes the reference pulses (subscript R) with quadrature values $\{X_{A_R}, P_{A_R}\}$, and signals (subscript S) with quadrature values $\{X_{A_S}, P_{A_S}\}$. The reference pulses and signals are then polarization-multiplexed and transmitted across atmospheric satellite-to-Earth channels to Bob (subscript B) at the ground station. The encoded reference pulse information is shared with Bob beforehand, while the signal information is unknown to Bob and contains the secure key information. At the receiver, Bob demultiplexes the reference pulses and signals, measuring their quadrature values $\{X_{B_R}, P_{B_R}\}$ and $\{X_{B_S}, P_{B_S}\}$, respectively.

The encoded quadratures are distorted by the channel, resulting in a *reference pulse phase error* $\Delta\phi_R$, calculated as,

$$\Delta\phi_R = \tan^{-1}\left(\frac{P_{B_R}X_{A_R} - X_{B_R}P_{A_R}}{X_{B_R}X_{A_R} + P_{B_R}P_{A_R}}\right). \tag{4.1}$$

Likewise, the signal is distorted by the channel and detector, accumulating a *signal phase error* $\Delta\phi_S$, calculated as,

$$\Delta\phi_S = \tan^{-1}\left(\frac{P_{B_S}X_{A_S} - X_{B_S}P_{A_S}}{X_{B_S}X_{A_S} + P_{B_S}P_{A_S}}\right). \tag{4.2}$$

In our scheme, an RLO at the receiver is split into the RLO_1 and RLO_2, where a heterodyne detector uses the RLO_1 to measure the reference pulse quadratures, such that $\Delta\phi_R$ can be derived. Phase shifting the RLO_2 by $\Delta\phi_R$ for each pulse provides a first approximation of the true signal phase error $\Delta\phi_S$, $\Delta\phi_R \approx \Delta\phi_S$, increasing the coherence between the signal and RLO_2. However, we assume that $\Delta\phi_R$ measured by mixing the reference pulses and RLO_1 does not correspond exactly to $\Delta\phi_S$ measured by mixing the signals and RLO_2, such that a relative phase error between $\Delta\phi_R$ and $\Delta\phi_S$ exists, $\Delta\phi_R \neq \Delta\phi_S$ (shown experimentally in [11], [12]).

Estimation of the true values of $\Delta\phi_S$ is, thus, an important aspect of RLO-based

CV-QKD, where a perfect estimation of $\Delta\phi_S$, used to phase correct the RLO_2, would result in ideal recovery of the encoded quadrature information, $\{X_{B_S}, P_{B_S}\} \approx \{X_{A_S}, P_{A_S}\}$. Estimation of $\Delta\phi_S$ is undertaken using a *signal phase error estimation algorithm.* In practice, a signal phase error estimation algorithm may be trained using known $\Delta\phi_S$ values; however, when being implemented for real-time CV-QKD, $\Delta\phi_S$ is inherently unknown, thus only $\Delta\phi_R$ can be used to output an estimate $\Delta\tilde{\phi}_S$ of $\Delta\phi_S$. Figure 4.1 illustrates our RLO-based CV-QKD scheme, where $\Delta\phi_R$ measurements are input into a NN-based signal phase error estimation algorithm (described in Section 4.3), which outputs an estimate $\Delta\tilde{\phi}_S$, used to phase correct the RLO_2.

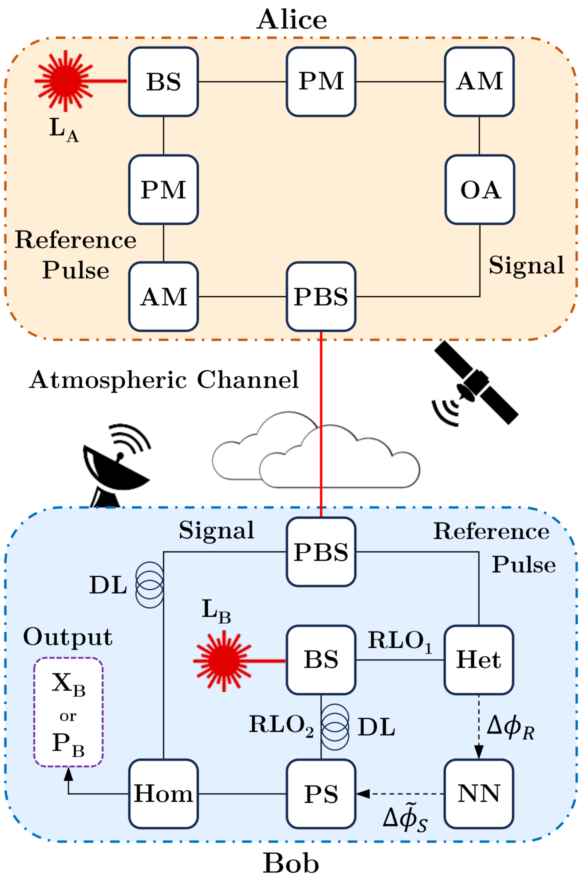

Figure 4.1: Schematic outlining the preparation and measurement of coherent states with $\Delta\phi_S$ estimation. L is a laser source, BS is a beamsplitter, AM is an amplitude modulator, PM is a phase modulator, OA is an optical attenuator, PBS is a polarized beamsplitter, Het is a heterodyne detector, Hom is a homodyne detector, PS is a phase shifter, DL is a delay line, and NN is neural network.

Previous works [15], [17], [72], [73] have described the relationship between $\Delta\phi_R$ and $\Delta\phi_S$ as being represented by a Gaussian function, which applies to every measurement of the reference pulses and signals. We term this the *Gaussian phase noise*, which is related to the reference pulse and signal measurements being shot-noise limited (representing the standard quantum limit), as well as experiencing different interactions with the atmospheric channel and receiver hardware.

Considering the Gaussian phase noise, simply taking the expectation of the previous N measurements of $\Delta\phi_R$ can result in a more precise estimate of the current $\Delta\phi_S$ value, which we define as the *expectation estimator* $\langle\Delta\phi_R\rangle_N$. However, the relationship between $\Delta\phi_R$ and $\Delta\phi_S$ is complex, involving many environmental and system-based factors. Therefore, a signal phase error estimation algorithm should be able to adapt to *any* relationship between $\Delta\phi_R$ and $\Delta\phi_S$, such that $\Delta\phi_S$ can be estimated using the information from the $\Delta\phi_R$ measurements. Therefore, we propose a NN-based signal phase error estimation algorithm capable of adapting to varying relationships between $\Delta\phi_R$ and $\Delta\phi_S$, including for the Gaussian phase noise. Note that any change in channel conditions will result in initial poor $\Delta\phi_S$ estimation performance while the signal phase error estimation algorithm adapts to the new conditions.

To demonstrate the robustness of our $\Delta\phi_S$ estimation approach, we present two different relationships between $\Delta\phi_R$ and $\Delta\phi_S$. Case 1 relates to the Gaussian phase noise (Section 4.4.1), and Case 2 includes both the Gaussian phase noise of Case 1 *and* an additional source of error between $\Delta\phi_R$ and $\Delta\phi_S$, which remains constant for each coherence time (Section 4.4.2). To demonstrate the advantage of our NN-based approach, we compare the $\Delta\phi_S$ estimation performance of the NNs to the approximation $\Delta\phi_R \approx \Delta\phi_S$ and the expectation estimator $\langle\Delta\phi_R\rangle_N$.

4.2.1 Phase Error Simulation

Previous works have described several potential sources contributing to the Gaussian phase noise [15], [17], [72], [73], which we adopt in our work. Specifically, we assume that the total Gaussian phase error variance σ^2_{error} is calculated as,

$$\sigma^2_{error} = \sigma^2_{det} + \sigma^2_{drift}. \tag{4.3}$$

Here, the detector phase error variance σ^2_{det} is introduced by the shot-noise limited and experimental noise of the reference pulse heterodyne detection [17], [72], which is calculated as [72],

$$\sigma_{det}^2 = \frac{\xi_{ch} + 2\frac{1+\xi_{det}}{\eta_{det}T}}{|\alpha_R|^2}, \tag{4.4}$$

where it is assumed that $|\alpha_R|^2$, ξ_{ch}, and η_{det} are constant (defined in Table 4.1). The transmissivity T is calculated using a satellite-to-Earth phase screen model (outlined in Section 4.2.2) for the channel conditions defined in Table 4.2. When considering the detector noise term ξ_{det} in Eq. 4.4, an additional error arises from the distortion of the reference pulse two-dimensional phase wavefront across the channel. The coherent efficiency γ quantifies the coherence between the reference pulse and RLO_1 wavefronts, where zero represents no coherence and one represents full coherence. Detector noise ξ_{det} is calculated using the coherent efficiency as,

$$\xi_{det} = \frac{((1-\gamma) + \xi_{el})\eta_{det}}{\gamma}, \tag{4.5}$$

where the coherent efficiency is sampled from the PDF $\mathcal{N}(0.8430, 0.0025)$. See [78] for a more in-depth explanation of the coherent efficiency.

Table 4.1: Noise parameters.

Parameter	**Value**
Reference pulse intensity ($|\alpha_R|^2$)	$20V_{mod}$ [15]
Detector efficiency (η_{det})	95% [72]
Detector electronic noise (ξ_{el})	0.0100 SNU [96]
Channel noise (ξ_{ch})	0.0172 SNU [72]
Phase drift error variance (σ_{drift}^2)	0.1012 SNU [15], [72]

Phase drift error variance σ_{drift}^2 is associated with both time-of-arrival fluctuations across the channel and imperfections in delay lines at the receiver [15], [72], defined in Table 4.1. Note that the choice of noise values in Table 4.1 are selected from previous works [15], [72], [96]. While it is assumed that these values would fluctuate between CV-QKD setups, they provide a reasonable representation for the purpose of the work in this chapter.

4.2.2 The Satellite-to-Earth Channel

While previous works have investigated $\Delta\phi_S$ estimation for optical fiber channels (see Chapter 2), free-space optical satellite-to-Earth connections remain vital to the establishment of a global QKD network.

A phase screen model based on [90] simulates the effects of the atmospheric channel on the reference pulses and signals, modified to account for the variation of atmospheric conditions as a function of altitude for the satellite-to-Earth channel. The atmosphere is conceptually separated into layers (as described in [91]), where the width of each layer is varied as a function of altitude. The phase screens are placed at the center of each layer to simulate the distortion of the intensity and phase of the beam wavefront as if it had passed through an equivalent atmospheric volume. Note that the phase screen model adopted in this chapter is validated against experimental data in Appendix B.

To model turbulence, the refractive index structure parameter of turbulence C_n^2 is calculated as a function of altitude h in m, root mean square wind speed v_{rms} in m/s, and ground-level C_n^2 in m$^{-2/3}$, $C_n^2(0)$ as [91],

$$\begin{aligned} C_n^2(h) = (&0.00594(v_{rms}/27)^2\ (h \times 10^{-5})^{10}\ \exp(-h/1000) \\ &+ 2.7 \times 10^{-16}\ \exp(-h/1500) + C_n^2(0)\exp(-h/100)), \end{aligned} \tag{4.6}$$

where the root mean square wind speed is given by,

$$v_{rms} = \left[\frac{1}{15 \times 10^3}\int_{5\times10^3}^{20\times10^3} v^2(h)\ dh\right]^{1/2}, \tag{4.7}$$

with $v(h)$ representing altitude-dependent wind speed in m/s. The Bufton wind profile [91] is adopted to simulate $v(h)$, derived as a function of altitude as,

$$v(h) = v_g + 30\exp\left[-\left(\frac{h-9400}{4800}\right)^2\right], \tag{4.8}$$

where v_g is the ground-level wind speed in m/s. Note that slew rate is not considered. The system parameters for our simulations are defined in Table 4.2.

Coherence time τ represents the duration of time for which channel conditions can be

Table 4.2: Simulation parameters.

Parameter	**Value**
Satellite altitude (H)	500 km
Receiver altitude (h_0)	2 km
Outer scale of turbulence (L_0)	5 m
Inner scale of turbulence (l_0)	0.025 m
Satellite zenith angle (θ_z)	0°
Laser wavelength (λ)	1550 nm
Beam waist radius (w_0)	0.15 m
Receiver radius (R_r)	1.25 m
Pulse transmission rate ($\mathcal{T}_R$)	1 MHz
Modulation variance (V_{mod})	10 SNU
Ground-level wind speed (v_g)	2.3 - 5.0 m/s
Ground-level C_n^2 ($C_n^2(0)$)	1.7×10^{-15} - 10^{-14} m$^{-2/3}$

considered as constant. For satellite-to-Earth atmospheric channels, the coherence time is modeled as a function of $C_n^2(h)$ (Eq. 4.6), the transverse wind profile $v(h)$ (Eq. 4.8), and optical wavelength λ in m, and is calculated as [100],

$$\tau = \left(118\lambda^{-2} \int_0^{+\infty} C_n^2(h)\ v^{5/3}(h)\ dh\right)^{-3/5}, \tag{4.9}$$

where v_g and $C_n^2(0)$ are randomly sampled from uniform distributions (defined in Table 4.2) to generate a phase screen model for each coherence time. Transmissivity is calculated from the results as $T = (P_R/P_T)$, where P_T and P_R represent the power at the transmitter and receiver, respectively.

4.3 Neural Network Architecture

4.3.1 Long Short-Term Memory Model

The signal phase error estimation algorithm needs to map the time-varying relationship between $\Delta\phi_R$ and $\Delta\phi_S$, estimating $\Delta\phi_S$ for each pulse by taking N $\Delta\phi_R$ measurements as input. An LSTM NN is a type of recurrent NN, which incorporates both long and short-term memory components. We provide only a brief overview here of LSTM NNs; for a detailed description see [101]. The N previous $\Delta\phi_R$ measurements can be used to output an estimate, $\Delta\tilde{\phi}_S$, of $\Delta\phi_S$ for the current pulse, where N also defines the number of LSTM units within the NN. An LSTM unit is a structure that contains three different gate types - the combination of which controls the information stored in the long-term and short-term memories. Normally, the long-term and short-term memories are abstracted as the vectors c_t and h_t, respectively, both vectors of length z_{dim} [101].

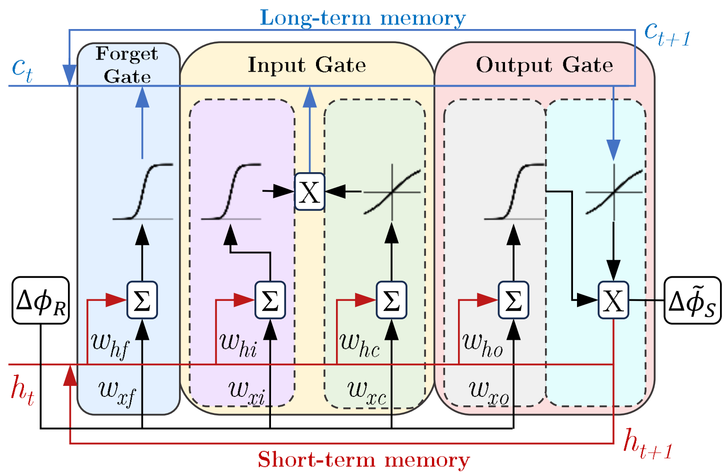

Figure 4.2: LSTM unit architecture, showing the forget, input, and output gates. Summations are represented by Σ, dot-products by X, and the time steps are defined by the subscript t. Sigmoid and tanh activation functions are shown within the gates.

A schematic of the LSTM unit designed for this work is given in Figure 4.2. In Figure 4.2, w_{xf}, w_{xi}, w_{xc}, and w_{xo} are weight vectors of length z_{dim}, and w_{hf}, w_{hi}, w_{hc}, and w_{ho} are weight matrices of dimension $z_{dim} \times z_{dim}$. During run-time, each element of the weight vectors is multiplied by $\Delta\phi_R$, and each element of the weight matrices is multiplied by h_t. For each pulse, the measurement $\Delta\phi_R$ is input into the unit. The Forget Gate multiplies each value in c_t by a number between 0 and 1, where the number is determined via a sigmoid activation function that takes as input w_{xf} and w_{hf}. The Input Gate updates c_t in a two-step process involving both a sigmoid and the tanh function, as illustrated. The Output Gate updates h_t, also through a two-step process (as illustrated), except with the incorporation of information from c_t. The Output Gate also passes c_t and h_t to the next time step, and outputs an estimate $\Delta\tilde{\phi}_S$ via a fully-connected layer. Essentially, the NN updates itself in real-time to the previous N measurements of $\Delta\phi_R$, learning long-term and short-term dependencies of the fluctuating relationship with $\Delta\phi_S$. The output estimate $\Delta\tilde{\phi}_S$ for each pulse is then used to phase correct the RLO_2 for coherent measurement of the signal.

4.3.2 Scale and Training

Given that only z_{dim} and N are varied between NN models, we compare their complexity in terms of the number of free parameters in the model (e.g. such as in [102]), where we define the NN architecture scale S as $S = 4N\left[(z_{dim}+1)z_{dim} + z_{dim}\right]$ [101].

All NNs are trained using approximately a million measurements of input $\Delta\phi_R$ and output $\Delta\phi_S$, representing approximately a hundred coherence times, where $\Delta\phi_S$ is sampled from the PDF $\mathcal{N}(0.2618, 0.0524)$ for each coherence time, and each measurement of $\Delta\phi_R$ is calculated using the phase errors defined in Section 4.2.1. The trained NNs are then tested on approximately a million input $\Delta\phi_R$ measurements, outputting an estimate $\Delta\tilde{\phi}_S$ for each pulse.

4.3.3 Quantum Cramér-Rao Bound

In testing the performance of our NNs, it will be useful to consider the optimal performance, which is attained via the quantum Cramér-Rao bound (QCRB). The QCRB is the ultimate estimation uncertainty bound with which a parameter ($\Delta\phi_S$) can be measured if all technical noise sources are eliminated, the measurements are completely ideal, and the states are perfectly prepared each time [103].

For a single-mode coherent state, the quantum Fisher information $\mathcal{I}_Q$ for phase estimation is calculated as [62],

$$\mathcal{I}_Q = \left(\frac{\partial d}{\partial\Delta\phi'}\right)^T \Sigma^{-1}\frac{\partial d}{\partial\Delta\phi'} + \mathrm{Tr}\left[\left(\frac{\partial\Sigma}{\partial\Delta\phi'}\Sigma^{-1}\right)^2\right], \tag{4.10}$$

where $\Delta\phi' = \Delta\phi_R - \Delta\phi_S$, d represents the displacement vector,

$$d = \frac{1}{2}|\alpha_S|\{\cos\Delta\phi', -\sin\Delta\phi', \sin\Delta\phi', \cos\Delta\phi'\}^T, \tag{4.11}$$

and Σ is the covariance matrix,

$$\Sigma_{ij} = \frac{1}{2}\langle\mathcal{X}_i\mathcal{X}_j + \mathcal{X}_j\mathcal{X}_i\rangle - \langle\mathcal{X}_i\rangle\langle\mathcal{X}_j\rangle, \tag{4.12}$$

for the quadratures vector $\mathcal{X} = [X_S, P_S]$. The QCRB is then calculated as

QCRB $\geq 1/(N\mathcal{I}_Q)$. For a single-mode coherent state used for phase estimation, the QCRB becomes [62],

$$\text{QCRB} \geq \frac{1}{2N|\alpha_S|^2}, \tag{4.13}$$

where $|\alpha_S|$ is the signal amplitude, $|\alpha_S| = \sqrt{X_S^2 + P_S^2}$.

4.4 Neural Network Analysis

4.4.1 Case 1

Case 1 focuses on the Gaussian phase noise (described in Section 4.2.1), where the NN essentially learns to estimate $\Delta\phi_S$ by taking the mean of the previous N measurements of $\Delta\phi_R$.

The performance of the NN-based $\Delta\phi_S$ estimation is quantified by the estimation error variance between the estimate $\Delta\tilde{\phi}_S$ and true $\Delta\phi_S$, $\text{Var}(\Delta\tilde{\phi}_S - \Delta\phi_S)$, where $\text{Var}(\Delta\tilde{\phi}_S - \Delta\phi_S) = 0$ would represent perfect $\Delta\phi_S$ estimation. When the approximation $\Delta\phi_R \approx \Delta\phi_S$ is applied, then the estimation error variance is $\text{Var}(\Delta\phi_R - \Delta\phi_S)$, while for the expectation estimator, the estimation error variance is $\text{Var}(\langle\Delta\phi_R\rangle_N - \Delta\phi_S)$.

Table 4.3 outlines the different NN architectures investigated for Case 1. # represents the assigned NN model numbers, with Model 0 representing the approximation $\Delta\phi_R \approx \Delta\phi_S$ (included for comparison). Taking the results of Model 3 as an example, $\text{Var}(\Delta\phi_R - \Delta\phi_S) = 0.1123$ rad^2 is reduced to $\text{Var}(\Delta\tilde{\phi}_S - \Delta\phi_S) = 0.0033$ rad^2 after the NN output $\Delta\tilde{\phi}_S$ is applied to the RLO_2. The NN estimation performance matches the expectation estimator at $\text{Var}(\langle\Delta\phi_R\rangle_N - \Delta\phi_S) = 0.0033$ rad^2, indicating that the NN is able to learn to take the mean of the previous N measurements of $\Delta\phi_R$ for the Gaussian phase noise.

Figure 4.3 shows the number of $\Delta\phi_R$ measurements N versus $\text{Var}(\Delta\tilde{\phi}_S - \Delta\phi_S)$ for each of the Case 1 NN models, as well as the QCRB. The models are defined by their scale, with Case 1 results shown with solid markers, while the NN model Case 2 results are shown with hollow markers, with the expectation estimator results for Case 2 plotted as hollow blue star markers (discussed in Section 4.4.2). It can be seen that none of the $\Delta\phi_S$ estimation models are able to reach the QCRB given the imperfections of the receiver,

Table 4.3: Case 1 neural network architecture performance.

#	N	z_{dim}	S	**Est. Error Variance** (rad^2)
0	-	-	-	0.1123 rad^2
1	20	4	2,020	0.0055 rad^2
2	20	32	87,700	0.0052 rad^2
3	40	4	4,040	0.0033 rad^2
4	60	4	6,060	0.0022 rad^2
5	80	4	8,080	0.0016 rad^2
6	100	4	10,100	0.0015 rad^2
7	100	32	438,500	0.0014 rad^2

though the results for $N = 20$ approach it. The estimation error variances of the Case 1 NN models appear to follow the same trend as the QCRB.

First we fix the number of $\Delta\phi_R$ measurements at $N = 100$, then vary the number of hidden units from $z_{dim} = 4$ to $z_{dim} = 32$. The increased estimation performance of Model 6, compared to Model 7, is negligible, as featured in the Figure 4.3 inset (b), while the scale increases from $S = 10,100$ to $S = 438,500$. The same range of z_{dim} is also tested for $N = 20$, with a slightly greater difference in estimation performance, as featured in Figure 4.3 inset (a). The number of $\Delta\phi_R$ measurements is also varied between $N = 20$ and $N = 100$, fixing $z_{dim} = 4$. Overall, it is shown that increasing the number of $\Delta\phi_R$ measurements has a greater increase in $\Delta\phi_S$ estimation performance than increasing z_{dim} within each cell.

The increase in $\Delta\phi_S$ estimation performance for Model 3 (outlined with the green oval in Figure 4.3), compared to Models 1-2, while only having a scale of $S = 4,040$, is more significant than the increase in estimation performance for Models 4-7. As such, Model 3 presents a low-complexity signal phase error estimation algorithm, which satisfies the performance requirements of a practical CV-QKD system.

The real-time performance of the NN-based signal phase error estimation algorithms depends on several factors, including the computational hardware, whether the model is pre-compiled, and the programming language used. For example, using the LSTM encoded in Python with Tensorflow and a single GeForce RTX 4090 GPU, the latency of a single signal phase error estimation is 0.52 ms for $N = 20$ and $z_{dim} = 4$, and 0.87 ms for $N = 100$ and $z_{dim} = 32$. However, we assume that implementing the signal phase

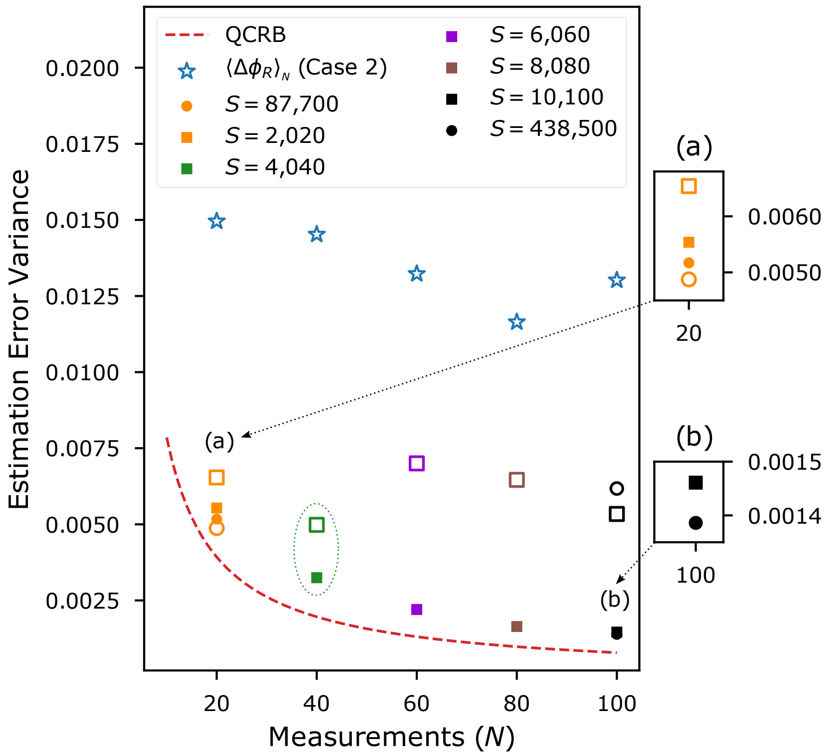

Figure 4.3: N measurements of $\Delta\phi_R$ versus estimation error variance for NN architectures tested, defined by their scale S, as well as the QCRB. Case 1 NN results are shown with solid markers, Case 2 NN results are shown with hollow markers, and Case 2 expectation estimator $\langle\Delta\phi_R\rangle_N$ results are shown with hollow blue star markers. Inset (**a**) highlights results for Models 1-2 and 9-10, and inset (**b**) highlights the results for Models 6-7.

error estimation algorithms in an operational CV-QKD system would involve encoding it into a lower-level language, where estimation latencies would drop significantly, and the increased complexity of the models would have a greater effect.

4.4.2 Case 2

In Case 2, we introduce an additional relative phase error component between $\Delta\phi_R$ and $\Delta\phi_S$, modeling a relationship between $\Delta\phi_R$ and $\Delta\phi_S$ that remains constant during each coherence time. It is assumed that the relationship between $\Delta\phi_R$ and $\Delta\phi_S$ could be represented by any function, where the LSTM should be able to map $\Delta\phi_R$ to $\Delta\phi_S$ for varying relationships. For demonstrative purposes, we introduce a Gaussian mixture model for the $\Delta\phi_R$ measurements using the additional PDF $\mathcal{N}(0.0000, 0.0524)$. The Gaussian phase noise from Case 1 still applies to every $\Delta\phi_R$ measurement. The same NN architectures as in Case 1 are tested for Case 2, outlined in Table 4.4. Model 8 represents the approximation $\Delta\phi_R \approx \Delta\phi_S$.

Table 4.4: Case 2 neural network architecture performance.

#	N	z_{dim}	S	**Est. Error Variance**
8	-	-	-	0.1294 rad^2
9	20	4	2,020	0.0065 rad^2
10	20	32	87,700	0.0049 rad^2
11	40	4	4,040	0.0050 rad^2
12	60	4	6,060	0.0070 rad^2
13	80	4	8,080	0.0065 rad^2
14	100	4	10,100	0.0053 rad^2
15	100	32	438,500	0.0062 rad^2

Figure 4.4 gives an example of the NN output $\Delta\tilde{\phi}_S$ for Model 11, with $N = 40$ and $z_{dim} = 4$ resulting in a scale of $S = 4,040$. The $\Delta\phi_R$ measurements are shown as the grey line, the $\Delta\phi_S$ values as the red line, the $\Delta\tilde{\phi}_S$ estimations by the black line, and the expectation estimator results by the blue line. The estimation performance of Model 11 reduces $\mathrm{Var}(\Delta\phi_R - \Delta\phi_S) = 0.1294$ rad^2 to $\mathrm{Var}(\Delta\tilde{\phi}_S - \Delta\phi_S) = 0.0050$ rad^2. The expectation estimator results in $\mathrm{Var}(\langle\Delta\phi_R\rangle_N - \Delta\phi_S) = 0.0145$ rad^2, highlighting the advantage of our NN-based signal phase error estimation algorithm over standard approaches when the relationship between $\Delta\phi_R$ and $\Delta\phi_S$ becomes more complex, as can be seen in Figure 4.4 by comparing the black line ($\Delta\tilde{\phi}_S$) with the blue line ($\langle\Delta\phi_R\rangle_N$).

Returning to Figure 4.3, we find that the $\Delta\phi_S$ estimation performance for Case 2 (shown as the hollow markers) has a different trend than the $\Delta\phi_S$ estimation performance for Case 1. With the $\Delta\phi_R$ measurements fixed at $N = 20$, there is a larger increase in estimation performance when hidden units are increased from $z_{dim} = 4$ to $z_{dim} = 32$, as shown in Figure 4.3 inset (a). On the contrary, when the $\Delta\phi_R$ measurements are fixed at $N = 100$, Model 14 ($z_{dim} = 4$) actually performs better than Model 15 ($z_{dim} = 32$). When the $\Delta\phi_R$ measurements are varied from $N = 20$ to $N = 100$, it can be seen that the estimation performance initially increases with N, before reducing above $N = 40$. Thus, the NN is better able to learn the relationship between $\Delta\phi_R$ and $\Delta\phi_S$ for lower N. It is shown that the NN-based signal error estimation algorithm outperforms the expectation estimator (plotted as blue stars in Figure 4.3) across all values of N.

After comparing different architectures for the NN-based signal phase error estimation

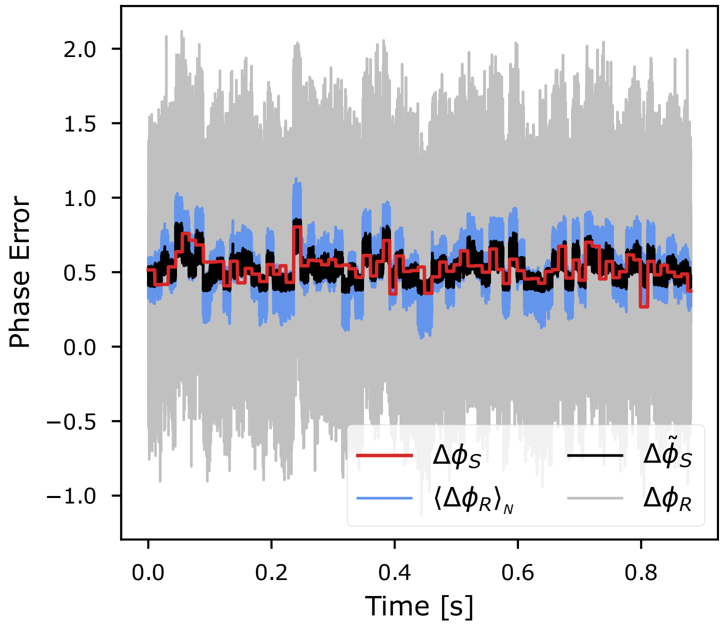

Figure 4.4: Signal phase error estimation results for Case 2 using a Model 11 example. The $\Delta\phi_R$ measurements are plotted as the grey line, the true $\Delta\phi_S$ values as a red line, the NN estimate $\Delta\tilde{\phi}_S$ as a black line, and the expectation estimator output as the blue line.

algorithms, we show that an efficient low-complexity model can be obtained. *This is the main result of this chapter* - the conclusion that NNs, an order of magnitude reduced in complexity, can provide for suitable performance levels. For context, the previous work [12] utilizes an LSTM for phase estimation, with an architecture of $N = 3$ and $z_{dim} = 100$, resulting in a scale of $S = 122,400$. While it is assumed that each CV-QKD setup will have different phase error relationships, similar estimation performance could potentially be achieved with a lower complexity model. For example, this chapter shows a limited increase in $\Delta\phi_S$ estimation performance for the higher complexity models with the same value of N (see Figure 4.3 insets). For Case 2, the $\Delta\phi_S$ estimation performance of Model 11 outperforms the higher complexity Models 12-15, while achieving similar performance to Model 10, which has an order of magnitude greater scale. We note that exact relative phase error reduction requirements will depend on the CV-QKD system and channel parameters. However, our low-complexity NN-based signal phase error estimation algorithm is capable of satisfying the requirements for a practical real-time CV-QKD system, improving the computational speed of the algorithm by reducing the NN scale, while achieving sufficient $\Delta\phi_S$ estimation. The results in this chapter reduce the computational requirements for relative phase error correction, which is essential for real-time CV-QKD.

4.5 Summary

This chapter analyzed long short-term memory neural network architectures for temporal signal phase error estimation, designed for a satellite-to-Earth continuous-variable quantum key distribution system, resulting in relative phase error correction. The results showed that adequate signal phase error estimation can be achieved using a low-complexity long short-term memory neural network architecture, with the machine learning-based approach outperforming standard models when the relationship between the reference pulse phase error and signal phase error became more complex. Decreasing the complexity of machine learning-based signal phase error estimation algorithms remains vital for achieving real-time continuous-variable quantum key distribution across atmospheric channels.

The temporal relative phase errors between the reference pulses and signals in this chapter were one-dimensional, resulting from homodyne/heterodyne detection of a real local oscillator and the signals. In Chapter 5, we instead investigate two-dimensional spatial relative wavefront errors between the reference pulses and signals, and develop machine learning-based algorithms to correct for them. Specifically, we simulate the use of multi-plane light conversion to analyze mode-specific relative phase errors, then analyze the resulting effects on secure key rate transfer, in the context of continuous-variable quantum key distribution across satellite-to-Earth channels.

Chapter 5

Mode Phase Error Correction via Machine Learning for Satellite-to-Earth CV-QKD

Machine learning was shown to assist in correcting one-dimensional temporal relative phase errors in continuous-variable key distribution in Chapter 4, where a neural network complexity analysis demonstrated that a relatively low-complexity signal phase error estimation model was capable of providing adequate relative phase error correction across satellite-to-Earth channels. In this chapter, we instead propose the existence of two-dimensional spatial relative wavefront errors between the reference pulses and signals in continuous-variable quantum key distribution across turbulent atmospheric channels. We discuss potential sources of relative wavefront errors, simulate several different cases of relative wavefront errors, and develop machine learning-based intelligent wavefront correction algorithms to mitigate their deleterious effects on secure key transfer. Note that the phase screen model adopted in this chapter is validated in Appendix B, and that the nomenclature used in this chapter applies solely to the content within this chapter.

Introduction

Satellite-Earth continuous-variable QKD (CV-QKD) involves encoding secret key information on the quadratures of the electric field of weak quantum signals (hereafter, referred to as "signals"), followed by transmission through turbulent atmospheric channels. In order to measure the signals, state-of-the-art CV-QKD protocols transmit reference pulses with the signals, which are measured at the receiver, and then the measured information is encoded onto a real local oscillator (RLO) generated at the receiver [15], [16], [104]. As a first approximation, it is assumed that the reference pulses and signals have the same phase after passing through an atmospheric channel; however, previous work reveals that there could exist a relative phase error between them after homodyne/heterodyne detection [15], [72] — an issue we addressed in Chapter 4. Although one-dimensional relative phase errors in the time-domain have been analyzed previously in this context, no work has explored spatial wavefront errors (WFEs) between the phase wavefronts of the reference pulses and signals. This topic is the focus of this chapter.

Shack-Hartmann wavefront sensors (SH-WFSs) are commonly used for optical wavefront detection. A major disadvantage of SH-WFSs is their sensitivity to non-uniform illumination, where turbulence regimes resulting in high-scintillation lead to uneven or insufficient light being received by a portion of the SH-WFS subapertures, resulting in inaccurate centroid determination [105], [106]. High levels of scintillation are possible in satellite-Earth channels (e.g. see [107]), such that SH-WFS performance may deteriorate. Further, sharp phase wavefront gradients can lead to ambiguities when using the slope-based wavefront reconstruction methods of SH-WFSs, which can also be present across satellite-Earth channels (e.g. see Chapter 3). Multi-plane light converters (MPLCs) are emerging as an alternative to SH-WFS as wavefront sensors for adaptive optics in free-space optical communications [108], where MPLCs are capable of decomposing an electric field into an arbitrary spatial mode basis. Modal sensing can minimize wavefront sensing errors, resulting from uneven illumination and phase gradient discontinuities. As such, we explore the use of an MPLC for wavefront detection, where MPLCs have previously been implemented for wavefront correction in free-space optical communications [109], [110]. Specifically, this chapter investigates the relative WFEs between Hermite-Gaussian (HG) modes (hereafter, referred to as "relative mode phase errors (MPEs)") by simulating the decomposition of reference pulse and signal wavefronts into the HG mode basis at the receiver.

Machine learning (ML), having been shown to be useful in estimating phase errors in the one-dimensional time-domain in CV-QKD (see Chapters 2 and 4), offers a potential solution to estimate relative MPEs. A neural network (NN) can be used to map the relationship between reference pulse MPEs and signal MPEs. The trained model can then take the reference pulse MPEs as input and output an estimate of the signal MPEs, without the need for assumption-based physical models or computationally intensive numerical search algorithms. Here, we design transformer NNs (also termed intelligent wavefront correction algorithms) to estimate the signal MPEs, which are then used to reconstruct the original signal wavefront. This corrected wavefront is mapped onto an RLO to compensate for the relative WFE between the reference pulses and the signals. The RLO is then mixed with the signal, using balanced homodyne detection, to recover the secure key information encoded on the signal quadratures.

The main aim of this chapter is to demonstrate how our intelligent wavefront correction algorithms can deliver improved secure key rates when a relative WFE between the reference pulses and signals is present, with no negative impact on the key rates otherwise. We argue that their inclusion in next-generation CV-QKD systems for the satellite-Earth channels will assist in the ongoing development of the Quantum Internet.

Note that in the following exposition, the signals are assumed to be in the form of weak pulses and the reference pulses in the form of strong pulses. Further, although our intelligent wavefront correction algorithms can be applied both in the downlink and the uplink of satellite-Earth channels, here we will focus entirely on the downlink channel, the satellite-to-Earth channel. The Earth-to-satellite uplink channel will have poorer transmissivity, and quantum communication through it could be better served via teleportation [111].

The rest of this chapter is presented as follows: a system overview for the CV-QKD protocol and MPLC hardware is given in Section 5.1, a description of satellite-Earth phase screen simulations is given in Section 5.2, the WFE theory is outlined in Section 5.3, a discussion of the NNs is given in Section 5.4, the NN estimation performance is shown in Section 5.5, wavefront correction results are summarized in Section 5.6, an analysis of the effects on secure key rates is presented in Section 5.7, and concluding remarks are made in Section 5.8.

5.1 System Overview

5.1.1 CV-QKD Protocol

In our RLO-based CV-QKD scheme, based on a Gaussian modulated coherent state CV-QKD protocol, secure key information is encoded on the signal quadratures on the satellite, which we call Alice (subscript A). The signals are polarization-multiplexed with reference pulses, then transmitted across satellite-to-Earth channels, where it has previously been assumed that the reference pulse phase WFE $\Delta\Phi_R(x, y)$ experiences the same distortion across the atmospheric channel as the signal phase WFE $\Delta\Phi_S(x, y)$, $\Delta\Phi_R(x, y) \approx \Delta\Phi_S(x, y)$ for each pulse. Bob (subscript B) demultiplexes the reference pulses and signals at the receiver, then measures $\Delta\Phi_R(x, y)$ using a wavefront sensor and maps $\Delta\Phi_R(x, y)$ onto an RLO, which is used to measure the signal through homodyne detection.

However, as discussed in more detail later (see Section 5.3), in real-world transmissions of multiplexed reference pulses and signals, the difference between $\Delta\Phi_R(x, y)$ and $\Delta\Phi_S(x, y)$ could be non-zero. Therefore, mapping an estimation of the true value of $\Delta\Phi_S(x, y)$ onto the RLO could result in greater coherence between the RLO and the signal, than with the *approximation* $\Delta\Phi_R(x, y) \approx \Delta\Phi_S(x, y)$, when measuring the quadrature values of the signal, leading to higher secure key rates through the channel.

Although several wavefront sensing technologies exist, we focus on the use of an MPLC, which is capable of decomposing an electric field into weighted HG modes (or any other basis transformation). The reference pulse HG mode phase measurements in each mode $\Delta\phi_{mn,R}$, termed the reference pulse MPEs, can then be used to *estimate* the signal HG mode phase measurements in each mode $\Delta\phi_{mn,S}$, termed the signal MPEs, where we construct NN-based intelligent wavefront correction algorithms to estimate $\Delta\phi_{mn,S}$ (rather than just using $\Delta\phi_{mn,R}$ as an approximation of $\Delta\phi_{mn,S}$). Estimates $\Delta\tilde{\phi}_{mn,S}$, termed the estimated signal MPEs of $\Delta\phi_{mn,S}$, are then used to reconstruct an estimation of $\Delta\Phi_S(x, y)$, which is mapped on the RLO using a deformable mirror, as shown in our schematic of the CV-QKD protocol in Figure 5.1.

In brief, the practical steps required to implement our intelligent wavefront correction algorithm, using the RLO-based CV-QKD protocol, are as follows.

Training:

1. Transmit multiplexed reference pulses and signals across the channel.

2. Split reference pulses and signals at receiver.

3. Measure reference pulses and signals independently using an MPLC.

4. Train neural network with $\Delta\phi_{mn,R}$ and $\Delta\phi_{mn,S}$ measurements.

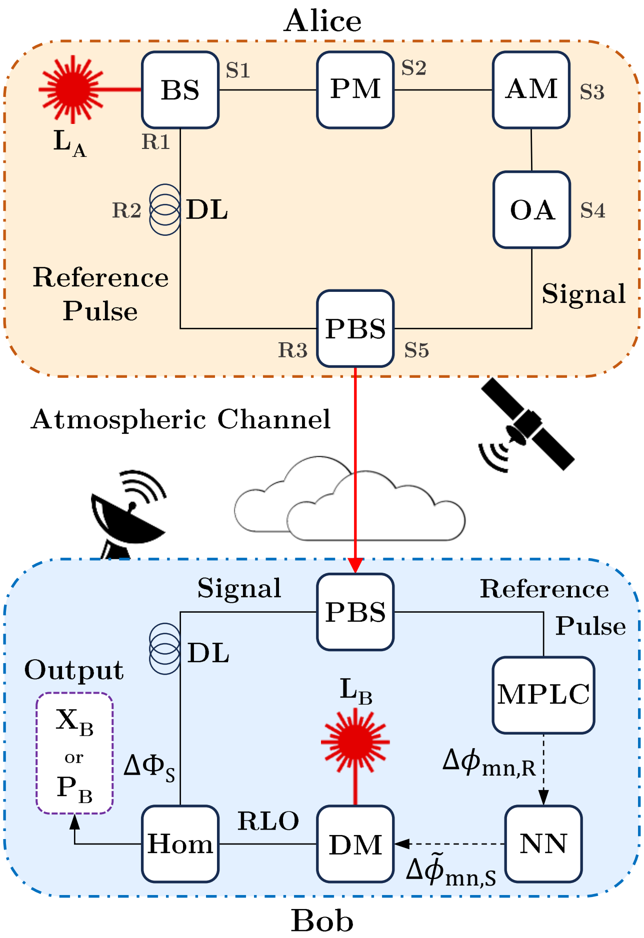

Figure 5.1: Schematic outlining the preparation and measurement of coherent states with our intelligent wavefront correction algorithm. L is a laser source, BS is a beamsplitter, AM is an amplitude modulator, PM is a phase modulator, OA is an optical attenuator, PBS is a polarized beamsplitter, MPLC is a multi-plane light converter, Hom is a homodyne detector, DM is a deformable mirror, DL is a delay line, and NN is a neural network. Zernike WFEs are applied to the signals at S1-S5 and reference pulses at R1-R3. The quadrature measurements X_B or P_B contain the secure key information.

Operation:

5. Transmit multiplexed reference pulses and signals across the channel.

6. Split reference pulses and signals at receiver.

7. Measure *only* reference pulses using an MPLC.

8. Input $\Delta\phi_{mn,R}$ to NN, outputting estimates $\Delta\tilde{\phi}_{mn,S}$.

9. Use $\Delta\tilde{\phi}_{mn,S}$ to map estimated signal phase WFE onto the RLO.

10. Measure signal via balanced homodyne detection using the RLO.

For clarity, we outline the full nomenclature of WFE types in this work in Table 5.1. Note that the MPEs $\Delta\phi_{mn,R}$ and $\Delta\phi_{mn,S}$ represent the phase measurements in each HG mode for the reference pulses and signals at the output of the MPLC, regardless of the type of WFE applied to the hardware before measurement (described in Section 5.3), while the estimated signal MPEs $\Delta\tilde{\phi}_{mn,S}$ represent the estimates produced by the intelligent wavefront correction algorithms.

5.1.2 Multi-Plane Light Conversion

An electric field can be described using any arbitrary spatial mode basis, where a beam passing through an atmospheric channel can be defined as a linear combination of the mode basis with a complex phase factor varying over time due to turbulence fluctuations in the channel.

An MPLC is a device capable of demultiplexing an electric field into N spatial modes. A series of phase masks, separated by free space, are used to transform the input electric field into N output spots of the fundamental Gaussian mode, which are then coupled to separate single-mode fibers (SMFs) [108]. While MPLCs can be designed to decompose an electric field into any spatial mode basis, we analyze decomposition into the HG basis, advancing previous works which have shown wavefront correction across turbulent atmospheric optical channels using an HG mode-based MPLC [109], [110].

HG modes represent solutions to the paraxial wave equation [112], and are a function of the wavelength λ, the beam waist w_0, and the location along the optical axis of the waist, z_0. HG modes can be described using beam radius $w(z)$, calculated as,

$$w(z) = w_0\sqrt{1 + \frac{(z - z_0)^2}{z_R^2}}, \tag{5.1}$$

where the radius of the curvature of the beam wavefront is $R(z) = (z - z_0) + z_R^2/(z - z_0)$, and the Gouy phase is $\psi(z) = \arctan([z - z_0]/z_R)$, representing an additional phase per mode due to its deviation from a plane wave. Both are calculated using the Rayleigh

Table 5.1: Wavefront error nomenclature.

Symbol	Name	Definition
$\Delta\Phi_R(x, y)$	Reference pulse phase WFE	Difference between the reference pulse phase at $\{x, y\}$ at the laser source and receiver.
$\Delta\Phi_S(x, y)$	Signal phase WFE	Difference between the signal phase at $\{x, y\}$ at the laser source and receiver.
$\Delta\phi_{mn}$	HG mode phase	Generic HG mode difference between the measured phase and zero phase, as defined in Eq. 5.5. Note that we replace $\Delta\phi_{mn}$ with $\Delta\phi_{mn,R}$, $\Delta\phi_{mn,S}$, $\Delta\theta_{mn,R}$, and $\Delta\theta_{mn,S}$ in Eq. 5.5 when calculating HG mode decompositions.
$\Delta\phi_{mn,R}$	Reference pulse MPEs	Difference between the reference pulse HG mode phase at the laser source and reference pulse MPLC output HG mode phase at the receiver (see Eq. 5.5), independent of the type of WFEs applied at transmitter or receiver.
$\Delta\phi_{mn,S}$	Signal MPEs	Difference between the signal HG mode phase at the laser source and signal MPLC output HG mode phase at the receiver (see Eq. 5.5), independent of the type of WFEs applied at transmitter or receiver.
$\Delta\tilde{\phi}_{mn,S}$	Estimated signal MPEs	Corrected (see Section 5.4), independent of the type of WFEs applied at transmitter or receiver.
$\Delta\phi_{mn,R} - \Delta\phi_{mn,S}$	Relative MPEs	Difference between the reference pulse and signal MPLC output HG mode phase at the receiver.

$\Delta\Phi(\rho, \beta)$	Zernike WFEs	2nd- and 3rd-order WFEs caused by imperfect optical hardware at the transmitter, defined using Zernike polynomials (see Section 5.3.2).
$\Delta\theta_{mn,R}$	Reference pulse transmit MPEs	Difference between the received reference pulse HG mode phase after Zernike WFEs are applied at the transmitter (without cross-leakage MPEs) (see Section 5.3.2).
$\Delta\theta_{mn,S}$	Signal transmit MPEs	Difference between the received signal HG mode phase after Zernike WFEs are applied at the transmitter (without cross-leakage MPEs) (see Section 5.3.2).
$\Delta\theta_{mn,X}$	Cross-leakage MPEs	Difference between the signal HG mode phase and the reference pulse HG mode phase caused by photon leakage from the reference pulses to signals, and including crosstalk from the MPLC (without reference pulse and signal transmit MPEs) (see Section 5.3.3).
$\mathrm{Var}(\Delta\phi_{mn,R} - \Delta\phi_{mn,S})$	Default variance	Variance of the relative MPEs (see Section 5.5).
$\mathrm{Var}(\Delta\tilde{\phi}_{mn,S} - \Delta\phi_{mn,S})$	Correction variance	Variance of the difference between the estimated signal MPEs and signal MPEs (see Section 5.5).

range $z_R = \pi w_0^2/\lambda$. The normalized non-astigmatic HG modes are defined as [108],

$$\mathrm{HG}_{mn}(x,y,z) = \frac{e^{-\frac{\rho^2}{w(z)^2} - i\left(\frac{k\rho^2}{2R(z)} + k(z-z_0) - (n+m+1)\psi(z)\right)}}{\sqrt{2^{n+m-1}\pi n! m!} w(z)} \cdot H_m\left(\frac{\sqrt{2}x}{w(z)}\right) \cdot H_n\left(\frac{\sqrt{2}y}{w(z)}\right), \quad (5.2)$$

where the wavenumber is $k = 2\pi/\lambda$ and the radial distance ρ is given by the Cartesian coordinates $\{x, y\}$ as $\rho^2 = x^2 + y^2$. The modes are spatially modulated using the Hermite polynomials H_m (or H_n),

$$H_m(x) = m! \sum_{j=0}^{\lfloor \frac{m}{2} \rfloor} \frac{(-1)^j}{j!(m-2j)} \frac{x^{m-2j}}{2^j}, \quad (5.3)$$

where $\lfloor\ \rfloor$ represents the floor function. The order of HG modes H_{mn} is $m+n$, resulting in indices $0, 1, \cdots, N$.

Any optical beam $E(x, y)$ can then be expressed in terms of an infinite linear combination of orthogonal HG modes [108],

$$E(x,y) = \sum_{mn}^{\infty} a_{mn} \mathrm{HG}_{mn} e^{i\Delta\phi_{mn}} = \sum_{mn}^{\infty} c_{mn} \mathrm{HG}_{mn}. \quad (5.4)$$

The HG mode complex coefficients c_{mn} are computed as follows,

$$c_{mn} = a_{mn} e^{i\Delta\phi_{mn}} = \frac{\iint E(x,y) \mathrm{HG}^*_{mn}(x,y) dxdy}{\iint \mathrm{HG}_{mn}(x,y) \mathrm{HG}^*_{mn}(x,y) dxdy}, \quad (5.5)$$

with amplitude a_{mn} and phase $\Delta\phi_{mn}$, where we generically define the phase here - later replacing $\Delta\phi_{mn}$ with $\Delta\phi_{mn,S}$, $\Delta\phi_{mn,R}$, $\Delta\theta_{mn,S}$, or $\Delta\theta_{mn,R}$ when simulating decomposition of the electric fields of the reference pulses and signals into the HG basis. Once each HG mode is coupled into a separate SMF, scaled by c_{mn}, then the voltage of each mode V_{mn} is measured using a photodetector, which is proportional to the power in each mode P_{mn},

$$V_{mn} \propto P_{mn} = |c_{mn}|^2. \quad (5.6)$$

Note that interferometry (e.g. heterodyne detection) of each mode would be required to measure $\Delta\phi_{mn}$.

Figure 5.2 shows five HG modes, where the transverse intensity profiles of each mode are shown on the top row, while the corresponding transverse phase profiles are shown on the bottom row. In this work, we examine CV-QKD performance for three different numbers of modes: $N = 10$, $N = 30$, and $N = 50$.

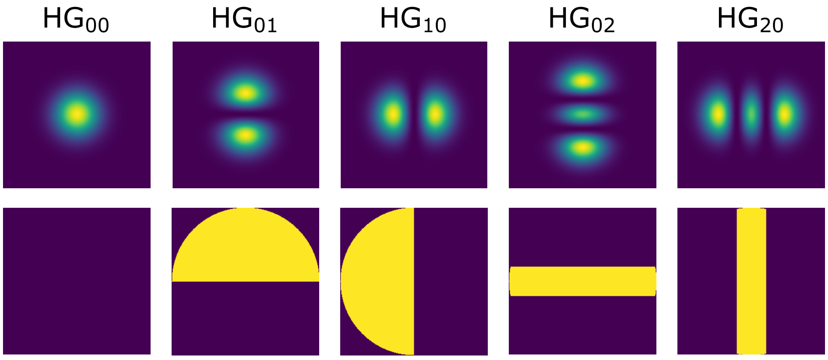

Figure 5.2: Hermite-Gaussian modes, with the transverse intensity profiles on the top row and the transverse phase profiles on the bottom row (for an undefined channel).

5.2 Satellite-to-Earth Channel

We implemented a phase screen model to simulate the propagation of the reference pulses and signals through atmospheric channels from the satellite to the Earth, based on [90], [113], [114]. A phase screen model utilizing Monte Carlo simulations with a Fresnel split-step propagator is detailed in the work [113]. Essentially, a series of phase screens are used to represent a turbulent atmospheric volume by assuming that an optical beam travels as if in a vacuum until it interacts with a phase screen at its center. The phase screen then perturbs the phase of the beam wavefront as it passes through it. The methodology of [113] is implemented as a Python package in [90]. Note that the phase screen model is validated against experimental data in Appendix B.

The works [113] and [90] both assume that the atmosphere has homogeneous turbulence properties along the propagation path. Although this assumption may be reasonable for beam propagation across short horizontal distances, the atmosphere changes substantially when considering the vertical distances from a satellite to the Earth [114].

For satellite-to-Earth channels, the distribution of phase screens is modified as the density of the atmosphere reduces with increased altitude [115], resulting in different turbulence properties, where we base the satellite-to-Earth phase screen modeling on [114] (as implemented in [116]). The atmosphere is divided into N_s slabs, bounded by the altitudes h_j and h_{j-1} in meters, where j increases in altitude from 1 to N_s. The thickness of each slab ΔL_j is estimated as $\Delta L_j = (h_j - h_{j-1})/\cos(\theta_z)$, where θ_z is the zenith angle of the satellite. The ground station is at an altitude h_0 and the satellite is at an altitude H.

The turbulence within each slab is defined by the scintillation index $\sigma^2_{I_j}$ and the Fried parameter r_{0_j}, where $\sigma^2_{I_j}$ for each slab is calculated as,

$$\sigma^2_{I_j} = \exp\left[\frac{0.49\sigma^2_{R_j}}{(1+1.11\sigma^{12/5}_{R_j})^{7/6}} + \frac{0.51\sigma^2_{R_j}}{(1+0.69\sigma^{12/5}_{R_j})^{5/6}}\right] - 1, \tag{5.7}$$

for the Rytov variance $\sigma^2_{R_j}$, calculated as a function of the altitude, θ_z, C^2_n and k,

$$\sigma^2_R = 2.25k^{7/6}\sec^{11/6}(\theta_z)\int_{h_{j-1}}^{h_j} C^2_n(h)(h-h_0)^{5/6}dh. \tag{5.8}$$

The altitude-dependent refractive index structure parameter $C^2_n(h)$ can be modeled as a function of the root-mean-square (RMS) wind speed v_{rms} in m/s and ground-level C^2_n in m$^{-2/3}$, $C^2_n(0)$ as [91],

$$\begin{aligned} C^2_n(h) &= 0.00594(v_{rms}/27)^2\ (h\times 10^{-5})^{10}\ e^{(-h/1000)} \\ &\quad + 2.7\times 10^{-16}\ e^{(-h/1500)} + C^2_n(0)e^{(-h/100)}. \end{aligned} \tag{5.9}$$

The RMS wind speed is calculated as a function of the altitude-dependent wind profile $v(h)$,

$$v_{rms} = \left[\frac{1}{15\times 10^3}\int_{5\times 10^3}^{20\times 10^3} v^2(h)\ dh\right]^{1/2}, \tag{5.10}$$

with $v(h)$ given in m/s. We adopt the Bufton wind profile [91] when simulating $v(h)$, calculated as,

$$v(h) = v_y + 30\exp\left[-\left(\frac{h-9400}{4800}\right)^2\right], \tag{5.11}$$

where v_y is the ground-level wind speed in m/s. Note that the slew rate is not considered. The Fried parameter for each slab is then given by,

$$r_{0_j} = \left[0.423k^2\sec(\theta_z)\int_{h_{j-1}}^{h_j} C^2_n(h)dh\right]^{-3/5}. \tag{5.12}$$

The distribution of phase screens for the satellite-to-Earth channel is optimized by determining N_s and h_j using a numerical search to solve for $\sigma^2_{I_j} < 0.1$ and $\sigma^2_{I_j} < 0.1\sigma^2_I$.

Figure 5.3 gives an example of how phase screens are distributed, with the distance between phase screens decreasing as the atmosphere becomes denser, where L_{ch} represents the total propagation distance from the transmitter to the receiver.

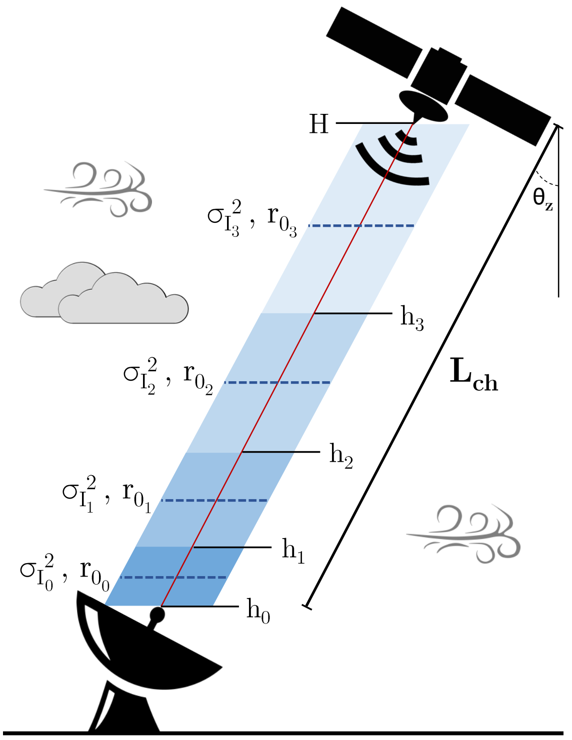

Figure 5.3: Satellite-to-Earth phase screen model for channel length L_{ch} and azimuth angle θ_z. Phase screens (dashed blue lines) simulate beam distortion using the scintillation index $\sigma^2_{I_j}$ and Fried parameter r_{0_j}, which are placed at the center of an atmospheric volume bounded by h_{j-1} and h_j.

The parameters used in our simulation of the satellite-to-Earth link are given in Table 5.2, where the inner scale l_0 and outer scale L_0 of turbulence are used to define turbulent eddy sizes in the phase screen model [113]. Note that we generate an independent $C^2_n(0)$ value (see range in Table 5.2) for each instance of the phase screen simulations, thus the phase screen vertical distribution optimization is undertaken for each simulation. Further, we assume that birefringence is negligible across the channel.

An example of the phase screen distribution altitudes and $C^2_n(h)$ values for $C^2_n(0) = 1.7 \times 10^{-14}$ m$^{-2/3}$ is provided in Table 5.3, resulting in $N_s = 15$ phase screens when using the channel parameters in Table 5.2. We note that the scintillation index-

Table 5.2: Simulation parameters.

Parameter	Value
Satellite altitude (H)	500 km
Receiver altitude (h_0)	2 km
Outer scale of turbulence (L_0)	5 m
Inner scale of turbulence (l_0)	0.025 m
Satellite zenith angle (θ_z)	0°
Laser wavelength (λ)	1550 nm
Beam waist radius (w_0)	0.15 m
Receiver radius (R_r)	1.25 m
RMS wind speed (ν_{rms})	21 m/s
Ground-level C_n^2 ($C_n^2(0)$)	1.7×10^{-15} - 10^{-14} m$^{-2/3}$

dependent phase screen distribution optimization used in this thesis does not produce a unique physical representation of atmospheric turbulence, rather, it aims to divide the atmosphere such that no single screen contributes too strongly to the total scintillation, while satisfying the altitude-dependent C_n^2(h) profile in Eq. 5.9.

Table 5.3: Phase screen distribution altitudes and $C_n^2(h)$ values for $C_n^2(0) = 1.7 \times 10^{-14}$ m$^{-2/3}$.

#	**Altitude**	$C_n^2(h)$
1	2.26 km	7.12 $\times 10^{-17}$ m$^{-2/3}$
2	2.99 km	5.01 $\times 10^{-17}$ m$^{-2/3}$
3	3.92 km	2.73 $\times 10^{-17}$ m$^{-2/3}$
4	4.84 km	1.57 $\times 10^{-17}$ m$^{-2/3}$
5	5.77 km	1.10 $\times 10^{-17}$ m$^{-2/3}$
6	6.70 km	1.05 $\times 10^{-17}$ m$^{-2/3}$
7	7.63 km	1.22 $\times 10^{-17}$ m$^{-2/3}$
8	8.56 km	1.45 $\times 10^{-17}$ m$^{-2/3}$
9	9.50 km	1.62 $\times 10^{-17}$ m$^{-2/3}$
10	10.45 km	1.67 $\times 10^{-17}$ m$^{-2/3}$
11	11.39 km	1.59 $\times 10^{-17}$ m$^{-2/3}$
12	12.38 km	1.41 $\times 10^{-17}$ m$^{-2/3}$
13	13.51 km	1.15 $\times 10^{-17}$ m$^{-2/3}$
14	14.97 km	8.37 $\times 10^{-18}$ m$^{-2/3}$
15	17.91 km	4.76 $\times 10^{-18}$ m$^{-2/3}$

5.3 Wavefront Errors

In this Section, we discuss the different forms of WFE that can occur in our system due to imperfections in optical hardware and system calibration.

5.3.1 Relative Wavefront Errors

Although previous work has investigated relative phase errors between reference pulses and post-heterodyne/homodyne detection signals in RLO-based CV-QKD (see Chapters 2 and 4), most have assumed that reference pulse phase WFEs equate to signal phase WFEs, $\Delta\Phi_R(x, y) \approx \Delta\Phi_S(x, y)$. Although this may be true in CV-QKD setups with near-perfect hardware and calibration, there are cases where $\Delta\Phi_R(x, y) \neq \Delta\Phi_S(x, y)$. Imperfections in optical hardware manufacturing, as well as post-manufacturing stresses and thermal fluctuations, can distort a wavefront passing through them [117]. Imperfect reference pulse and signal polarization multiplexing can cause photon leakage from reference pulses to signals [16], [118], and imperfect MPLC phase masks can lead to internal crosstalk between HG modes [119]. Although every CV-QKD setup is different, these imperfections can lead to $\Delta\Phi_R(x, y) \neq \Delta\Phi_S(x, y)$, which would affect CV-QKD performance (see Sections 5.5 to 5.7).

Additional differences in reference pulse and signal phase errors may occur through the channel. These issues are discussed in detail in [15]. In theory, no phase error difference is caused by the turbulent channel, provided the complex wavelength structure (width, amplitude, and phase distributions of all wavelengths) for the strong reference pulses and weak signal pulses are identical, no leakage across the polarized multiplexed signals is present, and all optical hardware and calibration are perfect. In reality, it is unlikely that all these conditions are met exactly.

We generalize our analysis of relative WFEs by investigating a variety of underlying functions, which define the relationship between the reference pulse WFEs and signal WFEs. The results are further generalized by varying the turbulence conditions for each simulation instance (see Section 5.2), thus improving the robustness of our wavefront correction approach (see Section 5.5).

In Figure 5.4, we provide a schematic of the end-to-end simulation process for each instance of the reference pulse and signal preparation, propagation across the atmosphere, and decomposition into the HG mode basis using an MPLC. For each instance, at Alice, the laser source generates the fundamental HG_{00} mode for the reference pulse and signal. The reference pulse and signal pass through different optical equipment, where reference pulse transmit WFEs $\Delta\theta_{mn,R}$ and signal transmit WFEs $\Delta\theta_{mn,S}$ are applied, respectively, using the Zernike WFEs discussed in Section 5.3.2. The reference pulse and signal are then transmitted across the channel, where the phase screen model introduces WFEs based on the turbulence conditions, as a function of the altitude-dependent scintillation index σ_I^2 and Fried parameter r_0 (discussed in Section 5.2). At Bob, the reference pulse and signal are injected into the MPLC, where the cross-leakage WFEs $\Delta\theta_{mn,X}$ are applied (see Section 5.3.3). The reference pulse WFEs and signal WFEs are first decomposed into the HG basis using Eq. 5.5, before the cross-leakage WFEs are applied. The outputs for each simulation instance are the reference pulse-mode WFEs $\Delta\phi_{mn,R}$ and the signal-mode WFEs $\Delta\phi_{mn,S}$. We refer the reader to Table 5.1 to clarify the WFE nomenclature used in the subsequent Sections.

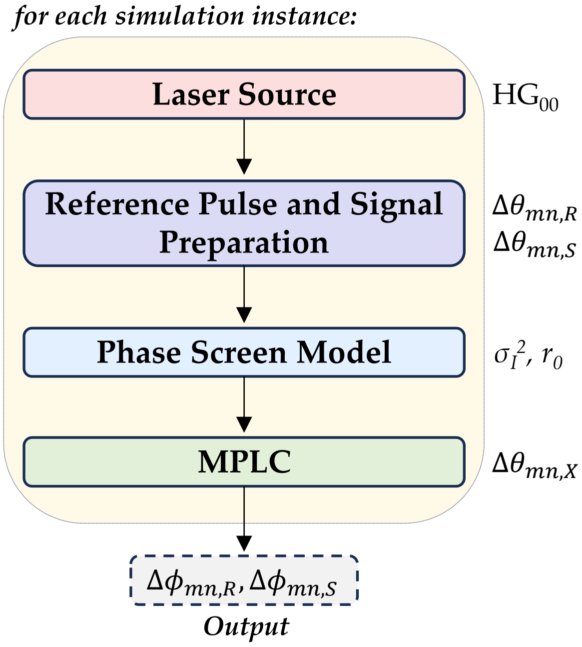

Figure 5.4: End-to-end simulation schematic outlining how each instance of the reference pulse-mode WFEs $\Delta\phi_{mn,R}$ and the signal-mode WFEs $\Delta\phi_{mn,S}$ are generated. HG_{00} represents the fundamental Gaussian mode, $\Delta\theta_{mn,R}$ are the reference pulse transmit WFEs, $\Delta\theta_{mn,S}$ are the signal transmit WFEs, σ_I^2 is the scintillation index, r_0 is the Fried parameter, and $\Delta\theta_{mn,X}$ are the cross-leakage WFEs.

5.3.2 Transmitter Wavefront Errors

We assume that the reference pulses and signals are Gaussian HG_{00} beams generated by the same laser source at the transmitter. The signals then pass through a beamsplitter (BS), phase modulator (PS), amplitude modulator (AM), optical attenuator (OA), and polarized beamsplitter (PBS) at the transmitter; while the reference pulses pass through a BS, delay line (DL), and PBS (see Figure 5.1). Imperfections in optical hardware can result in distortion of a beam's wavefront [117]. Given the different pieces of optical hardware through which the reference pulses and signals pass independently, their wavefronts could be distorted independently, resulting in a relative WFE between them at transmission.

For example, the properties of optical equipment can be affected by temperature gradients across the surfaces, absorption, coating materials, residual polishing compounds, and inhomogeneities in the optical material itself [117]. As such, changes in the refractive index within the material, due to thermal inhomogeneities and the state of stress of optical materials, can cause distortions to a beam passing through them, resulting in WFEs. This effect could be particularly prominent on satellites, given the large temperature fluctuations in space, as well as the stresses caused when launching a satellite into space.

Zernike polynomials are commonly used to describe the types of WFE induced by optical hardware, where we base the following theory on [120]. Zernike polynomials are an infinite set of polynomials, defined by the polar coordinates ρ and β for a unit circle aperture as,

$$Z_q^p(\rho,\beta) = \begin{cases} R_q^p(\rho)\cos(p\beta), & \text{if } p \geq 0, \\ R_q^{-p}(\rho)\sin(-p\beta), & \text{otherwise,} \end{cases} \tag{5.13}$$

given the integers p and q, where $q-p$ must be even, and $q \geq p \geq 0$. The radial polynomial term $R_q^p(\rho)$ is calculated as,

$$R_q^p(\rho) = \sum_{j=0}^{\frac{q-p}{2}} \frac{(-1)^j (q-j)!}{j!(\frac{p+q}{2}-j)!(\frac{p-q}{2}-j)!}\rho^{q-2j}. \tag{5.14}$$

We focus on the second-order WFE *tilt*; as well as the third-order WFEs *defocus*, *astig-*

matism, *coma*, and *spherical* (shown in Figure 5.5), defined by their Zernike polynomials in Table 5.4. The Zernike WFEs $\Delta\Phi(\rho, \beta)$ can then be calculated in terms of the Zernike polynomials using the coefficient a_q^p,

$$\Delta\Phi(\rho, \beta) = a_q^p Z_q^p(\rho, \beta), \tag{5.15}$$

where $0 \leq \rho \leq 1$ and $0 \leq \beta \leq 2\pi$.

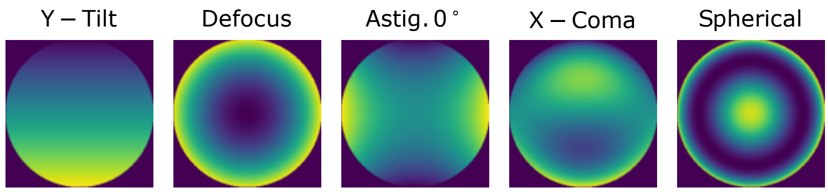

Figure 5.5: Wavefront errors due to imperfections in optical hardware: y-tilt, de-focus, 0° astigmatism, x-coma, and spherical (for an undefined RMS error).

Table 5.4: Zernike coefficients for wavefront errors.

n	m	**Zernike Polynomial**	**Error Type**
1	-1	$\rho \sin(\beta)$	Y-tilt (Z_1^{-1})
2	0	$2\rho^2 - 1$	De-focus (Z_2^0)
2	2	$\rho^2 \cos 2\beta$	Astigmatism 0° (Z_2^2)
2	-2	$\rho^2 \sin 2\beta$	Astigmatism 45° (Z_2^{-2})
3	1	$(3\rho^3 - 2\rho) \cos \beta$	X-coma (Z_3^1)
3	-1	$(3\rho^3 - 2\rho) \sin \beta$	Y-coma (Z_3^{-1})
4	0	$6\rho^4 - 6\rho^2 + 1$	Spherical (Z_4^0)

WFE specifications for optical hardware are often given in terms of their RMS error, calculated as RMS $= (\frac{1}{J}\sum_j \Delta\phi_j^2)^{1/2}$ for J discrete error measurements $\Delta\phi_j$ across the discretized wavefront. Therefore, for known Zernike WFE types (such as given in Table 5.4), which are defined by their RMS error by the manufacturers, we can solve Eq. 5.15 to calculate a_q^p so that $\Delta\Phi(\rho, \beta)$ can be derived for each Zernike WFE at the transmitter.

The Zernike WFEs $\Delta\Phi_{S1}(\rho, \beta)$ to $\Delta\Phi_{S5}(\rho, \beta)$ are applied to the signals and the Zernike WFEs $\Delta\Phi_{R1}(\rho, \beta)$ to $\Delta\Phi_{R4}(\rho, \beta)$ are applied to the reference pulses as they pass through each piece of hardware at the transmitter, as shown in Figure 5.1. The reference pulses and signals are then sent across the channel, as simulated by the phase screen model described in Section 5.2.

The RMS errors used to generate each of the Zernike WFEs are given in Table 5.5, where they approximate "precise" equipment, as defined by [121]. Note that $\Delta\Phi_{R2}(\rho,\beta)$ represents a phase shift throughout the wavefront caused by an imperfectly calibrated DL, added to account for the difference in paths taken by the reference pulses and signals.

Table 5.5: Transmitter wavefront error types and parameterization.

Source	**WFE Type**	**WFE Value**
$\Delta\Phi_{S1}(\rho,\beta)$	Astigmatism 0°	0.081λ (RMS)
$\Delta\Phi_{S2}(\rho,\beta)$	Spherical	0.100λ (RMS)
$\Delta\Phi_{S3}(\rho,\beta)$	Defocus	0.078λ (RMS)
$\Delta\Phi_{S4}(\rho,\beta)$	X-coma	0.091λ (RMS)
$\Delta\Phi_{S5}(\rho,\beta)$	Y-tilt	0.096λ (RMS)
$\Delta\Phi_{R1}(\rho,\beta)$	Y-coma	0.106λ (RMS)
$\Delta\Phi_{R2}(\rho,\beta)$	Phase Shift	0.869 rad
$\Delta\Phi_{R3}(\rho,\beta)$	Astigmatism 45°	0.098λ (RMS)

After passing the reference pulse and signal beams through the satellite-to-Earth channel, their distorted electric fields are deconstructed into N HG modes using Eq. 5.5. The resulting phase components of the complex coefficients c_{mn} for both the reference pulses and signals are defined by $\Delta\theta_{mn,R}$ (termed the reference pulse transmit MPEs) and $\Delta\theta_{mn,S}$ (termed the signal transmit MPEs) after propagating through the atmosphere.

5.3.3 Crosstalk and Leakage Wavefront Errors

Apart from second- and third-order WFEs at the transmitter, additional hardware imperfections can cause relative MPEs between the reference pulses and signals. Imperfect polarization multiplexing can lead to photon leakage from brighter reference pulses into weaker signals [16], [118]. In addition, crosstalk in the HG mode can occur due to imperfections in the design of the MPLC phase masks (e.g. [119]).

Imperfect polarization multiplexing would occur at the transmitter, where the transmit MPEs could leak from the reference pulses to the signals, which would then propagate over the atmosphere, before entering the MPLC at the receiver. The crosstalk between the MPLC HG modes is usually quantified in terms of the ratio of powers at the input to output of the MPLC [119], extracted from the real component of the coefficients c_{mn}, rather than the phase component (see Eq. 5.5). Imperfections in polarization multiplexing

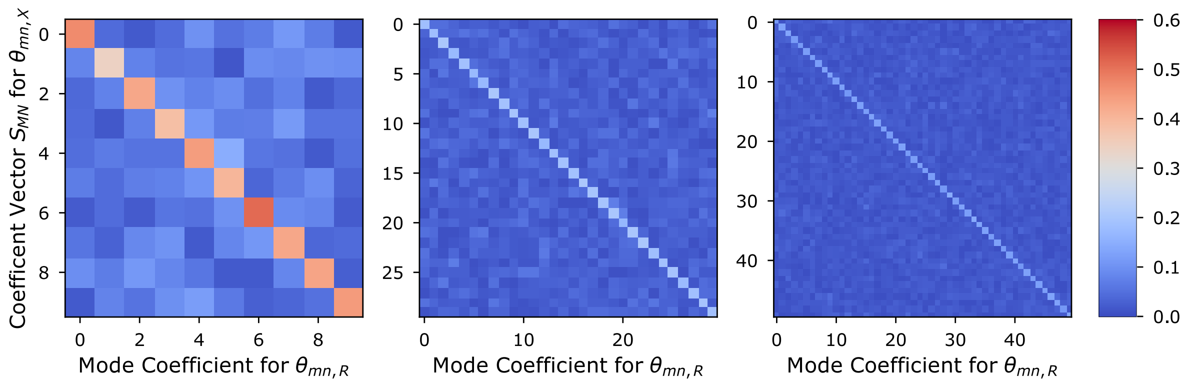

Figure 5.6: cross-leakage MPE coefficient matrices of vectors S_{MN} for (**a**) $N = 10$, (**b**) $N = 30$, and (**c**) $N = 50$.

and phase-mask design within an MPLC will vary for each CV-QKD set-up, leading to unique additional MPEs. However, as an example, we introduce a sinusoidal harmonic function to represent the additional MPEs in each mode, termed the cross-leakage MPEs $\Delta\theta_{mn,X}$, given by,

$$\Delta\theta_{mn,X} = \sum_{d=1}^{3} A_{mn,d} \sin(S_{MN} \cdot \Theta_{MN,R}), \tag{5.16}$$

where the subscript X represents the *cross*-leakage, d represents the harmonic number, $A_{mn,d}$ is a vector of amplitudes for each harmonic for each mode, S_{MN} is a vector of coefficients for each mode, and $\Theta_{MN,R} = [\Delta\theta_{00,R}, \Delta\theta_{01,R}, \cdots, \Delta\theta_{mn,R}]^T$. We calculate the vectors S_{MN} by approximating the MPLC power ratio crosstalk matrices in [119].

The cross-leakage MPE for each mode is a function of the amplitudes $A_{mn,d}$, the coefficient vector S_{MN}, and the reference pulse transmit MPEs $\Theta_{MN,R} = [\Delta\theta_{00,R}, \Delta\theta_{01,R}, \cdots, \Delta\theta_{mn,R}]^T$. Therefore, the cross-leakage MPE for each mode is a function of *all* reference pulse transmit MPE modes, weighted by a coefficient (see Eq. 5.16). We extract the coefficient vectors S_{MN} from the matrices shown in Figure 5.6 for (a) $N = 10$, (b) $N = 30$, and (c) $N = 50$, which approximate the MPLC crosstalk matrices in [119].

The vertical axis represents the cross-leakage MPE mode to which the coefficient vector S_{MN} is applied, while the horizontal axis represents the specific coefficients applied to the modes in $\Theta_{MN,R}$. As can be seen, the diagonal coefficients are greater than the off-diagonal ones. This is implemented as we assume that more leakage would occur from $\Delta\theta_{mn,R}$ to $\Delta\theta_{mn,S}$ in the same mode, where mn for $\Delta\theta_{mn,S}$ equals mn for $\Delta\theta_{mn,R}$, while less leakage would occur when $mn \neq mn$, as is true when considering MPLC power ratio crosstalk [119].

The amplitude vectors $A_{mn,d}$ for each harmonic for each mode are sampled from uniform distributions, $A_{mn,d} \in \mathcal{R}(0.00, 0.50)$. We further note the analogy between HG modes and harmonic oscillators [122], as well as the discussion of phase correction for harmonic distortions (e.g. [123]), in our choice of the cross-leakage MPE function, while emphasizing that the additional WFE function could be different for every CV-QKD setup.

5.4 Transformer Neural Network

In Chapter 3, we use a convolutional NN (CNN) for wavefront correction, which takes reference pulse intensity distributions as input, then outputs an estimation of the wavefront correction to apply to an RLO. Chapter 3 assumes that $\Delta\Phi_R(x, y) = \Delta\Phi_S(x, y)$, thus does not take relative WFEs into account. CNNs could potentially be used to map $\Delta\Phi_R(x, y)$ to $\Delta\Phi_S(x, y)$; however, the CNN kernels only extract relationships between localized areas of the wavefront distributions, while the complexity of a CNN designed to map $\Delta\Phi_R(x, y)$ to $\Delta\Phi_S(x, y)$ could be high (e.g. in Chapter 3, the CNN has 644,097 trainable parameters). Note that a similar approach is adopted in [95] to estimate turbulent phase screens with a CNN, using intensity distributions as input for orbital angular momentum-based QKD. Further, [124] demonstrated CNN-based real-time wavefront correction experimentally, using intensity distributions as input, and a DM to imprint the wavefront correction on the received signal, resulting in higher signal-to-noise ratios than without the corrections. However, [95], [124] do not take relative WFEs into account.

In this chapter, we instead build our intelligent wavefront correction algorithm architectures using transformer NNs (TNNs). TNNs are capable of processing an entire input vector (here, a vector of $\Delta\phi_{mn,R}$ for a single pulse) simultaneously, where patterns in the spatial relationship between *all* elements in the vector are derived and weighted, before being mapped to the desired output vector (here, a vector of $\Delta\phi_{mn,S}$ for a single pulse). TNNs permit global relationships between spatial elements to be extracted, rather than only local, while vector-to-vector mapping could reduce the complexity of the TNN, compared to mapping the relationship between two-dimensional distributions of the CNN. We give only a brief description of TNNs in the rest of this Section; refer to [125] for a more in-depth discussion of TNNs.

The input vector is first embedded in the TNN using a (fully connected) *dense layer*, resulting in a vector of length d_m. The positional encoding is then performed on each embedded mode, where their relative spatial positions are weighted, then summed with the embedded modes (shown at the bottom of Figure 5.7). The *positionally encoded modes* (PEMs), a matrix of dimension $N \times d_m$, are then passed into the transformer layers (represented as the light blue block in Figure 5.7), which are made up of multiple *attention heads* (shown as the yellow block in Figure 5.7) and two dense layers.

An attention head calculates three different matrices of weighted values, used to represent similarities between PEM elements. A *query matrix* and *key matrix* are calculated for each PEM, providing "context" and "relevance", respectively, then combined by taking the dot product between them. The dot product is then passed to a Softmax function [126]. The output of the Softmax function is combined with a separate *value matrix* (calculated for the PEMs) via dot product, then summed with the original PEMs to obtain the "similarity" between them, outputting a *self-attention vector*. The key, query and value matrices are of dimension $n_{attn} \times |d_m/n_{attn}|$, where n_{attn} is the number of attention heads in the TNN architecture. The self-attention vector has dimensions d_m.

Using multiple attention heads allows different relationships between the PEMs to be derived, improving the accuracy of the output estimations. The self-attention vector is then passed into a dense layer of dimension d_t and another dense layer of dimension d_m, where they are again summed with the PEMs (helping to preserve positionality). The results from the transformer layers are then passed into a final dense layer, resulting in a vector of dimension N, before outputting a vector of estimates $\Delta\tilde{\phi}_{mn,S}$ (shown for $N = 10$ in Figure 5.7).

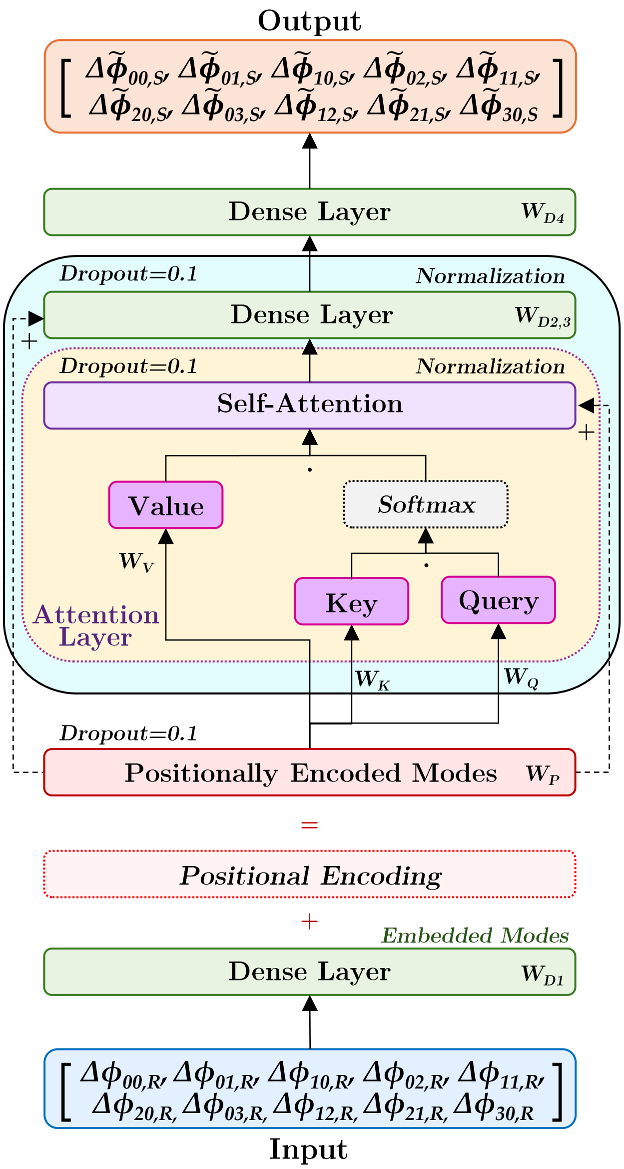

Figure 5.7: TNN architecture, with an input and output of $N = 10$ modes. The transformer layer architecture is shown as the light blue block, and the attention head architecture is shown in the yellow block. The weights W_P represent the weights for the positional encoding, the weights W_{D1}, W_{D2}, W_{D3}, W_{D4} represent the weights for the dense layers, while the weights W_K, W_Q, and W_V represent the weights for the key, query, and value matrices, respectively.

5.4.1 Architectures

We develop three different TNN architectures for the three different numbers of HG modes measured ($N = 10$, $N = 30$, and $N = 50$), with their hyperparameters given in Table 5.6.

The number of transformer layers, the number of attention heads, and the dimensions of the matrix (d_m and d_t) vary between the architectures. An additional transformer layer is added for $N = 30$ and $N = 50$, due to the additional information being processed by

Table 5.6: TNN architecture hyperparameters for $N = 10$, $N = 30$, and $N = 50$ HG modes.

Hyperparameter	$N = 10$	$N = 30$	$N = 50$
Matrix dimension (d_m)	64	64	96
Matrix dimension (d_t)	256	256	256
Attention heads (n_{attn})	6	6	10
Transformer layers #	3	4	4
Trainable parameters #	147,037	195,985	339,545
Dropout	0.1	0.1	0.1

the TNNs, while four additional attention heads are added for $N = 50$. The dimension of the matrix d_m was increased from 64 to 96 to account for the increase in the vector length of $N = 50$ that is mapped through the TNN. In general, the number of training parameters in the TNN architectures increases as N increases to map the increase in information from the input data $\Delta\phi_{mn,R}$ to the output data $\Delta\phi_{mn,S}$. Note that the hyper-parameterization of each TNN is achieved through an iterative process, aiming to achieve sufficient estimation performance, while limiting the number of trainable parameters.

5.4.2 Training

We simulate 24,000 instances of reference pulse and signal propagation through the satellite-to-Earth channels, followed by HG mode decomposition, which are split into 19,200 training instances and 4,800 testing instances.

During training, the known input values $\Delta\phi_{mn,R}$ are assigned to the known $\Delta\phi_{mn,S}$ values. The weights W_{D1}, W_{D2}, W_{D3}, W_{D4} represent the weights applied to the connections for the dense layers, while the weights W_K, W_Q, and W_V represent the weights applied to the key, query, and value matrices. All weights are optimized using the Adam optimizer [127] to minimize an RMS error loss function, training for 3,000 epochs using a batch size of 32. After training, all weight values remain fixed.

Further, we implement *dropout* for the PEM, self-attention, and dense layers within the transformer layers. Dropout helps prevent overfitting the TNN weights by replacing a percentage of data within each matrix with zeros, improving the robustness of the intelligent wavefront correction algorithms [128]. The self-attention and dense layers, within the transformer layers, are also normalized to improve estimation performance [129].

5.4.3 Operation

During operation, the TNNs take known input $\Delta\phi_{mn,R}$ measurements for each pulse, then output the estimates $\Delta\tilde{\phi}_{mn,S}$ of $\Delta\phi_{mn,S}$ for that pulse. Estimates $\Delta\tilde{\phi}_{mn,S}$ can then be used to reconstruct an estimated signal electric field $\tilde{E}_{mn,S}(x, y)$ (see Section 5.6.1), which can then be mapped onto the RLO to measure the signal with greater coherence.

Inference latency represents the time required for our encoder TNN-based intelligent wavefront correction algorithm to predict a single output of N $\Delta\tilde{\phi}_{mn,S}$ values [56]. Using an NVIDIA Tesla V100-SXM2-32GB GPU, the inference time for our intelligent wavefront correction algorithm is approximately 1.82 ms for $N = 10$, 3.54 ms for $N = 30$, and 8.25 ms for $N = 50$. The inference latency time scales are of the same order of magnitude as the expected coherence times for satellite-Earth channels (e.g. see [130]). Using the tested GPU, a wavefront correction could be applied for each coherence time for a range of channel conditions, while future hardware and software optimization should ensure the real-time applicability of our wavefront correction approach for all channel conditions.

We note that in a real-world implementation of a satellite-Earth CV-QKD network, a NN would need to be trained on each link (learning the particular relative WFEs associated with the specific hardware setup) for a wide variety of atmospheric conditions.

5.5 Estimation Performance

To analyze the performance of the intelligent wavefront correction algorithms in estimating $\Delta\phi_{mn,S}$, we calculate the *correction variance* between $\Delta\tilde{\phi}_{mn,S}$ and $\Delta\phi_{mn,S}$, where a perfect estimation performance would result in $\mathrm{Var}(\Delta\tilde{\phi}_{mn,S} - \Delta\phi_{mn,S}) = 0$. We benchmark the estimation performance against the reference pulse approximation by calculating the *default variance* between $\Delta\phi_{mn,R}$ and $\Delta\phi_{mn,S}$. When condition $\mathrm{Var}(\Delta\tilde{\phi}_{mn,S} - \Delta\phi_{mn,S}) < \mathrm{Var}(\Delta\phi_{mn,R} - \Delta\phi_{mn,S})$ is achieved, the estimates $\Delta\tilde{\phi}_{mn,S}$ give a better representation of $\Delta\phi_{mn,S}$ than $\Delta\phi_{mn,R}$, which leads to improved CV-QKD performance (shown in Section 5.7).

We present four different cases of relative MPEs to demonstrate the effects on CV-QKD performance, using the transmit and cross-leakage MPEs described in Section 5.3. Case 0 represents when there are no transmit or cross-leakage MPEs, such that the approximation

$\Delta\phi_{mn,R} \approx \Delta\phi_{mn,S}$ is true. Case 1 represents when there are only transmit MPEs, but no cross-leakage MPEs. Case 2 represents when there are transmit MPEs and cross-leakage MPEs. Case 3 represents when there are only cross-leakage MPEs, but no transmit MPEs.

5.5.1 Case 0

In the case of zero relative MPEs, the approximation $\Delta\phi_{mn,R} \approx \Delta\phi_{mn,S}$ is true. We term this Case 0, representing CV-QKD systems with perfect optical hardware. To show that the intelligent wavefront correction algorithms will *not* negatively affect CV-QKD performance for Case 0, TNNs are effectively trained to estimate $\Delta\tilde{\phi}_{mn,S} \approx \Delta\phi_{mn,R} \approx \Delta\phi_{mn,S}$.

For Case 0, the default variances are zero, $\mathrm{Var}(\Delta\phi_{mn,R} - \Delta\phi_{mn,S}) = 0$, for all modes. Likewise, the correction variances approach zero, $\mathrm{Var}(\Delta\tilde{\phi}_{mn,S} - \Delta\phi_{mn,S}) \to 0$, with the maximum and minimum variances for $N = 10$, $N = 30$, and $N = 50$ given in Table 5.7.

Table 5.7: Minimum and maximum correction variances $\mathrm{Var}(\Delta\tilde{\phi}_{mn,S} - \Delta\phi_{mn,S})$ for $N = 10$, $N = 30$, and $N = 50$ in Case 0.

N	**Min. Correction Variance**	**Max. Correction Variance**
10	4.29×10^{-8} rad^2	1.24×10^{-6} rad^2
30	3.63×10^{-5} rad^2	2.21×10^{-4} rad^2
50	6.34×10^{-4} rad^2	1.67×10^{-3} rad^2

As expected, the intelligent wavefront correction algorithms can learn when there are no relative MPEs, outputting the estimate $\Delta\tilde{\phi}_{mn,S} \approx \Delta\phi_{mn,R}$. Therefore, a CV-QKD setup with near-perfect hardware would not be disadvantaged by implementing the intelligent wavefront correction algorithms. However, if the hardware is disturbed or deteriorates over time, resulting in relative MPEs, then the intelligent wavefront correction algorithms would be able to adapt and correct for them (as we show in the following analysis).

5.5.2 Case 1

In the case of the relative MPEs only being associated with imperfections of the optics at the transmitter (i.e. the transmit MPEs), termed Case 1, the relative MPEs output from the MPLC are given by,

$$\Delta\phi_{mn,R} - \Delta\phi_{mn,S} = \Delta\theta_{mn,R} - \Delta\theta_{mn,S}. \tag{5.17}$$

Figure 5.8 shows the correction variance (red bars) and default variance (blue bars) for (a) $N = 10$, (b) $N = 30$, and (c) $N = 50$, while the purple sections show where the two variances overlap.

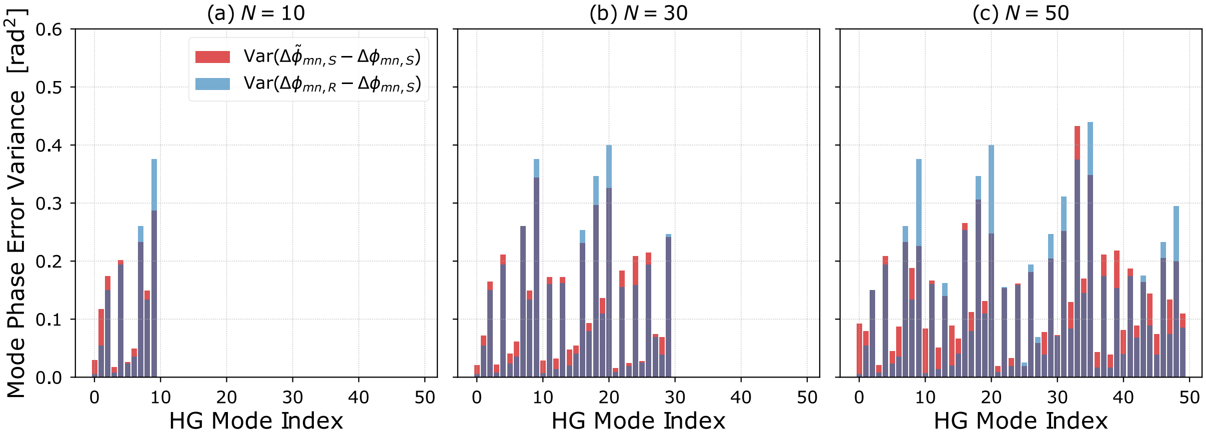

Figure 5.8: Correction variance $\mathrm{Var}(\Delta\tilde{\phi}_{mn,S} - \Delta\phi_{mn,S})$ (red bars) versus default variance $\mathrm{Var}(\Delta\phi_{mn,R} - \Delta\phi_{mn,S})$ (blue bars) for (**a**) $N = 10$, (**b**) $N = 30$, and (**c**) $N = 50$ in Case 1. The purple sections show where the two variances overlap.

The default variances are lower than the correction variances for some modes (where the red bars are greater than the blue bars), while for other modes the correction variances are lower than the default variances (where the blue bars are greater than the red bars), seen for each value of N. In general, there is no large difference between the default variances and the correction variances. Although there are more modes where the default variances are lower than the correction variances, the modes where the correction variances are lower show a greater variance reduction than the default variances. In general, the greater reduction in the correction variances compensates for the greater number of modes in which the default variances are lower.

The estimation performance of the intelligent wavefront correction algorithms could be attributed to the phase wavefront perturbations caused by the channel (using the phase screen model [113]). The transmitted reference pulse and signal wavefronts (after Zernike WFEs are applied) are affected by Gaussian random processes, resulting in $\Delta\phi_{mn,R}$ - $\Delta\phi_{mn,S}$ for each mode being Gaussian distributions. However, while the correction variances do not provide a more accurate representation of $\Delta\phi_{mn,S}$ than the default variances, they do not provide a considerably worse representation. As such, incorporating our intelligent wavefront correction algorithms into CV-QKD protocols should not reduce CV-QKD performance when relative MPEs are purely random.

5.5.3 Case 2

In the case of the relative MPEs being associated both with imperfections of the optics at the transmitter (transmit MPEs) *and* also with imperfections causing photon leakage and MPLC crosstalk (cross-leakage MPEs) - termed Case 2 - the relative MPEs output from the MPLC are given by,

$$\Delta\phi_{mn,R} - \Delta\phi_{mn,S} = \Delta\theta_{mn,R} - \Delta\theta_{mn,S} - \Delta\theta_{mn,X}. \tag{5.18}$$

As in Case 1, Figure 5.9 plots the correction variance (red bars) and the default variance (blue bars) for (a) $N = 10$, (b) $N = 30$, and (c) $N = 50$.

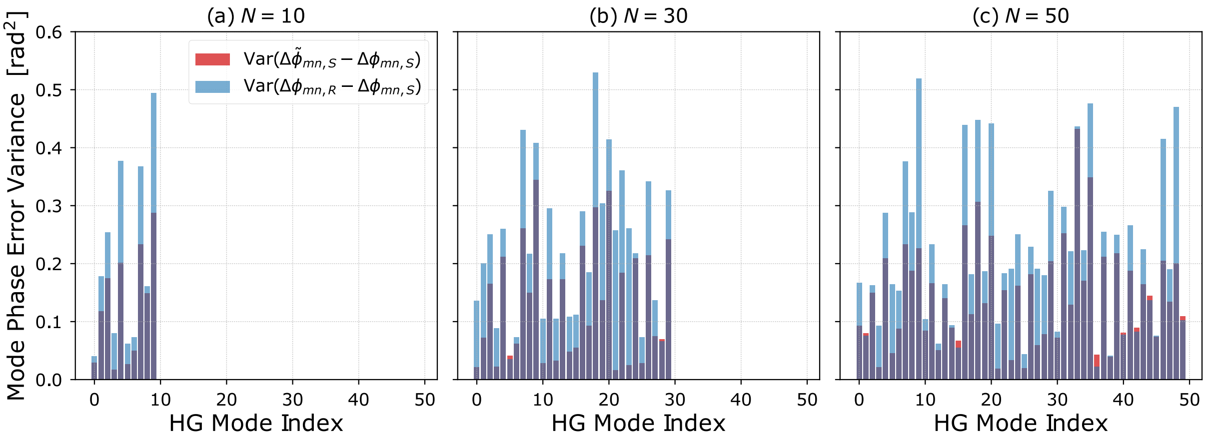

Figure 5.9: Correction variance $\mathrm{Var}(\Delta\tilde{\phi}_{mn,S} - \Delta\phi_{mn,S})$ (red bars) versus default variance $\mathrm{Var}(\Delta\phi_{mn,R} - \Delta\phi_{mn,S})$ (blue bars) for (**a**) $N = 10$, (**b**) $N = 30$, and (**c**) $N = 50$ in Case 2. The purple sections show where the two variances overlap.

For almost all modes, the correction variances are lower than the default variances (where the blue bars are greater than the red bars). The differences between the correction variances and the default variances in Case 2 are also generally greater than the differences in Case 1, with default variances even exceeding 0.50 rad^2 (mode 18 for $N = 30$ and mode 9 for $N = 50$), after introducing cross-leakage MPEs.

The cross-leakage MPEs make $\Delta\phi_{mn,S}$ a function of $\Delta\phi_{mn,R}$, so that the intelligent wavefront correction algorithms can learn the relationship between them. Although the introduced cross-leakage MPEs will not represent every CV-QKD setup, the results for Case 2 highlight the utility of our intelligent wavefront correction algorithms - when a CV-QKD setup induces a relationship between $\Delta\phi_{mn,S}$ and $\Delta\phi_{mn,R}$, then estimation of $\Delta\phi_{mn,S}$ leads to improved CV-QKD performance (shown in Sections 5.6 and 5.7). In addition, TNNs are particularly useful when there exists a relationship between the individual $\Delta\phi_{mn,S}$ and $\Delta\phi_{mn,R}$ modes.

5.5.4 Case 3

In the case of the relative MPEs being associated only with hardware imperfections causing photon leakage and MPLC crosstalk (cross-leakage MPEs) - termed Case 3 - the relative MPEs output from the MPLC are given by,

$$\Delta\phi_{mn,R} - \Delta\phi_{mn,S} = -\Delta\theta_{mn,X}. \tag{5.19}$$

Overall, the default variances are lower for Case 3 than Case 1 or Case 2, with the correction variances being even lower, where the intelligent wavefront correction algorithms output estimates $\Delta\tilde{\phi}_{mn,S}$ with lower correction variances than the default variances (as in Case 2). We provide an example of the estimation performance in Figure 5.10, which plots the correction variance (red bars) and the default variance (blue bars) for $N = 50$. However, we assume that Case 2 is more representative than Case 3 in the presence of WFEs, given the noisier representation of the relative MPEs in Eq. 5.18 than for Eq. 5.16. For this reason, and given that the correction variances for Case 1 generally do not outperform the default variances, we focus on Case 2 when analyzing wavefront correction (Section 5.6) and subsequent performance on secure key rates (Section 5.7) for CV-QKD.

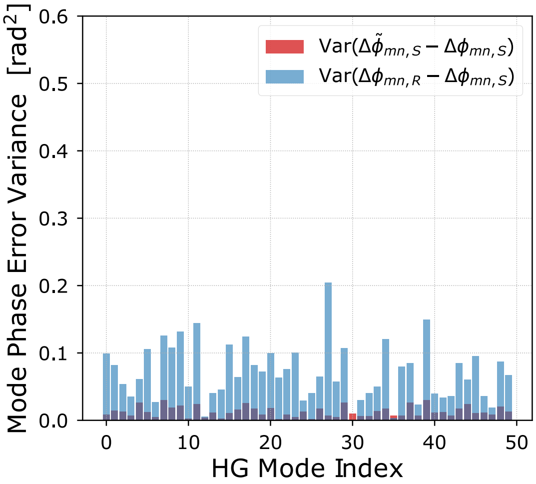

Figure 5.10: Correction variance $\mathrm{Var}(\Delta\tilde{\phi}_{mn,S} - \Delta\phi_{mn,S})$ (red bars) versus default variance $\mathrm{Var}(\Delta\phi_{mn,R} - \Delta\phi_{mn,S})$ (blue bars) for $N = 50$ in Case 3. The purple sections show where the two variances overlap.

5.6 Wavefront Correction

5.6.1 Wavefront Reconstruction

Any electric field can theoretically be reconstructed using an infinite number of HG modes (see Eq. 5.4); however, given that MPLCs measure a limited number of modes (N), the wavefront reconstructions are imprecise.

We consider three scenarios to understand the performance of the $\Delta\tilde{\phi}_{mn,S}$ estimations for Case 2. First, we reconstruct the electric field of the signal using the "perfect" MPLC measurement case, where the signal HG coefficients are used to reconstruct the signal electric field,

$$E_{mn,S}(x, y) = \sum_{mn}^{N} a_{mn,S} \mathrm{HG}_{mn} e^{i\Delta\phi_{mn,S}},$$

representing the greatest amount of information we could theoretically extract from the MPLC about the signal. Next, we consider the scenario where the reference pulse HG coefficients are used to reconstruct the electric field,

$$E_{mn,R}(x, y) = \sum_{mn}^{N} a_{mn,R} \mathrm{HG}_{mn} e^{i\Delta\phi_{mn,R}},$$

which represents the approximation of $E_R(x, y) \approx E_S(x, y)$. Finally, we reconstruct the electric field using the estimates $\Delta\tilde{\phi}_{mn,S}$,

$$\tilde{E}_{mn,S}(x,y) = \sum_{mn}^{N} a_{mn,R}\mathrm{HG}_{mn}e^{i\Delta\tilde{\phi}_{mn,S}}.$$

Note that we ignore the difference in the distortion of the reference pulse amplitudes $a_{mn,R}$ and signal amplitudes $a_{mn,S}$ in this work, where we use the reference pulse amplitudes when reconstructing the estimated signal WFEs using $\Delta\tilde{\phi}_{mn,S}$. The improvements in coherent efficiency are $< 1\%$, when the signal amplitudes are used in place of the reference pulse amplitudes, thus there is no major impact on the signal WFE reconstructions. However, the relative intensity noise of the reference pulses, caused by atmospheric turbulence, is a known phenomenon and can be captured using a separate noise term when analyzing secure key rates in CV-QKD (e.g. see [131]).

Figure 5.11 gives an example of the wavefront reconstruction for $\Delta\phi_{mn,S}$, $\Delta\phi_{mn,R}$, and $\Delta\tilde{\phi}_{mn,S}$ using $N = 30$ modes, as well as the true signal phase WFE $\Delta\Phi_S(x,y)$. It can be seen that the estimated signal wavefront reconstruction $\tilde{E}_{mn,S}(x,y)$ more closely matches the signal wavefront reconstruction using $E_{mn,S}(x,y)$ than the reference pulse wavefront reconstruction using $E_{mn,R}(x,y)$.

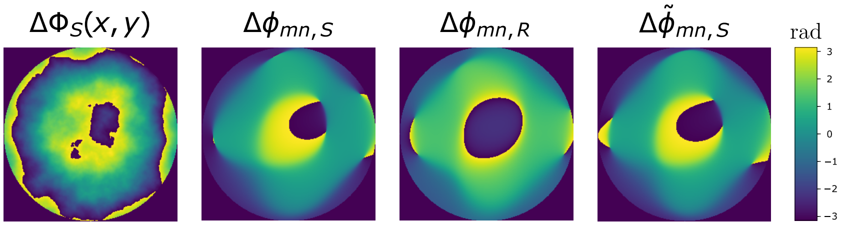

Figure 5.11: True signal wavefront $\Delta\Phi_S(x,y)$ compared to wavefront reconstructions using $\Delta\phi_{mn,S}$, $\Delta\phi_{mn,R}$, and $\Delta\tilde{\phi}_{mn,S}$ for $N = 30$ modes.

5.6.2 Coherent Efficiency

To quantify the difference between the true signal electric field $E_S(x,y)$ and reconstructed electric fields $E_{mn}(x,y)$, we calculate the coherent efficiency γ as [132],

$$\gamma = \frac{|\frac{1}{2}\iint_{\mathcal{D}_R}\left(E_S^*(x,y)E_{mn}(x,y) + E_S(x,y)E_{mn}^*(x,y)\right)\ dr|^2}{\left(\iint_{\mathcal{D}_R}|E_S(x,y)|^2\ dr\right)\left(\iint_{\mathcal{D}_R}|E_{mn}(x,y)|^2\ dr\right)}, \tag{5.20}$$

across the aperture with diameter $\mathcal{D}_R$ for the radial integral dr. A coherent efficiency of one represents perfect mode matching between the electric fields, e.g. $E_S(x,y) = E_{mn}(x,y)$, while a coherent efficiency of zero represents complete decoherence.

Table 5.8 gives the results for the average coherent efficiencies for each of the three reconstruction scenarios presented, where we replace $E_{mn}(x,y)$ with $E_{mn,S}(x,y)$, $E_{mn,R}(x,y)$, or $\tilde{E}_{mn,S}(x,y)$ in Eq. 5.20.

Table 5.8: Coherent efficiencies between electric fields of true signal $E_S(x,y)$ and reconstructed electric fields of $E_{mn,S}(x,y)$, $E_{mn,R}(x,y)$, and $\tilde{E}_{mn,S}(x,y)$ for different numbers of HG modes N.

N	$E_{mn}(x,y)$	**Avg. Coherent Efficiency**
10	$E_{mn,S}(x,y)$	0.329
10	$E_{mn,R}(x,y)$	0.229
10	$\tilde{E}_{mn,S}(x,y)$	0.313
30	$E_{mn,S}(x,y)$	0.507
30	$E_{mn,R}(x,y)$	0.321
30	$\tilde{E}_{mn,S}(x,y)$	0.484
50	$E_{mn,S}(x,y)$	0.553
50	$E_{mn,R}(x,y)$	0.349
50	$\tilde{E}_{mn,S}(x,y)$	0.523

Firstly, comparing the perfect signal reconstructions $E_{mn,S}(x,y)$ with the reference pulse reconstructions $E_{mn,R}(x,y)$, we see that the coherent efficiency decreases from $\gamma = 0.329$ to $\gamma = 0.229$ for $N = 10$, $\gamma = 0.507$ to $\gamma = 0.321$ for $N = 30$, and from $\gamma = 0.553$ to $\gamma = 0.349$ for $N = 50$. The estimated signal reconstructions $\tilde{E}_{mn,S}(x,y)$ show coherent efficiencies more closely matched to the signal wavefront reconstructions, with $\gamma = 0.329$ to $\gamma = 0.313$ for $N = 10$, $\gamma = 0.507$ to $\gamma = 0.484$ for $N = 30$, and from $\gamma = 0.553$ to $\gamma = 0.523$ for $N = 50$, thus showcasing the benefit of our intelligent wavefront correction algorithm when there is a relationship between $\Delta\phi_{mn,S}$ and $\Delta\phi_{mn,R}$. The effects of coherent efficiency on secure key rates in CV-QKD follow in Section 5.7.

5.7 Secure Key Rates

In CV-QKD, signals are affected by excess noise contributions from channel noise ξ_{ch} and detector noise ξ_{det}, both of which reduce the secure key rates attainable across a channel [13]. In this work, we adopt the GG02 CV-QKD protocol with reverse reconciliation [8]. Note that we apply the same security analysis as the entanglement-based GG02 protocol (given in [10]). This means we assume that the ground station is trusted and an eavesdropper (Eve) does not have knowledge of the training process, nor is even aware that our new system is deployed at the ground station. That is, we take the reduced excess noise that our system can deliver to be a consequence of the correction to relative MPEs between the reference pulses and signals. A formal security proof of the impact of our system when Eve is aware of its presence (and can interfere with the training process) is clearly a substantial task and should be the subject of future work. We also note that we have ignored finite key effects, so our secure key rates are valid only in the limit of large block lengths. Conversion of rates into bits/s would require a full reconciliation analysis, including the low-density parity-check (LDPC) decoding timescale for decoding large block-length LDPC codes. This reconciliation will likely be the bottleneck for all CV-QKD protocols - the impact of our NN-runtime would be insignificant in comparison.

The coherent efficiency between the corrected RLO and signal contributes to the detector noise, where [78] derives the detector efficiency as a function of coherent efficiency as,

$$\xi_{det} = \frac{((1-\gamma) + \xi_{el})\eta_{det}}{\gamma}, \tag{5.21}$$

for electronic noise ξ_{el} and detector efficiency η_{det}. The detector noise is used to calculate the shared information between Alice and Bob I_{AB}, whereby increasing the coherent efficiency reduces ξ_{det}, increasing the shared information as follows [78]. The shared information is calculated as,

$$I_{AB} = \frac{1}{2}\log_2(1 + \frac{T_f V_{mod}}{1 + T_f \xi_f}), \tag{5.22}$$

for the modulation variance V_{mod}. Shared information is a function of transmissivity T, calculated as $T = (P_R/P_T)\eta_{det}$, where P_T and P_R represent the total power at the transmitter and receiver, respectively. Here, we apply an *effective transmissivity* and an *effective excess noise* term $T_f\xi_f$ to evaluate the secure key rates, given by,

$$T_f\xi_f = \xi_{ch}\langle T\rangle + \langle\xi_{det}\rangle + \left(\langle T\rangle - \langle\sqrt{T}\rangle^2\right) V_{mod}. \tag{5.23}$$

The average transmissivity $\langle T \rangle = 0.652$ is calculated using the signal propagation through the phase screen simulations, while the average detector noise $\langle \xi_{det} \rangle$ is calculated using the coherent efficiencies described in Section 5.6.2 for $N = 10$, $N = 30$, and $N = 50$. The noise and efficiency parameters used to calculate the secure key rates are given in Table 5.9.

Table 5.9: Noise parameters.

Parameter	**Value**
Detector efficiency (η_{det})	95% [72]
Reverse reconciliation efficiency (β_r)	95% [72]
Detector electronic noise (ξ_{el})	0.010 SNU [96]
Channel noise (ξ_{ch})	0.0172 SNU [72]

The covariance matrix M_{AB} for the GG02 protocol, using homodyne detection of the signal, is given by [10], [78],

$$M_{AB} = \begin{pmatrix} a\mathbb{1} & c\sigma_z \\ c\sigma_z & b\mathbb{1} \end{pmatrix} = \begin{pmatrix} (V_{mod}+1)\mathbb{1} & \sqrt{T_f(V_{mod}^2 + 2V_{mod})}\sigma_z \\ \sqrt{T_f(V_{mod}^2 + 2V_{mod})}\sigma_z & (T_f V_{mod} + 1 + \xi_{ch})\mathbb{1} \end{pmatrix},$$

where σ_z is the Pauli-z matrix and $\mathbb{1}$ is the identity matrix of dimension 2×2.

Holevo information χ_{BE} is the amount of information about a quantum state ρ_{AB} attainable by Eve, calculated as $\chi_{BE} = S_{AB} - S_{A|B}$, for the von Neumann entropy S_{AB} of the quantum state accessible by Eve and $S_{A|B}$ is the entropy after homodyne measurement by Bob [10]. The subscripts A, B, and E represent Alice, Bob, and Eve, respectively. If Eve has a purification of Alice and Bob's shared quantum state, $S(\rho_{AB}) = \sum_i g(\nu_i)$ for the function $g(x)$, given by,

$$g(x) = \frac{x+1}{2}\log_2\left(\frac{x+1}{2}\right) - \frac{x-1}{2}\log_2\left(\frac{x-1}{2}\right). \tag{5.24}$$

The first two symplectic eigenvalues $\nu_{1,2}$ (the modulus of any eigenvalue) of M_{AB} are derived as $\nu_{1,2} = \frac{1}{2}(z \pm [b-a])$, where $z = ([a+b]^2 - 4c^2)^{1/2}$ [98]. A third symplectic eigenvalue ν_3 is calculated for the conditional matrix $M_{A|B}$, found by,

$$\tilde{M}_{A|B} = i\Omega M_{A|B} = i \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a - \frac{c^2}{b} & 0 \\ 0 & a \end{pmatrix} = i \begin{pmatrix} 0 & a \\ -a + \frac{c^2}{b} & 0 \end{pmatrix},$$

which leads to $\nu_3 = (a[a - \frac{c^2}{b}])^{1/2}$ [10]. The Holevo information can be derived as $\chi_{BE} = g(\nu_1) + g(\nu_2) - g(\nu_3)$.

Secure key rates R_{sec} are calculated from shared information and Holevo information, $R_{sec} = \beta_r I_{AB} - \chi_{BE}$, with β_r representing reverse reconciliation efficiency. Our secure key rate analysis assumes the asymptotic limit of the signal under a collective attack.

Figure 5.12 plots the secure key rates for Case 0 and Case 2 against modulation variance for the wavefront reconstructions outlined in Section 5.6.1, where the secure key rates for $N = 50$ (colored orange) are higher than for $N = 30$ (colored purple). Note that null key rates are achieved for $N = 10$.

For Case 0, where $\Delta\tilde{\phi}_{mn,S} \approx \Delta\phi_{mn,R} \approx \Delta\phi_{mn,S}$, secure key rates will be equivalent for estimations $\Delta\tilde{\phi}_{mn,S}$ and approximations $\Delta\phi_{mn,R}$ (represented as dotted lines for $N = 30$ and $N = 50$), where shared information is the highest attainable for N modes, given equal coherent efficiencies for $\Delta\tilde{\phi}_{mn,S}$, $\Delta\phi_{mn,R}$ and $\Delta\phi_{mn,S}$. As such, if the assumption $\Delta\phi_{mn,R} \approx \Delta\phi_{mn,S}$ is true (optical hardware is high precision and perfectly calibrated, resulting in negligible relative MPEs), then the intelligent wavefront correction algorithms output $\Delta\tilde{\phi}_{mn,S} \approx \Delta\phi_{mn,R}$, attaining equivalent secure key rates.

Importantly, when considering Case 2, null key rates are found for the $\Delta\phi_{mn,R}$ reconstructions when $\Delta\phi_{mn,R} \neq \Delta\phi_{mn,S}$ for the channel and hardware specifications outlined in this work. The decrease in shared information, resulting from the increased detector noise, is caused by the low coherent efficiencies between the signal wavefront and the reconstructed reference pulse wavefront. However, after our intelligent wavefront correction algorithms are implemented, positive key rates are achieved (dashed lines) for $N = 30$ and $N = 50$. We again note that these results assume asymptotic key rates and idealized channel estimations, which would lead to improved secure key rate calculations. However, the point we exemplify here is the impact of our intelligent wavefront correction algorithms - when large WFEs occur in a CV-QKD setup, then the intelligent wavefront correction algorithms could prove crucial in enabling practical CV-QKD, providing a step

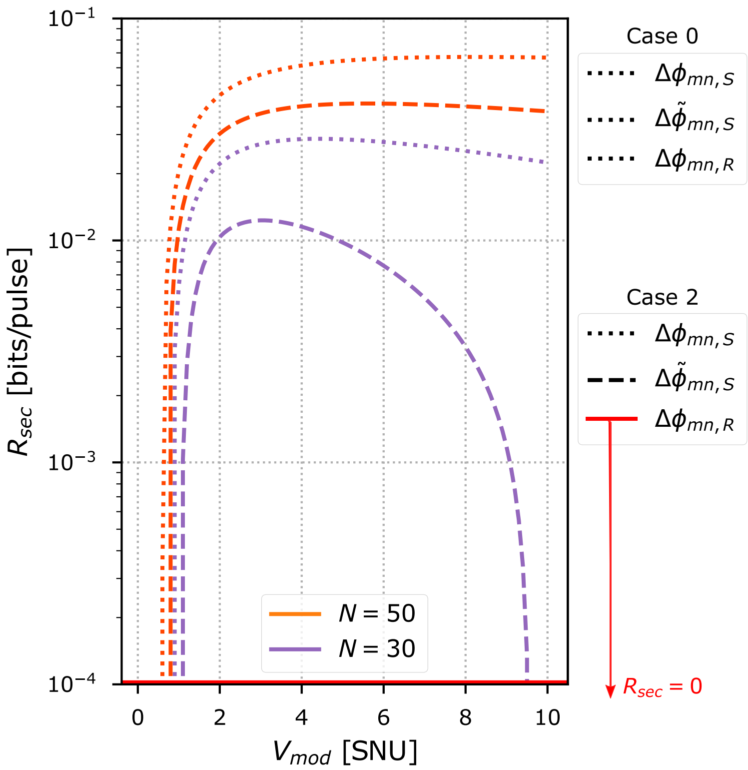

Figure 5.12: Secure key rates versus modulation variance, measured in shot noise units (SNU), for $N = 30$ (purple) and $N = 50$ (orange) modes. Both Case 0 and Case 2 results are presented, where $\Delta\tilde{\phi}_{mn,S} \approx \Delta\phi_{mn,R} \approx \Delta\phi_{mn,S}$ results in equivalent secure key rates for Case 0, such that the dotted lines represent the secure key rates attained for the $\Delta\tilde{\phi}_{mn,S}$, $\Delta\phi_{mn,R}$, and $\Delta\phi_{mn,S}$ wavefront reconstructions. In Case 2, the secure key rates attained for the wavefront reconstructions using $\Delta\tilde{\phi}_{mn,S}$ are plotted as dashed lines and $\Delta\phi_{mn,S}$ are plotted as dotted lines, while null key rates are attained for $\Delta\phi_{mn,R}$. The curve for $\Delta\tilde{\phi}_{mn,S}$ at $N = 30$ is caused by the trade-off between shared information and excess noise as V_{mod} increases.

forward in establishing a realizable Quantum Internet.

5.8 Summary

We presented an analysis of relative wavefront errors associated with real local oscillator-based continuous-variable quantum key distribution protocols across satellite-to-Earth channels, including the potential harm that relative wavefront errors could have on secure key rates. Focusing on the use of a multi-plane light converter for wavefront sensing, we designed machine learning-based intelligent wavefront correction algorithms to correct for relative mode phase errors, increasing secure key rates across the simulated channels, thereby increasing the practical feasibility of implementing a satellite-based quantum communications network. The intelligent wavefront correction algorithms provide another practical example of machine learning assisting continuous-variable quantum key distribution across turbulent atmospheric channels.

After assessing the potential sources, impact, and correction methodologies of relative wavefront errors via simulation in this chapter, we undertake a series of experimental campaigns, providing preliminary evidence of their existence in Chapter 6. An optical reference beacon and a weaker optical signal are polarization-multiplexed, then transmitted across a 2.4 km free-space optical link, before their electric fields are measured using a multi-plane light converter at the receiver. The intelligent wavefront correction algorithms in this chapter provide a framework for the development of intelligent wavefront correction algorithms designed to correct for relative wavefront errors in the experimental data in Chapter 6. Theoretical effects of the relative wavefront errors on excess noise in continuous-variable quantum key distribution are then explored, where the intelligent wavefront correction algorithms are shown to reduce the phase noise introduced by the relative wavefront errors.

Chapter 6

Relative Wavefront Error Correction Over a 2.4 km Free-Space Optical Link via Machine Learning

In Chapter 5, we proposed the existence of two-dimensional spatial relative wavefront errors between multiplexed reference pulses and quantum signals, after transmission across turbulent atmospheric channels. Relative wavefront errors were simulated, then machine-learning based intelligent wavefront correction algorithms were designed to correct for them. In this chapter, we conduct a series of experimental campaigns to test for the existence of relative wavefront errors using an optical reference beacon and a weaker optical signal, which are polarization-multiplexed, then transmitted across a 2.4 km free-space optical link. Intelligent wavefront correction algorithms are developed to estimate the relative mode phase errors derived from multi-plane light converter measurements at the receiver. The intelligent wavefront correction algorithms are designed by combining the machine learning strategies from Chapters 3 to 5. The algorithms are built using the transformer neural network framework from Chapter 5, which harnessed spatial relationships between modes, with the addition of a temporal dimension, as investigated in Chapter 4, while harnessing an intensity-only phase retrieval approach, as applied in Chapter 3. The effects of the relative mode phase errors on theoretical excess noise, in the context of continuous-variable quantum key distribution, are then explored. Note that preliminary lab-based experiments investigating relative wavefront errors are presented in Appendix C, and that the nomenclature used in this chapter applies solely to the content within this chapter.

6.1 Introduction

In coherent optical communication across turbulent atmospheric channels, information can be encoded on the electric field quadratures of a signal, which are then polarization-multiplexed with a transmitted local oscillator (TLO) before transmission. It is commonly assumed that the distortion experienced by the spatial phase wavefront of the TLO is equivalent to the distortion experienced by the spatial phase wavefront of the signal, such that the information encoded on the signal electric field quadratures can be measured using balanced homodyne/heterodyne detection [133].

However, various factors could lead to the TLO and signal wavefronts experiencing differential distortion, resulting in relative wavefront errors (WFEs) between them. For example, imperfections in optical hardware used at the transmitter and receiver, imperfect polarization multiplexing, photon leakage between the stronger TLO and weaker signals, and differential interactions with atmospheric turbulence (e.g. see the discussion in Chapter 5).

We undertake a series of experiments to investigate and characterize relative WFEs, where an optical reference beacon (hereafter, "the reference") and a weaker optical signal (hereafter, "the signal") are polarization-multiplexed before transmission, then sent across a 2.4 km free-space optical (FSO) link. The reference and signal are demultiplexed at the receiver, before their distorted electric fields are measured.

Shack-Hartmann wavefront sensors are often adopted for optical wavefront sensing, particularly in the context of adaptive optics (e.g. [134]; however, they are sensitive to scintillation, which can result in non-uniform illumination [105], [106], where scintillation is often greater across longer channels (e.g. satellite-Earth links [107]). Recently, multi-plane light converters (MPLCs) have been proposed as an alternative to adaptive optics in FSO communications [108], where MPLCs can decompose an electric field into any arbitrary spatial mode basis, which could better capture highly distorted beam profiles. In this work, we focus on the Hermite-Gaussian (HG) spatial mode basis, where the reference and signal electric fields are decomposed in the HG basis, and the relative WFEs between them in each mode (hereafter, referred to as "relative mode phase errors (MPEs)") are derived from the resulting measured voltages.

We then develop machine learning (ML)-based intelligent wavefront correction algorithms, which are designed to estimate the relative MPEs, such that a wavefront correction can be applied to a real local oscillator (RLO) at the receiver. The signal quadratures can then be measured, using the corrected RLO and homodyne/heterodyne detection, with increased coherence. We expand upon the intelligent wavefront correction algorithms in Chapter 5, by designing transformer neural networks (TNNs), which incorporate a temporal dimension to estimate the relative MPEs, in addition to the spatial information, as investigated in Chapter 4.

Further, we present an analysis of the effects of relative WFEs on continuous-variable quantum key distribution (CV-QKD) across turbulent atmospheric channels. In CV-QKD, strong optical references and very weak optical signals (the "quantum" signals) are often polarization-multiplexed before transmission, then demultiplexed at the receiver, where the references are measured and their phase information is encoded onto an RLO. As such, our classical optical experimental work provides the basis for similar intelligent wavefront correction algorithm implementation in CV-QKD protocols. We theoretically analyze the contributions of relative WFEs to excess noise, which can affect the uncertainty of the quantum signal state. The results in Chapter 5 indicate that the additional excess noise can have a deleterious effect on secure key rates, potentially reducing or destroying secure key transfer. Our ML-based intelligent wavefront correction algorithms could prove vital in reducing excess noise to acceptable levels in CV-QKD across turbulent atmospheric channels.

We outline the theory of an MPLC and our experimental setup in Section 6.2, estimate the channel conditions in Section 6.3, present our analysis of the relative WFEs in Section 6.4, examine our intelligent wavefront correction algorithm architectures and performance in Section 6.5, and discuss the effects of relative WFEs on CV-QKD in Section 6.6, before providing a conclusion in Section 6.7.

6.2 System Setup

6.2.1 Multi-Plane Light Conversion

Any electric field can be defined by an arbitrary spatial mode basis, where a fluctuating optical beam can be described using a linear combination of the mode basis with a varying complex phase factor, after passing through turbulent atmospheric channels.

An MPLC is a piece of optical hardware that is able to decompose any arbitrary electric field into N HG spatial modes (note that the subsequent description of an MPLC is based on [108], [112]). A series of phase plates transform the N HG modes into N spatially separated output HG_{00} modes, which are coupled into individual single-mode fibers (SMFs). The HG modes are an orthogonal spatial mode basis, defined by their wavelength λ, the beam waist w_0, and the location along the optical propagation axis z_0. The beam radius $w(z)$ is calculated as,

$$w(z) = w_0\sqrt{1 + \frac{(z - z_0)^2}{z_R^2}}, \tag{6.1}$$

for the beam wavefront radius of curvature $R(z) = (z - z_0) + z_R^2/(z - z_0)$, and the Gouy phase $\psi(z) = \arctan([z - z_0]/z_R)$, both calculated as a function of the Rayleigh range $z_R = \pi w_0^2/\lambda$. The complex fields $\text{HG}_{mn}(x, y, z)$ of the HG modes are derived as,

$$\text{HG}_{mn}(x, y, z) = \frac{e^{-\frac{\rho^2}{w(z)^2} - i\left(\frac{k\rho^2}{2R(z)} + k(z-z_0) - (n+m+1)\psi(z)\right)}}{\sqrt{2^{n+m-1}\pi n! m!}w(z)} \cdot H_m\left(\frac{\sqrt{2}x}{w(z)}\right) \cdot H_n\left(\frac{\sqrt{2}y}{w(z)}\right), \tag{6.2}$$

for the radial distance ρ and the wavenumber $k = 2\pi/\lambda$. Hermite polynomials H_m and H_n are used to spatially modulate the HG modes, calculated as,

$$H_m(x) = m! \sum_{j=0}^{\lfloor \frac{m}{2} \rfloor} \frac{(-1)^j}{j!(m - 2j)} \frac{x^{m-2j}}{2^j}, \tag{6.3}$$

using the floor function $\lfloor\ \rfloor$. Figure 6.1 shows seven HG mode electric fields, with the transverse intensity profiles on the top row and the wavefronts (transverse phase profiles) on the bottom row, for each mode.

The electric field $E(x, y)$ of an optical beam can then be represented using an infinite

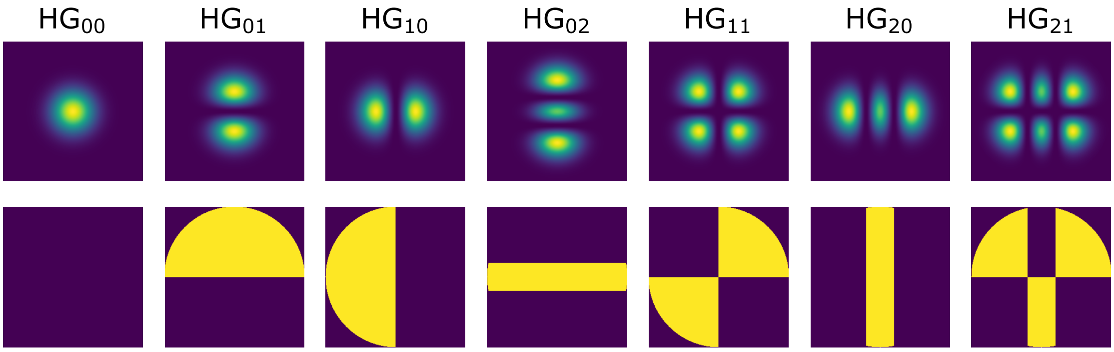

Figure 6.1: Transverse intensity profiles (top row) and transverse phase profiles (bottom row) representation of seven Hermite-Gaussian modes (for an undefined channel).

linear combination of the orthogonal HG modes (at propagation distance z), as,

$$E(x,y) = \sum_{mn}^{\infty} a_{mn} e^{i\Delta\theta_{mn}} \mathrm{HG}_{mn}(x,y) = \sum_{mn}^{\infty} c_{mn} \mathrm{HG}_{mn}(x,y), \tag{6.4}$$

for the weighted amplitudes a_{mn} and phases $\Delta\theta_{mn}$, where the phase term $\Delta\theta_{mn}$ is generically defined in Eq. 6.4. The HG mode complex weighted coefficients c_{mn} can be derived as,

$$c_{mn} = \frac{\iint E(x,y)\mathrm{HG}^*_{mn}(x,y)\ dxdy}{\iint \mathrm{HG}_{mn}(x,y)\mathrm{HG}^*_{mn}(x,y)\ dxdy}, \tag{6.5}$$

where the HG mode complex coefficient in each mode describes the electric field output into each SMF.

6.2.2 Experimental Setup

In order to investigate the presence of relative WFEs between a reference and signal, we conduct a series of experimental campaigns, whereby we transmit a reference beacon (the reference) and a weaker signal (the signal) across a 2.4 km retroreflected FSO link. The retroreflected link is between two buildings at the University of Western Australia in Perth, Australia (shown on the map in Figure 6.2), where the transmitter/receiver is at an altitude of ∼25 m and the corner-cube retroreflector is at an altitude of ∼40 m. We repeat this experiment three times in July 2025, represented as a morning campaign (at approximately 9:00 am), a noon campaign (at approximately 12:00 pm), and an evening campaign (at approximately 5:00 pm).

A schematic of our experimental setup is given in Figure 6.2. The reference and signal

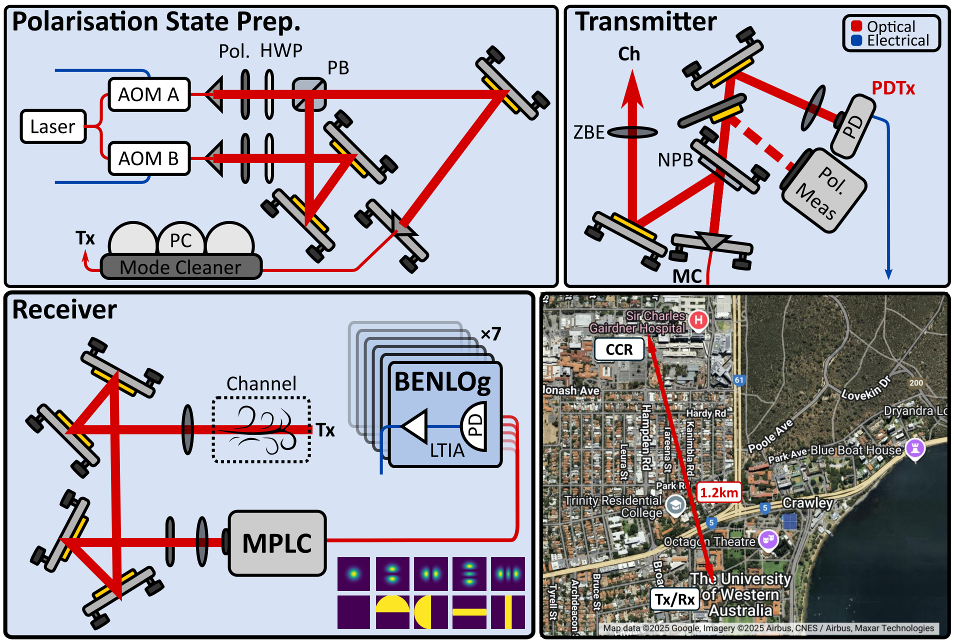

Figure 6.2: Experimental setup for preparation and measurement of a reference and signal, with the 2.4 km link shown at the bottom right, and HG mode intensity and phase profiles shown at the receiver. AOM is an acousto-optic modulator, Pol is a polarizer, PC is a polarization controller, PD is a photodetector, ZBE is a zoom beam expander, MC is the mode cleaner, Tx is the transmitter, Rx is the receiver, Ch is the channel, CCR is the corner-cube retroreflector, and BENLOg represents our logarithmic photodetectors.

are generated using the same 1550 nm laser source, split, then attenuated and frequency-shifted by 50 kHz using separate acousto-optic modulators (AOMs), AOM A and AOM B. The reference and signal are launched into free-space, then orthogonally polarized using a polarizer (Pol) and half-wave plate (HWP), before being polarization-multiplexed using a polarizing beamsplitter (PBS). A mode cleaner, a single-mode fiber (SMF), is used to increase coherence between the reference and signal, before the multiplexed beam is passed through a 90:10 NPB, such that the multiplexed beam powers and the relative phase error between the reference and signal can be derived before transmission across the 2.4 km link.

At the receiver, a polarizer is used to align the polarizations of the reference and signal (to remove any polarization-dependence of the MPLC), before the mixed beam is injected into a Cailabs Tilba-R MPLC [110], which outputs to N SMFs, corresponding to N HG modes. The voltage V_{mn} in each SMF is measured using N separate custom-built "BENLOg" logarithmic photodetectors, then converted to powers P_{mn} in each mode using the empirically derived relation,

$$P_{mn} = |c_{mn}|^2 = 50V_{mn} - 114.543, \tag{6.6}$$

in dBm. The experimental parameters used in this work are given in Table 6.1. Note that in our experiments, we measure the following $N = 7$ modes (shown in Figure 6.1),

$$HG_{MN} = [HG_{00}, HG_{01}, HG_{10}, HG_{02}, HG_{11}, HG_{20}, HG_{12}].$$

Table 6.1: Experimental parameters.

Variable	**Value**
Wavelength (λ)	1550 nm
Channel length (L_{ch})	2.4 km
Transmitter beam waist ($w_{0,T}$)	1.15 mm
Retroreflector diameter (D_{ccr})	63.50 mm
MPLC focal plane beam waist radius (w_R)	300 μm
Signal transmit power ($P_{T,S}$)	-8.1 dBm
Reference signal transmit power ($P_{T,R}$)	-3.1 dBm
Sample rate	10 kHz (kSa/s)

Given that the references and signals are frequency-shifted by the difference frequency $\Delta f = 50$ kHz, we are able to derive the resulting reference mode powers $P_{mn,R}$, signal mode powers $P_{mn,S}$, and *relative MPEs* $\Delta\phi_{mn}$ (described in further detail in Section 6.4), from the resulting heterodyne beat in the photocurrent i_{mn},

$$i_{mn} \propto P_{mn,R} + P_{mn,S} + 2\sqrt{P_{mn,R}P_{mn,S}}\cos(2\pi\Delta f t + \Delta\phi_{mn}), \tag{6.7}$$

via Hilbert transformation for each HG mode (see [135] for a more in-depth description of Hilbert transformations), resulting in approximately one million determinations of the reference mode powers, signal mode powers, and relative MPEs. Note that the voltages V_{mn} in Eq. 6.6 represent the actual measurements in our experiments - all other metrics we refer to in this work are derived from these voltages.

6.3 Channel Estimation

We include a separate transmitter/receiver as a laser beacon across the same link, where the received beam is coupled into a dual-clad fiber for an approximate measure of the free-space power, such that we can estimate the refractive index structure parameter of turbulence C_n^2 across the channel. C_n^2 represents the variation of the refractive index fluctuations in the atmosphere (in $\mathrm{m}^{-2/3}$).

Intensity fluctuations represent "higher-order" disturbances acting on an optical beam as it passes through an atmosphere, caused by a beam's interaction with turbulent eddies with varying refractive indices, resulting in power fluctuations at the receiver. The variance about the mean intensity is known as scintillation [91], which can be represented by the variance of the log-intensity $\sigma^2_{ln(P)}$, calculated as,

$$\sigma^2_{ln(P)} = \ln\left(1 + \frac{\sigma_P^2}{\mu_P^2}\right), \tag{6.8}$$

for the mean μ_P and variance σ_P^2 of the measured power at the photodetector. The log-intensity variance can be calculated for a horizontal channel with a uniform C_n^2 as,

$$\sigma^2_{ln(P)} = 0.496 C_n^2 k^{7/6} L_{ch}^{11/6}, \tag{6.9}$$

where k represent the wavenumber and L_{ch} is the channel length. Therefore, the power measurements can be used to fit an estimate of C_n^2 for the channel (validated in [136]), where the scintillation of the retroreflected two-way link is expected to be equivalent to that of an equivalent-length one-way channel (2.4 km), used to calculate C_n^2. The resulting C_n^2 estimations, for the morning, noon, and evening campaigns, are shown in Table 6.2, which are all of a similar magnitude. The uniformity of the C_n^2 estimations could be attributed to the altitude of the link, where turbulence strength at ground level rapidly decreases with altitude [114].

We further estimate coherence times τ_0 using the relation [137],

$$\tau_0 = 0.314 \frac{r_0}{v_y^{5/3}}, \tag{6.10}$$

for the transverse wind speed v_y (given in Table 6.2) and Fried parameter r_0, calculated as $r_0 = (0.423k^2 C_n^2 L_{ch})^{-3/5}$. The estimated coherence times, based on the estimated C_n^2 for each campaign, are given in Table 6.3.

Table 6.2: C_n^2 and τ_0 estimation for the morning, noon, and evening campaigns.

Campaign	C_n^2	v_y	τ_0
1	2.68×10^{-15} m$^{-2/3}$	3.3 m/s	4.4 ms
2	2.57×10^{-15} m$^{-2/3}$	3.3 m/s	4.3 ms
3	2.70×10^{-15} m$^{-2/3}$	0.8 m/s	45.6 ms

6.4 Relative Wavefront Errors

Most works assume that multiplexed reference spatial phase WFEs $\Delta\Phi_R(x,y)$ equate the signal spatial phase WFEs $\Delta\Phi_S(x,y)$, $\Delta\Phi_R(x,y) \approx \Delta\Phi_S(x,y)$, after transmission across turbulent atmospheric channels. However, relative WFEs between the references and signals could arise due to a number of factors, such that $\Delta\Phi_R(x,y) \neq \Delta\Phi_S(x,y)$. Firstly, imperfections in, and imperfect calibration/alignment of, optical hardware could result in differential distortion of the reference and signal wavefronts (e.g. due to post-manufacturing stresses and thermal fluctuations [117]). Imperfect polarization-multiplexing could lead to photon leakage between the references and weaker signals [16], [118], and imperfections in MPLC phase mask manufacturing could lead to cross-talk between the HG modes, when using an MPLC for wavefront sensing [119]. Further, differential interactions with the atmosphere, such as via weak birefringence, could result in further relative WFEs.

We previously simulated relative WFEs between the references and signals for satellite-Earth channels in Chapter 5, in the context of CV-QKD. We also proposed ML-based intelligent wavefront correction algorithms as a solution, where wavefront corrections are applied to the RLO, increasing secure key rates across the simulated channels. We then conducted preliminary experiments indicating the presence of relative WFEs across turbulent channels in a lab-based environment (see Appendix C), showing normalized mode power differentials between references and weaker signals after measurement using an MPLC.

In this chapter, we further the body of work on relative WFEs by emulating a CV-

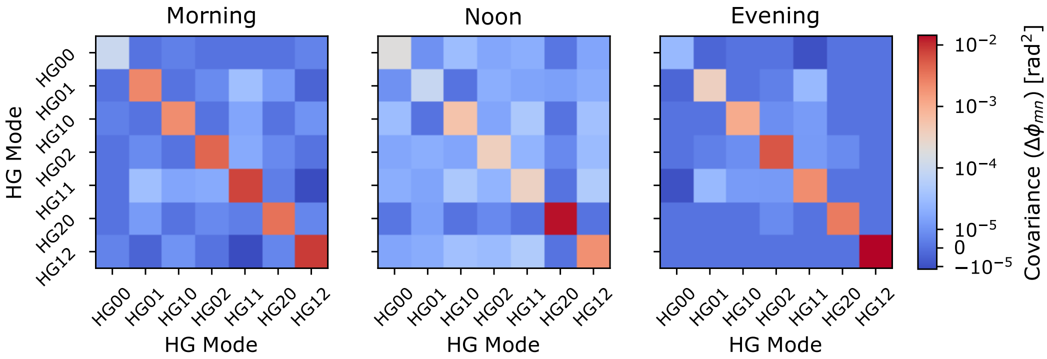

Figure 6.3: Covariances of relative mode phase errors $\Delta\phi_{mn}$ between reference and signal in each HG mode for the morning, noon, and evening campaigns.

QKD setup across a 2.4 km retroreflected FSO link, showing the presence of relative WFEs across turbulent atmospheric channels. We represent the relative WFEs using the relative MPEs $\Delta\phi_{mn}$, defined as $\Delta\phi_{mn} = \Delta\phi_{mn,R} - \Delta\phi_{mn,S}$, where $\Delta\phi_{mn,R}$ and $\Delta\phi_{mn,S}$ are the phase components of the reference and signal HG complex coefficients, respectively (replacing $\Delta\theta_{mn}$ in Eq. 6.5).

6.4.1 Covariance

To understand how the relative MPEs evolve between modes, we calculate the covariances of the relative MPEs $\mathrm{Cov}(\Delta\phi_{mn})$ [138], which gives an indication of how relative MPEs vary with each other, as shown in Figure 6.3 for the morning, noon, and evening campaigns. Firstly, the relative MPE variances are given along the diagonal for each campaign. The relative MPE variances are the lowest for the HG_{00} mode in the morning and evening campaigns, while they are lowest for the HG_{01} mode in the noon campaign. The noon campaign has low relative MPE variances across the modes HG_{00} to HG_{11}, with the greatest variance spiking for the HG_{20} mode, which we believe can be attributed to the low signal-mode power observed in the HG_{20} mode for that campaign. The evening campaign has the greatest relative MPE variance in the HG_{12} mode, while the morning campaign appears to have more evenly distributed relative MPE variances than the noon and evening campaigns. Overall, the relative MPE variances are different for each campaign, indicating that relative MPEs will not only vary between experimental setups, but also between campaigns using the same setup.

Focusing on the off-diagonal covariance terms, it can immediately be seen that there is a weak relationship between the variation of relative MPEs in one mode and the variation of relative MPEs in another. The noon campaign appears to have slightly greater covariances between relative MPEs in different modes; however, the results could be purely random, given the low covariance values in general. Crucially, here we show that relative WFEs *can exist* between the references and signals.

6.4.2 Significance

To further understand the relationship between relative MPEs in each mode, we show their probability density functions (PDFs) in Figure 6.4 for all campaigns. We subtract the relative MPEs for the HG_{00} mode to give an indication of how the transmitted HG_{00} mode evolves into the higher-order relative MPEs. It can immediately be seen that the higher-order relative MPEs are different from the HG_{00} mode ($\Delta\phi_{mn} - \Delta\phi_{00} \neq 0$). Examining the differences between the higher-order relative MPE PDFs, it appears that they vary the most between modes for the evening campaign, with the relative MPE PDFs appearing more similar between higher-order relative MPE modes for the morning and noon campaigns.

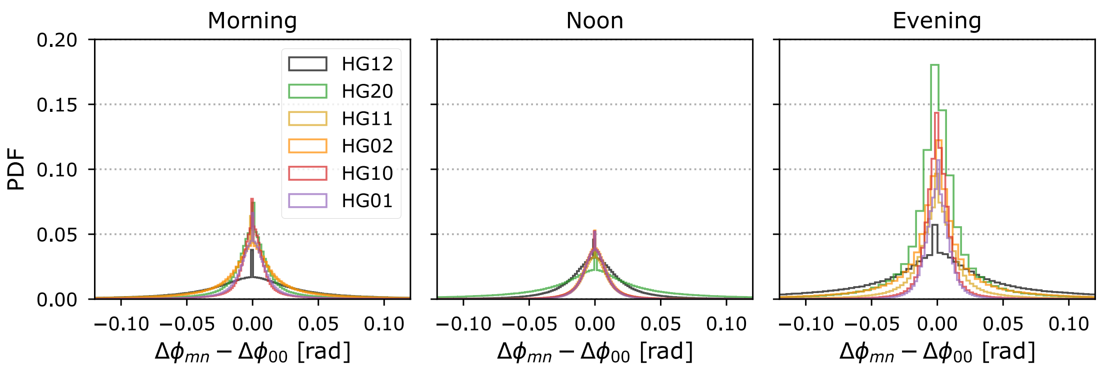

Figure 6.4: Relative mode phase error $\Delta\phi_{mn}$ probability density functions (PDFs) in each HG mode for the morning, noon, and evening campaigns, with $\Delta\phi_{00}$ subtracted.

Here, we aim to determine if the relative MPE distributions are statistically different between modes, indicating that the optical hardware and atmosphere evolve the transmitted reference and signal HG_{00} mode into higher-order HG modes in a unique manner. To quantify whether the relative MPEs in each mode can be considered as from the same distribution, we perform a Friedman test. The Friedman test is a non-parametric test for differences between groups (here, the relative MPEs in each HG mode), without assuming independence of the groups [139].

The Friedman test determines if the null hypothesis of the relative MPE determinations in each mode being from the same distributions is valid. The Friedman parameter F is compared to a critical chi-squared value χ^2, determined by a confidence level α and the number of degrees of freedom (DoF) in the test, where the null hypothesis is rejected if $F > \chi^2$. F is calculated as,

$$F = \frac{12}{S \cdot N \cdot (N+1)} \sum_{mn}^{N} R_{mn}^2 - [3 \cdot S \cdot (N+1)], \tag{6.11}$$

where S is the number of relative MPE determinations and R_{mn} represents the total rank score for each mode (an accumulative weighting of each determination across the modes). The DoF in our Friedman test is DoF$= N - 1 = 6$, such that χ^2 for a confidence level $\alpha = 0.05$ is $\chi^2 = 12.592$ [139]. The Friedman test is discussed in greater detail in [139].

Comparing F for the morning campaign ($F = 41.37$), noon campaign ($F = 41.35$), and evening campaign ($F = 84.95$) with χ^2, we find that $F > \chi^2$, thus the null hypothesis is rejected for all campaigns, and the relative MPEs in each mode are not considered as all coming from the same distribution. Therefore, our results indicate that the various factors affecting the transmitted HG_{00} mode reference and signal wavefronts cause variational leakage into the higher-order HG modes, rather than a consistent transformation into all higher-order HG modes. Note that F for the evening campaign is the highest, which corresponds to the visual interpretation of the relative MPE PDFs being the most different for the evening campaign in Figure 6.4.

When considering the CV-QKD application, relative MPEs between the references and signals can destroy secure key information (as explored in Chapter 5). Therefore, increasing our understanding of the relationship between the relative MPEs, after transmission across turbulent atmospheric channels, could aid in the design of wavefront correction techniques to mitigate their deleterious effects on quantum information transfer between parties.

6.5 Transformer Neural Network

Given the difference between the reference and signal phase WFEs, $\Delta\Phi_R(x,y) \neq \Delta\Phi_S(x,y)$, we propose the use of a NN to estimate the correction $\Delta\Phi_R(x,y) - \Delta\Phi_S(x,y)$, which should be applied to the RLO at the receiver for coherent measurement of the signal.

In Chapter 5, we implemented NNs for wavefront correction using independent simulated reference MPE $\Delta\phi_{mn,R}$ inputs, which mapped to independent simulated signal MPE outputs $\Delta\phi_{mn,S}$ for each determination. However, our experiments derived only the relative MPEs (i.e., $\Delta\phi_{mn,R} - \Delta\phi_{mn,S}$); thus, in this work, we investigate phase retrieval for wavefront correction. The problem of phase retrieval involves retrieving the phase wavefront information of an electric field using only intensity information [140]. Previous works have used a variety of algorithms for phase retrieval, ranging from the Gerchberg–Saxton algorithm [141] to ML-based phase retrieval algorithms [142]. One work proposed a phase retrieval algorithm for the HG basis [143], while we implemented a convolutional NN to estimate the wavefront of reference pulses using only their intensity distribution, for the application of CV-QKD, in Chapter 3. In this work, we develop ML-based intelligent wavefront correction algorithms, which output an estimate ("retrieval") of the relative MPEs using only the reference mode powers as input, such that the relative MPEs can be used to reconstruct an RLO wavefront correction. Note that the RLO wavefront first needs to be shaped using the reference MPEs, which can be measured in a practical coherent FSO setup, before the relative MPEs are applied.

We base our intelligent wavefront correction algorithm architectures on encoder TNNs. TNNs are capable of building spatial relationships between input matrix elements, here the reference mode powers, then mapping them to a vector of output elements, here the relative MPEs. We describe TNNs in-brief; please refer to [125] for a more comprehensive discussion of TNNs. We expand the information available to the intelligent wavefront correction algorithms in Chapter 5 by including a temporal dimension to the input, such that the previous M_t reference mode power time-steps are used to estimate the relative MPEs for the current time-step (at time t), as in the temporal analysis in Chapter 4. As such, the reference mode powers are input as a matrix of dimension $M_t \times N$.

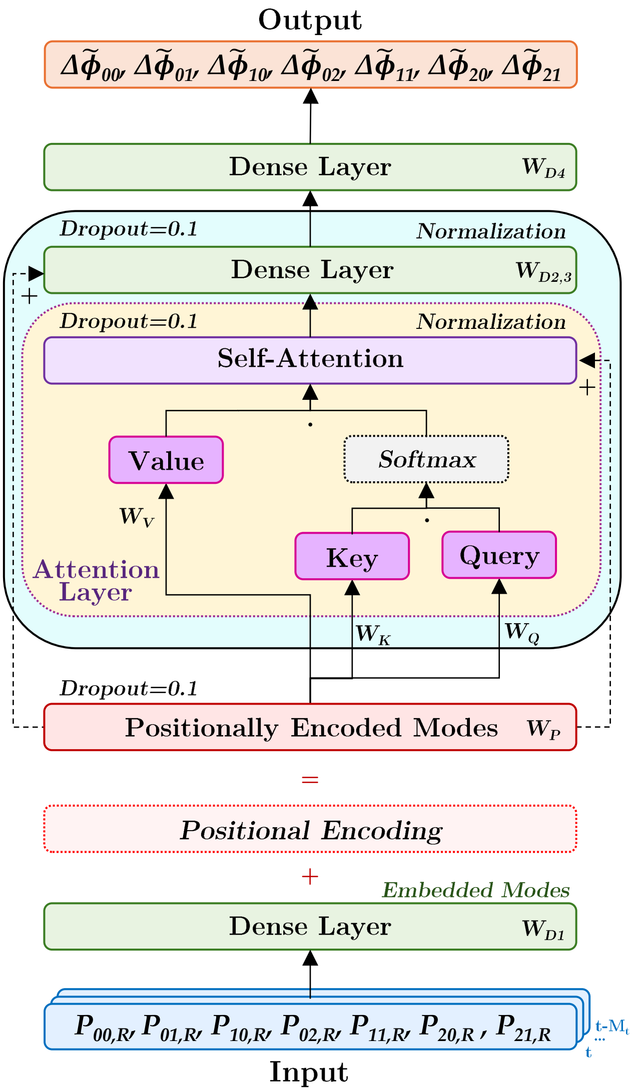

Figure 6.5: TNN architecture, taking reference mode powers for the previous M_t time-steps as input, then outputting relative MPEs for the current time-step. Transformer layer architecture is depicted in the light blue block, while the attention head architecture is depicted in the yellow block. The weights W_P represent the weights for the positional encoding, the weights W_{D1}, W_{D2}, W_{D3} and W_{D4} represent the weights for the dense layers, and the weights W_K, W_Q and W_V represent the weights for the key, query, and value matrices, respectively. The time-step at time t is shown.

The input matrix is embedded into the TNN via a (fully-connected) *dense layer* of dimension d_m, where the same dense layer weightings W_{D1} are applied to each determination of the reference mode powers in the input matrix, resulting in an embedded mode matrix of dimension $M_t \times d_m$. Positional encoding is then performed, where the spatial positions of the embedded mode matrix elements are weighted and summed with the embedded modes (as shown at the bottom of Figure 6.5), resulting in the *positionally* *encoded modes* (PEMs), a matrix of dimension $M_t \times d_m$. The PEMs are then passed into the transformer layers (shown as the light blue block in Figure 6.5), which are comprised of multiple *attention heads* (depicted as the yellow block in Figure 6.5), and two dense layers.

Each attention head derives three representations of the relationships between PEM elements. A *key matrix* (providing "relevance") and a *query matrix* (providing "context") are derived for each PEM, before their dot product is taken, and passed through a Softmax function [126]. A dot product of the Softmax function output and a separate *value matrix* (applied to the PEMs) is taken, which is then summed with the original PEMs to evaluate a "similarity" between them, resulting in a *self-attention matrix* of dimension $M_t \times d_m$. The key, query, and value matrices are of dimension $n_{attn} \times M_t \times |d_m/n_{attn}|$, where n_{attn} represents the number of attention heads in the TNN architecture.

Multiple attention heads can identify different relationships between the PEM elements, aiding estimation performance. The self-attention matrix is passed to a dense layer of dimension d_t and another of dimension d_m, which are then summed with the PEMs (to preserve positionality information). The transformer layer results are then passed into a final dense layer of dimension d_t, outputting a vector of relative MPE estimations for that time-step (of length $N = 7$), as is depicted in Figure 6.5.

The architecture hyperparameters for the TNN used in this work are given in Table 6.3, as applied to all campaigns, which are obtained via an iterative process initialized by the simulation work in Chapter 5, with the objective of achieving sufficient estimation performance, while limiting the total number of trainable parameters in the model.

Table 6.3: TNN architecture hyperparameters.

Hyperparameter	**Value**
Time dimension (M_t)	100
Matrix dimension (d_m)	32
Matrix dimension (d_t)	256
Attention heads (n_{attn})	8
Transformer layers (#)	8
Trainable parameters (#)	166,583
Dropout	0.1

6.5.1 Training

We reduce the sample rate from 10 kHz to 1 kHz when estimating the relative MPEs, where we assume that the fluctuations of the reference mode powers and relative MPEs, caused by varying turbulent atmospheric conditions, are sufficiently captured, given the estimated coherence times in Table 6.2. The 1 kHz samples result in approximately 100,000 determinations of the reference mode powers and relative MPEs, in each mode for each campaign. We split the data into ∼85,000 training determinations and ∼15,000 testing determinations.

During training, the known reference mode powers from the previous M_t time-steps are mapped to known relative MPEs for the current time-step. The weights W_{D1}, W_{D2}, W_{D3} and W_{D4} are applied to the dense layer connections, and the weights W_K, W_Q and W_V are applied to the key, query and value matrices, respectively. The weights are all optimized by minimizing a mean squared error loss function, using the Adam optimizer [127]. We trained the TNNs using a batch size of 16, for 100 epochs, where the batch size represents the number of samples (here, $M_t \times N$) that were mapped from the input to output before the internal weights were updated, while the epochs represent how many times the entire training dataset (the ∼85,000 training determinations) were passed through the TNN during optimization. All weights are fixed after training. *Dropout* is implemented for the PEM, self-attention, and dense layers within the transformer layers, which helps to prevent overfitting by replacing a percentage of the data in each matrix with zeros [128]. The self-attention and dense layers are also normalized to improve estimation performance [129].

If implemented into a CV-QKD protocol, the TNNs take the known previous M_t reference mode power determinations as input, then output an estimate of the relative MPEs for the current time-step. The relative MPEs can then be used to modulate the RLO for improved coherence during signal measurement (discussed in Sections 6.5.2 and 6.6).

6.5.2 Estimation Performance

We analyze the TNN estimation performance by comparing the *derived variance* $\mathrm{Var}(\Delta\phi_{mn})$ in each mode, as would be the case without any wavefront correction, with the *correction variance* $\mathrm{Var}(\Delta\phi_{mn} - \Delta\tilde{\phi}_{mn})$, which is achieved after the TNN estimations are applied to the RLO. In the case of perfect estimation performance of the TNNs, $\Delta\phi_{mn} - \Delta\tilde{\phi}_{mn} = 0$, while as long as $\mathrm{Var}(\Delta\phi_{mn} - \Delta\tilde{\phi}_{mn}) < \mathrm{Var}(\Delta\phi_{mn})$, coherence between the signal and RLO should be increased.

The derived variances (blue bars) and correction variances (red bars) are shown in Figure 6.6 for all campaigns, where the purple sections show where the derived variances and correction variances overlap. Overall, it can be seen that the correction variances are generally lower than the derived variances (where the blue is greater than the red), indicating improved coherence between the RLO and signal. The correction variances are lower than the derived variances for all modes in the evening campaign, and lower than all modes except the HG_{02} mode for noon campaign, while the derived variance is lower than the correction variance for the modes HG_{00} and HG_{12}, and approximately equal for mode HG_{01}, in the morning campaign. We analyze the impact of the derived variance and correction variance on CV-QKD in Section 6.6.

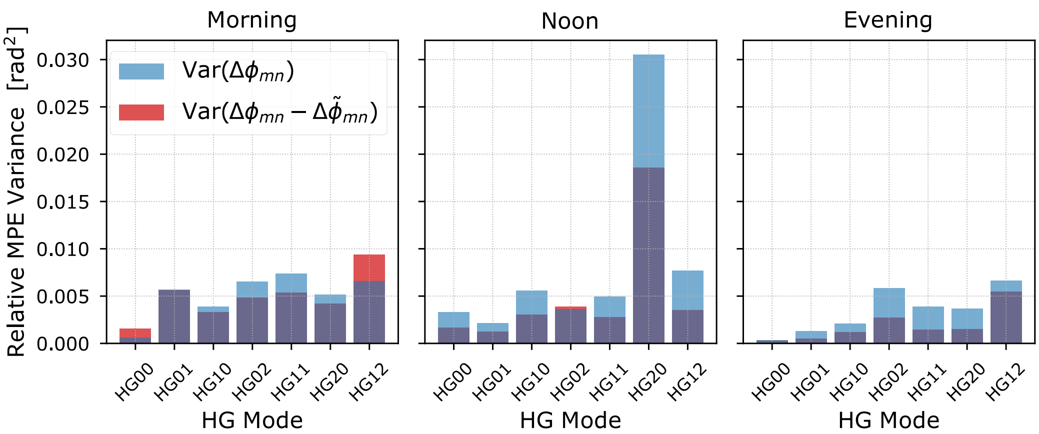

Figure 6.6: Derived variance $\mathrm{Var}(\Delta\phi_{mn})$ (blue bars) versus correction variance $\mathrm{Var}(\Delta\phi_{mn} - \Delta\tilde{\phi}_{mn})$ (red bars) for the morning, noon, and evening campaigns, where purple shows were the derived variances and correction variances overlap.

6.6 Implications for CV-QKD

As in classical FSO communication, CV-QKD often involves encoding secure key information on the electric field quadratures of weak quantum signals, before transmission across turbulent atmospheric channels, where a "truly" quantum signal is comprised of up to tens of photons [144]. State-of-the-art CV-QKD protocols multiplex reference pulses with the quantum signals, such that the reference pulse phase error is then encoded onto an

RLO at the receiver, which is used for coherent measurement of the quantum signals [15], [16]. In Chapter 5, we simulated relative WFEs between the reference pulses and quantum signals across satellite-Earth links, indicating that the resulting incoherence between the reference pulse and signal wavefronts could lead to reduced or null key rates for the channels.

In this work, we assume that the derived relative WFEs across the 2.4 km FSO link could be representative of relative WFEs between reference pulses and quantum signals, in the context of CV-QKD. We note that the greater difference in optical powers could result in greater photon leakage between the reference pulses and quantum signals, which could have an effect on the relative WFEs. While several factors could contribute to their existence, ultimately, relative WFEs would have an effect on the overall excess noise affecting the quantum signals, and should be taken into consideration when undertaking reverse reconciliation during FSO CV-QKD protocols across turbulent atmospheric channels. We present the CV-QKD protocol in this section, then discuss how the relative WFEs could contribute to the excess noise affecting the quantum signal measurements. The total excess noise ξ_{tot} has contributions from both detector noise ξ_{det} and channel noise ξ_{ch}, $\xi_{tot} = \xi_{det} + \xi_{ch}$, which leads to reduced secure key rates across a channel [13].

6.6.1 CV-QKD Protocol

The schematic in Figure 6.7 outlines the Gaussian modulated coherent state CV-QKD protocol adopted in this chapter. At the transmitter (Alice), a laser source is split into the reference and quantum signal using a non-polarizing beamsplitter, the quantum signal is phase and amplitude modulated, attenuated, then recombined with the reference using a PBS before transmission. At the receiver (Bob), the reference and quantum signal are split using a PBS. Here, we use an MPLC as a wavefront sensor to measure the reference mode powers in the HG basis, then use the reference mode power information to estimate the relative MPEs, which are used to reconstruct an estimate of the quantum signal wavefront, which is encoded onto the RLO wavefront using a deformable mirror. The RLO is then mixed with the quantum signal using a homodyne detector to measure the quantum signal quadratures X_B or P_B, where the subscript B represents Bob.

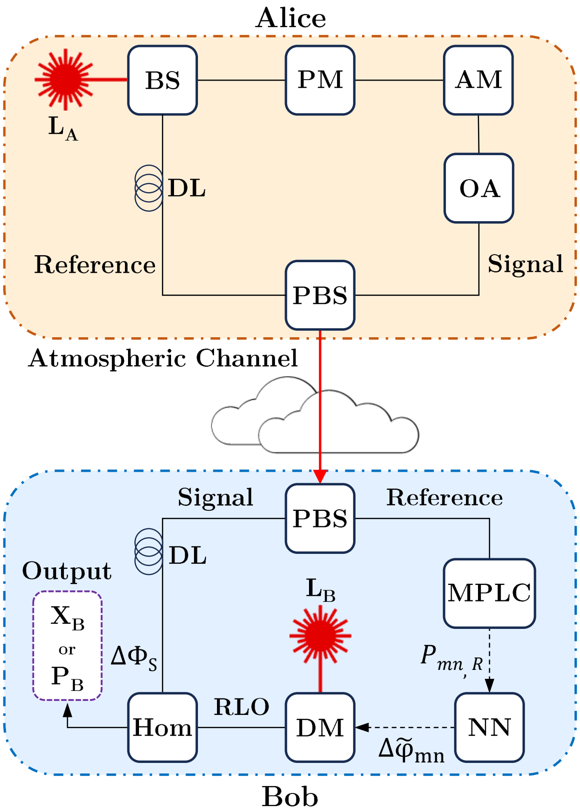

Figure 6.7: Schematic outlining the preparation and measurement of coherent states in the CV-QKD protocol. L is a laser source, BS is a non-polarizing beamsplitter, AM is an amplitude modulator, PM is a phase modulator, OA is an optical attenuator, PBS is a polarizing beamsplitter, MPLC is a multi-plane light converter, Hom is a homodyne detector, DM is a deformable mirror, DL is a delay line, and NN is the neural network (intelligent wavefront correction algorithm). The quadrature measurements X_B or P_B contain the secure key information.

6.6.2 Detector Noise

To quantify the coherence between the reference electric field $E_R(x, y)$ and signal electric field $E_S(x, y)$, the coherent efficiency γ can be calculated as [132],

$$\gamma = \frac{|\iint_{\mathcal{D}_R} E_S^*(x,y) E_R(x,y)\ dr|^2}{\left(\iint_{\mathcal{D}_R} |E_S(x,y)|^2\ dr\right)\left(\iint_{\mathcal{D}_R} |E_R(x,y)|^2\ dr\right)}, \tag{6.12}$$

across the aperture with diameter $\mathcal{D}_R$ for the radial integral dr. A coherent efficiency of one represents perfect mode matching between the electric fields, e.g. $E_R(x, y) = E_S(x, y)$, while a coherent efficiency of zero represents complete decoherence.

Given that we have independent HG mode measurements, we restructure Eq. 6.12

using Eq. 6.4, where $E_S^*(x,y)E_R(x,y)$ becomes,

$$E_S^*(x,y)E_R(x,y) = \sum_{mn}^{N}\sum_{m'n'}^{N}\left(HG_{mn}^*(x,y)\sqrt{P_{mn,S}}e^{-i\Delta\phi_{mn,S}}\right) \cdot \left(HG_{m'n'}(x,y)\sqrt{P_{m'n',R}}e^{i\Delta\phi_{m'n',R}}\right). \tag{6.13}$$

The orthonormality of the HG basis is defined by [108],

$$\iint HG_{m'n'}^*(x,y)HG_{mn}(x,y)\ dxdy = \delta_{mm'}\delta_{nn'}, \tag{6.14}$$

where δ_{ij} is the Kronecker delta. Therefore, we can simplify the numerator of Eq. 6.12 to only the terms where $mn = m'n'$ as,

$$|\iint_{\mathcal{D}_R} E_S^*(x,y)E_R(x,y)\ dr|^2 = |\sum_{mn}^{N}\sqrt{P_{mn,S}P_{mn,R}}\ e^{i\Delta\phi_{mn}}|^2,$$

noting the relation $\Delta\phi_{mn} = \Delta\phi_{mn,R} - \Delta\phi_{mn,S}$. We can also simplify the denominator terms as,

$$\iint_{\mathcal{D}_R} |E_S(x,y)|^2\ dr = \sum_{mn}^{N} P_{mn,S},$$
$$\iint_{\mathcal{D}_R} |E_R(x,y)|^2\ dr = \sum_{mn}^{N} P_{mn,R},$$

resulting in the updated coherent efficiency calculation,

$$\gamma = \frac{|\sum_{mn}^{N}\sqrt{P_{mn,S}P_{mn,R}}\ e^{i\Delta\phi_{mn}}|^2}{\left(\sum_{mn}^{N}P_{mn,S}\right)\left(\sum_{mn}^{N}P_{mn,R}\right)}, \tag{6.15}$$

as a function of the reference mode powers, signal mode powers, and relative MPEs. Similarly, the corrected coherent efficiency γ_{cor}, after applying the wavefront correction, becomes,

$$\gamma_{cor} = \frac{|\sum_{mn}^{N}\sqrt{P_{mn,S}P_{mn,R}}\ e^{i(\Delta\phi_{mn}-\Delta\tilde{\phi}_{mn})}|^2}{\left(\sum_{mn}^{N}P_{mn,S}\right)\left(\sum_{mn}^{N}P_{mn,R}\right)}. \tag{6.16}$$

Here, we calculate Eqs. 6.15 and 6.16 for the 15,000 test determinations of the reference mode powers, signal mode powers, and relative MPEs for each campaign.

Considering that an infinite number of HG modes theoretically comprise an electric field, the finite number of modes used to calculate the coherent efficiency in Eqs. 6.15 and 6.16 captures only a portion of the total coherent efficiency. We consider the effect of N on the average calculated coherent efficiencies in Figure 6.8(a) for all campaigns, by summing an increasing number of modes (in the order given on the x-axis). It can be seen that increasing N decreases the average coherent efficiency between the reference and signal for each campaign, where summing additional reference and signal modes leads to a greater difference in their wavefronts. Extrapolating on our results, we would assume that the average coherent efficiency would reduce further as the number of measured modes increases. Taking the $N = 7$ HG modes measured in this work, the average coherent efficiencies are $\langle\gamma\rangle = 0.977$ for the morning campaign, $\langle\gamma\rangle = 0.987$ for the noon campaign, and $\langle\gamma\rangle = 0.986$ for the evening campaign, while the average corrected coherent efficiencies are $\langle\gamma_{cor}\rangle = 0.977$ for the morning campaign, $\langle\gamma_{cor}\rangle = 0.988$ for the noon campaign, and $\langle\gamma_{cor}\rangle = 0.986$ for the evening campaign. As such, it can be seen that there is only a small increase in average coherent efficiency when the wavefront corrections are applied, on the order of 10^{-3} to 10^{-4}.

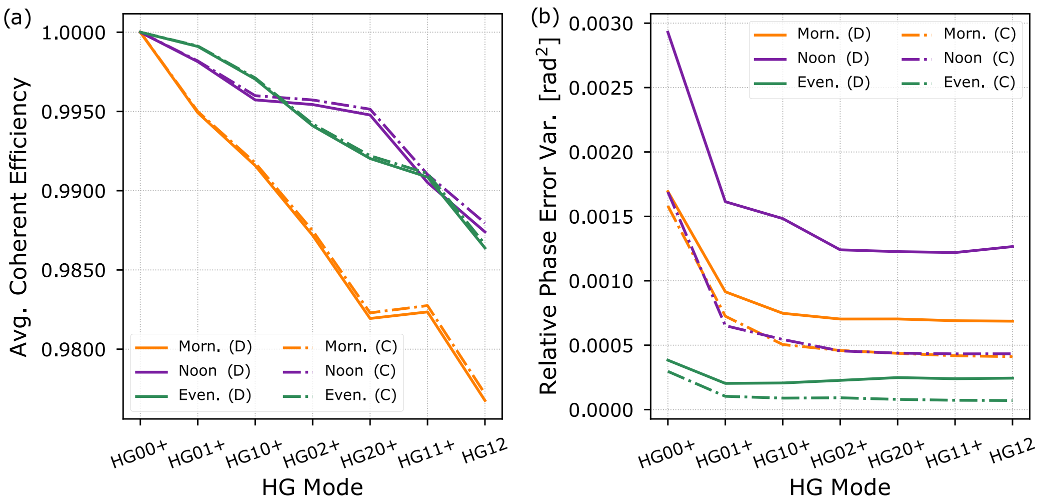

Figure 6.8: (**a**) Average coherent efficiencies and (**b**) relative phase error variances for accumulating HG modes for the morning, noon, and evening campaigns, where D represents derived (plotted as solid lines) and C represents corrected (plotted as dot-dashed lines).

Detector noise is calculated as a function of the coherent efficiency as [78],

$$\xi_{det} = \frac{((1-\gamma) + \xi_{el})\eta_{det}}{\gamma}, \tag{6.17}$$

$$\xi_{det,cor} = \frac{((1-\gamma_{cor}) + \xi_{el})\eta_{det}}{\gamma_{cor}}, \tag{6.18}$$

for the detector efficiency η_{det} and electronic noise ξ_{el} (given in Table 6.4), where $\xi_{det,cor}$ is the corrected detector noise. The effects of detector noise on effective excess noise are shown in Section 6.6.4.

Table 6.4: Noise parameters.

Parameter	Value
Modulation variance (V_{mod})	5 SNU
Detector efficiency (η_{det})	95% [72]
Detector electronic noise (ξ_{el})	0.01 SNU [96]
Background noise (ξ_{back})	0.0002 SNU [131]
Modulation noise (ξ_{mod})	0.0005 SNU [131]
Time-of-arrival fluctuations (ξ_{ta})	0.0012 V_{mod} [72]
RIN of RP from atmosphere ($\xi_{RIN,R}$)	0.002 V_{mod} [72]
RIN of signal from atmosphere ($\xi_{RIN,S}$)	$<$ 0.0001 [72]
RIN of RLO ($\xi_{RIN,LO}$)	0.00035 V_{mod} [72]

6.6.3 Channel Noise

In Gaussian modulated coherent state CV-QKD protocols, information is encoded on the electric field quadratures of the quantum signal. As such, the signal quadrature(s) are measured by mixing the RLO and signal using balanced homodyne/heterodyne detection [13]. Relative MPEs will ultimately result in a relative phase error $\Delta\phi$ in the quadrature phase space. The relative phase error can be calculated by taking the complex amplitude Ξ of the resulting mixed signal, given by [145], [146],

$$\Xi = \iint_{\mathcal{D}_R} E_S^*(x,y)E_R(x,y)\ dxdy, \tag{6.19}$$

which we then simplify in a similar manner to the numerator in Eq. 6.15, as,

$$\Xi = \sum_{mn}^{N} \sqrt{P_{mn,S}P_{mn,R}}\ e^{i\Delta\phi_{mn}}. \tag{6.20}$$

The resulting relative phase error of the mixed signal at each independent determination becomes,

$$\Delta\phi = \arg\left(\Xi\right). \tag{6.21}$$

Once the wavefront correction is applied, the corrected relative phase error $\Delta\phi_{cor}$ is calculated as,

$$\Xi_{cor} = \sum_{mn}^{N} \sqrt{P_{mn,S}P_{mn,R}}\; e^{i(\Delta\phi_{mn} - \Delta\tilde{\phi}_{mn})}, \tag{6.22}$$

$$\Delta\phi_{cor} = \arg\left(\Xi_{cor}\right). \tag{6.23}$$

The relative phase error variance $\mathrm{Var}(\Delta\phi)$ and corrected relative phase error variance $\mathrm{Var}(\Delta\phi_{cor})$ are then calculated for the 15,000 test determination of the reference mode powers, signal mode powers, and relative MPEs for the morning, noon, and evening campaigns.

Given the finite number of HG modes measured, we consider the effect of N on the calculated relative phase error variance in Figure 6.8(b) for all campaigns, by summing an increasing number of modes (in the order given on the x-axis). It can be seen that increasing N initially decreases the relative phase error variance. However, the reduction in relative phase error variances appears to asymptote as the number of modes increases. As such, given our experimental results, we assume that the asymptoted relative phase error variances are representative of the true relative phase error variances (given an infinite number of modes).

When considering the corrected relative phase error variances in Figure 6.8(b), it can be seen that the proportional improvements are much greater when the wavefront correction is applied, than for the proportional increase in coherent efficiency. For $N = 7$ modes, the relative phase error variances decrease after wavefront correction from $\mathrm{Var}(\Delta\phi) = 0.0007$ rad^2 to $\mathrm{Var}(\Delta\phi_{cor}) = 0.0004$ rad^2 for the morning campaign, $\mathrm{Var}(\Delta\phi) = 0.0013$ rad^2 to $\mathrm{Var}(\Delta\phi_{cor}) = 0.0004$ rad^2 for the noon campaign, and $\mathrm{Var}(\Delta\phi) = 0.0002$ rad^2 to $\mathrm{Var}(\Delta\phi_{cor}) = 0.0001$ rad^2 for the evening campaign. As such, given $N = 7$ modes, we can expect up to a 2/3 decrease in relative phase error variance after the wavefront correction is applied, as achieved in the noon campaign.

Phase noise ξ_ϕ is calculated as a function of the relative phase error variance, where phase noise is one contribution to channel noise, calculated as,

$$\xi_\phi = 2V_{mod}(1 - e^{-Var(\Delta\phi)/2}), \tag{6.24}$$

$$\xi_{\phi_{cor}} = 2V_{mod}(1 - e^{-Var(\Delta\phi_{cor})/2}), \tag{6.25}$$

for the modulation variance V_{mod} (given in Table 6.4), where $\xi_{\phi_{cor}}$ represents the corrected phase noise. The resulting phase noise, calculated for $N = 7$ modes before the wavefront correction is applied, is $\xi_\phi = 0.0034$ for the morning campaign, $\xi_\phi = 0.0063$ for the noon campaign, and $\xi_\phi = 0.0012$ for the evening campaign.

Note that several sources are known to contribute to the relative phase error variance [15], [17], including detector phase error variance introduced by the shot-noise limited and experimental noise of reference pulse heterodyne detection [17], [72], as well as phase drift variance, caused by the relative drift of the laser source at Alice and laser source for the RLO at Bob. In this chapter, we focus only on the relative phase error variance when calculating phase noise, to specifically understand the effects of relative MPEs on effective excess noise.

Total channel noise accumulates from several additional sources to phase noise, including background noise ξ_{back}, modulation noise ξ_{mod}, time-of-arrival fluctuations ξ_{ta}, and the relative intensity noise (RIN) of the signal $\xi_{RIN,S}$, reference $\xi_{RIN,R}$ and RLO $\xi_{RIN,LO}$ (given in Table 6.4). The total channel noise is then a summation of each noise contribution, calculated for the derived and corrected phase noise terms as,

$$\xi_{ch} = \xi_{back} + \xi_{mod} + \xi_{ta} + \xi_\phi + \xi_{RIN,R} + \xi_{RIN,S} + \xi_{RIN,LO}, \tag{6.26}$$

$$\xi_{ch,cor} = \xi_{back} + \xi_{mod} + \xi_{ta} + \xi_{\phi_{cor}} + \xi_{RIN,R} + \xi_{RIN,S} + \xi_{RIN,LO}, \tag{6.27}$$

where $\xi_{ch,cor}$ represents the corrected channel noise. The effects of channel noise on the effective excess noise are examined in Section 6.6.4. Note that time-of-arrival fluctuations would be one mechanism that impacts the reference and signals differently, which has a similar magnitude to the derived phase noises. We do not claim that this is the source of the relative WFEs observed — we simply point out that the levels we see in the experiment are consistent with those anticipated for time-of-arrival effects [96].

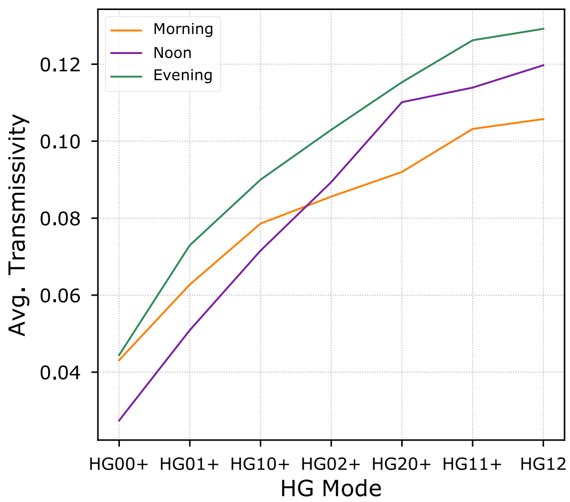

Figure 6.9: Average transmissivities for accumulating HG modes for the morning, noon, and evening campaigns.

6.6.4 Effective Excess Noise

Here, we present an analysis of the effects of relative MPEs on the *effective excess noise*, before and after the wavefront corrections are applied, for both a trusted and untrusted detector, where the theoretical effective excess noise would affect the uncertainty of quantum signals in a CV-QKD system. A trusted detector assumes that any eavesdropper (Eve) does not have access to the detector, while an untrusted detector assumes that Eve does have access to the detector.

To understand the effects of the number of modes on effective excess noise, we calculate the mode-dependent transmissivity T as,

$$T = \frac{\sum_{mn}^{N} P_{mn,R}}{P_{T,R}}, \tag{6.28}$$

for the reference transmit power $P_{T,R}$. We assume that the power received at the MPLC in $N = 7$ modes is equal to the total power received, while we analyze only the power in the reference to calculate the mode-dependent transmissivity. The average transmissivities for an increasing number of modes (in the order given on the x-axis) are shown in Figure 6.9.

The effective excess noise for a trusted detector ξ_t and untrusted detector ξ_{ut}, are given by,

$$\xi_t = \xi_{ch} T, \quad \xi_{t,cor} = \xi_{ch,cor} T, \tag{6.29}$$

$$\xi_{ut} = \xi_{ch} T + \xi_d, \quad \xi_{ut,cor} = \xi_{ch,cor} T + \xi_{d,cor}, \tag{6.30}$$

where $\xi_{t,cor}$ and $\xi_{ut,cor}$ are the corrected excess noise terms for a trusted and untrusted detector, respectively, after the wavefront corrections are applied. The effective excess noise terms applied in this chapter are modeled on [74]. Note that transmissivity variance does not contribute to effective excess noise, as instantaneous transmissivities are known.

We consider the effect of N on the average effective excess noise $\langle \xi_t \rangle$ for all campaigns, by summing an increasing number of modes. Here, two variables are changing as N increases: the transmissivity and the total relative phase error variance. Overall, the average effective excess noise *increases* as N increases, even though the total relative phase error variance decreases as N increases. For an untrusted detector, the morning campaign is found to have the highest average effective excess noise across all of the summed modes. Using $N = 7$ modes, the noon campaign has the greatest decrease in the average effective excess noise after wavefront correction, reducing from $\xi_{ut} = 0.0254$ SNU to $\xi_{ut,cor} = 0.0251$ SNU. However, average effective excess noise is dominated by the increase in detector noise as N increases for the untrusted detector. Therefore, we focus on a trusted detector for the rest of our analysis of effective excess noise.

Figure 6.10 shows the average effective excess noise for a trusted detector for all campaigns as N increases, by summing an increasing number of modes (in the order given on the x-axis), where the trend of increasing average effective excess noise as N increases can be seen. For a trusted detector, the average effective excess noise is dominated by the increase in transmissivity as N increases. Without considering the total impact of transmissivity on secure key rates in CV-QKD, this would indicate that shaping the RLO with only the HG_{00} mode would result in the best performance, where higher-order orthogonal HG modes should be discarded. However, clearly the signal-to-noise ratio improves as transmissivity increases, leading to increased key rates (e.g. see the secure key rate calculations in Chapter 5).

Implementation of our wavefront correction algorithm, which leads to a reduction in effective excess noise, helps to compensate for the additional effective excess noise caused by the increased transmissivity. For $N = 7$ modes, the effective excess noise was reduced by 0.0002 SNU for the morning campaign, 0.0005 SNU for the noon campaign, and 0.0001 SNU for the evening campaign, representing up to a 17% decrease in the

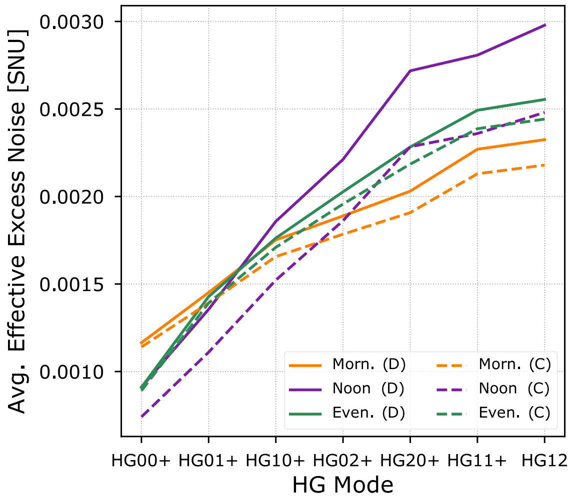

Figure 6.10: Effective excess noise for a trusted detector for accumulating HG modes for the morning, noon, and evening campaigns, where D represents derived (plotted as solid lines) and C represents corrected (plotted as dot-dashed lines).

effective excess noise (for the noon campaign). Therefore, implementation of our wavefront correction algorithms could help mitigate the destructive impact of relative WFEs on effective excess noise, which can have a detrimental effect on secure key rates, in the context of CV-QKD across atmospheric turbulent channels. For example, previous work has shown nearly an order of magnitude increase in secure key rates for excess noise reductions of 15% [147].

Further, our experimental campaign across the 2.4 km FSO link represents only a section of the satellite-Earth links required to develop a truly global Quantum Internet. Assuming that transformation of the transmitted HG_{00} mode into higher-order modes would increase across the satellite-Earth links, our intelligent wavefront correction algorithms could prove vital to reducing effective excess noise to tolerable levels, potentially enhancing secure key rates, or even mitigating secure key destruction (as seen in Chapter 5).

Note that we have not identified the source(s) of the relative WFEs and whether the channel itself is contributing to the observed WFEs. However, it is important to note that identification of the WFE source(s) is not a necessary component of our wavefront correction scheme.

6.7 Summary

In this chapter, we presented experimental evidence of relative wavefront errors between multiplexed optical references and weaker signals, after transmission across a 2.4 km turbulent free-space optical link. We developed machine learning-based intelligent wavefront correction algorithms to correct for the relative wavefront errors, reducing the total relative phase error variance by up to a factor of 2/3. The impact of such a phase error reduction on theoretical excess noise, in the context of continuous-variable quantum key distribution, was presented, showing that an order-of-magnitude increase in quantum key distribution secure key rates is potentially achievable using these machine learning-based corrections.

In the final chapter, we present the conclusions from the results and analyses undertaken in this thesis, where machine learning has been shown to assist in the practical implementation of free-space optical continuous-variable quantum key distribution across turbulent atmospheric channels.

Conclusion

This thesis explored several applications of machine learning (ML) in assisting free-space optical (FSO) continuous-variable quantum key distribution (CV-QKD) across turbulent atmospheric channels. With a focus on reducing phase noise in CV-QKD, ML was able to reduce CV-QKD hardware complexity, provide low-complexity signal phase error estimation, and correct for relative wavefront errors (WFEs) between optical references beacons and signals, both via simulation and using experimental data. Overall, given the findings in this thesis, we suggest that future CV-QKD protocols should employ ML algorithms to mitigate the potential negative impact of phase noise on CV-QKD performance, thereby supporting the practical implementation of a global CV-QKD network.

7.1 Summary

In Chapter 2, the literature on ML-assisted CV-QKD was surveyed, where ML was found to augment several core challenges in CV-QKD, including phase error estimation, excess noise estimation, state discrimination, parameter estimation and optimization, key sifting, and key rate estimation. The survey showed that ML was generally found to outperform traditional algorithms designed for the same challenges. Overall, ML was demonstrated as providing valuable assistance in the practical application of CV-QKD. A critical discussion of the literature was then undertaken, where key research gaps were identified, motivating the research undertaken in this thesis.

In Chapter 3, a convolutional neural network (NN) was designed to estimate phase wavefront corrections to apply to a real local oscillator (RLO), using only the intensity distribution of reference pulses as input. Positive key rates were achieved for several satellite-to-Earth channels, thereby mitigating the need for complex wavefront sensing hardware. Implementation of the developed ML algorithms could hasten the practical implementation of a global quantum communication network.

In Chapter 4, a NN complexity analysis was undertaken, where the scale of long short-term memory NN-based signal phase error estimation algorithms was compared with estimation performance, using the quantum Cramér-Rao bound as a benchmark. Results showed that relatively low-complexity NNs could perform satisfactory estimation performance across satellite-to-Earth channels. As such, ML could prove to be a vital resource in achieving real-time relative phase error correction, reducing phase noise in practical FSO CV-QKD systems, hence progressing the establishment of a global Quantum Internet.

In Chapter 5, an analysis of the potential sources of relative WFEs in RLO-based FSO CV-QKD protocols was presented, and several cases were simulated for satellite-to-Earth channels. The effects of relative WFEs on secure key rates were explored, which were found to have a deleterious effect on key rates, even leading to null key rates in some scenarios. ML-based intelligent wavefront correction algorithms were designed to correct for the relative WFEs, resulting in enhanced secure key rates, thus increasing the practical feasibility of developing a satellite-based CV-QKD network.

In Chapter 6, the results from a series of experimental campaigns were presented, where relative WFEs between a polarization-multiplexed optical reference beacon and a weaker optical signal were found to exist in every campaign, after transmission across a 2.4 km FSO link. ML-based intelligent wavefront correction algorithms were designed to correct for the relative WFEs, which resulted in a relative phase variance reduction of up to 2/3. The contributions of the derived relative WFEs on excess noise were investigated, in the context of FSO CV-QKD, where relative WFEs could represent a major contribution to phase noise if left uncorrected. The intelligent wavefront correction algorithms, therefore, could prove crucial to the realization of a truly global Quantum Internet.

7.2 Suggested Future Work

Given that ML-assisted CV-QKD remains a burgeoning field of research, there is currently a wide scope for future work, considered beyond the scope of this thesis. In terms of ML algorithm types and structures, there has been limited contrasting of different ML algorithms used for the same application thus far, making it difficult to know whether the ML algorithms used in the current literature could have been improved upon with alternative state-of-the-art ML algorithms. For example, no classification-based NN models have been used to estimate discretely-modulated CV-QKD states, which could potentially match or improve upon k-nearest neighbor or Bayesian-based models.

Works with ML architectures constructed using physics-informed architectures remain limited. For example, the application of the Schrödinger equation as the activation function in NNs could be further explored to determine if it can more effectively model quantum signal evolution, such as applying the recurrent quantum NN designed in [33] to relative phase error correction in Gaussian modulated coherent state CV-QKD.

While ML has shown potential to enhance several aspects of CV-QKD, its application needs to be carefully scrutinized with respect to the introduction of security vulnerabilities (e.g. [55], [56]), where existing proofs may no longer hold. Further, ML also has potential for QKD protocol selection (as was investigated in [148] for discrete-variable QKD) and could additionally be expanded to protocol optimization and design.

Focusing on the use of ML for FSO CV-QKD and relative phase error correction (the primary focus of this thesis), there remain several important directions for future work, in pursuit of the development of a global CV-QKD network.

Experimental implementation of FSO CV-QKD has thus far been limited to reported channel lengths of up to 9.6 km [149]. Therefore, real-world implementation of CV-QKD across longer channels could be demonstrated, which would help to establish the feasibility of developing a practical FSO CV-QKD network. To that effect, while discrete-variable QKD has been achieved over satellite-Earth links [150], [151], CV-QKD has yet to be performed over a true satellite-Earth link, the realization of which would represent a major milestone in the development of a Quantum Internet.

In relation to the experimental work undertaken in this thesis, there remains a possible next step. Our collaborators at the International Centre for Radio Astronomy Research (ICRAR) have a background clearly centered on astronomical observation, with their foundations in radio astronomy. The Astrophotonics Group within ICRAR has expanded into optical fiber-based phase synchronization, free-space laser timing and positioning, and free-space laser communications (e.g. [152], [153]). However, our collaboration represents their first step into the realm of CV-QKD. As such, due to budgetary constraints, we were primarily limited to using pre-existing optical hardware for classical laser communications, where achieving quantum signal-level preparation and detection was beyond the current infrastructure at our disposal. We assume that the relative WFEs between the polarization-multiplexed optical reference beacon and weaker optical signal were representative of those that would occur between a reference pulse and a quantum pulse. Nonetheless, due to the greater difference in optical powers, there could be alternative, or even greater, sources of relative WFEs when using a very weak quantum signal. For example, higher photon leakage between the reference pulses and quantum signals could have an effect on the relative WFEs. Therefore, we note that future experiments could be undertaken to explore any additional effects a truly quantum signal could have on relative WFEs, in the context of FSO CV-QKD, where very weak optical pulses (comprised of up to tens of photons) are multiplexed with strong optical reference pulses.

Further experiments could also be designed to isolate the source(s) of relative WFEs, which would require a detailed methodical approach, particularly to determine if the atmospheric turbulence itself contributes to the relative WFEs. Experiments could also be undertaken under varying conditions, such as varying atmospheric turbulence strengths, varying channel lengths, and varying environmental conditions (such as atmospheric temperature and pollutant composition). Understanding how different conditions affect relative WFEs could inform future wavefront correction algorithm design. Ultimately, relative WFEs could be investigated over true satellite-Earth links, where wavefront correction algorithms could prove vital in reducing the phase noise affecting the quantum signals.

A complete RLO-based FSO CV-QKD setup could be established, where the speed and precision of a control and actuation system (e.g. an adaptive optics unit), used to modulate the wavefront of an RLO, could further be taken into account. A feedback loop between the control and actuation system and the wavefront correction algorithms could help to improve the RLO wavefront imprinting accuracy, thereby increasing coherence between the RLO and quantum signals.

Finally, in a practical FSO CV-QKD setup, a NN designed for relative phase error correction should be capable of adapting to any channel conditions. As such, a very large dataset of relative phase errors could be collected over a very long period of time, which could be used to train the algorithm. This is particularly true for low-Earth orbit satellite-Earth links, given the rapidly changing channel conditions as the satellite orbits the Earth. Overall, gathering very large and varied datasets would help to remove unwanted bias once the model is trained, and improve the robustness of the relative phase error correction in real-world CV-QKD implementations.

Appendix A

Quantum Optics

This appendix outlines the fundamental quantum optics theory underpinning the foundations of quantum information encoding on quantized light. We begin by providing a brief discussion on the quantization of the electromagnetic field, followed by the derivation of Fock states, ending with a description of coherent states. Coherent states are the basis used to encode secure key information in Gaussian modulated coherent state continuous-variable quantum key distribution protocols (analyzed throughout this thesis). The following quantum optics theory is adapted from [98], [154]. Note that the nomenclature used in this appendix applies solely to the content within this appendix.

In the following analysis in this appendix, **bold font** represents a vector, $\mathbf{r}$ represents the spatial coordinates $\mathbf{r} = \{x, y, z\}$, and t represents time. The vector potential $\mathbf{A}(\mathbf{r}, t)$ of a free electromagnetic field satisfies the electromagnetic wave equation,

$$\nabla^2 \mathbf{A} - \frac{1}{c^2}\frac{\partial^2 \mathbf{A}}{\partial t^2} = 0, \tag{A.1}$$

where c is the speed of light. The electric field $\mathbf{E}(\mathbf{r}, t)$ can be derived as a function of $\mathbf{A}(\mathbf{r}, t)$ as,

$$\mathbf{E}(\mathbf{r}, t) = -\frac{\partial}{\partial t}\mathbf{A}(\mathbf{r}, t), \tag{A.2}$$

while the magnetic field $\mathbf{B}(\mathbf{r}, t)$ is calculated as $\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t)$. Considering a finite (though very large) cubic cavity with volume $V = L^3$, $\mathbf{A}(\mathbf{r}, t)$ can be described using the Fourier expansion of plane-wave modes as,

$$\mathbf{A}(\mathbf{r}, t) = \sum_{\mathbf{k}} \mathbf{A}_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{A.3}$$

The cubic cavity imposes periodic boundary conditions, which restricts $\mathbf{k}$ to,

$$\mathbf{k} = \left\{ \frac{2\pi l}{L_x}, \frac{2\pi m}{L_y}, \frac{2\pi n}{L_x} \right\}, \tag{A.4}$$

with $\{l, m, n\} \in \{0, \pm 1, \pm 2, \cdots\}$. Considering the Coulomb gauge, we can determine that each of the Fourier series expansion amplitudes $\mathbf{A_k}(t)$ can be solved by,

$$\left(\frac{\partial^2}{\partial t^2} + \omega^2 \right) \mathbf{A_k}(t) = 0, \tag{A.5}$$

where ω represents the angular frequency of the harmonic mode. By considering only a single polarization $\hat{\mathbf{e}}$, a solution to Eq. A.5 is,

$$\mathbf{A}(\mathbf{r}, t) = \sum_{\mathbf{k}} \hat{\mathbf{e}} \mathcal{A}_{\mathbf{k}} \left[a_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{r}} + a^*_{\mathbf{k}}(t) e^{-i\mathbf{k}\cdot\mathbf{r}} \right], \tag{A.6}$$

for the mode-specific coefficient $\mathcal{A}_{\mathbf{k}}$ (determined by the initial conditions of the electromagnetic field), and $a_{\mathbf{k}}$ is the amplitude for the mode $\mathbf{k}$. Note that $a_{\mathbf{k}}(t) = a_{\mathbf{k}} e^{-i\omega t}$ and $a^*_{\mathbf{k}}(t) = a^*_{\mathbf{k}} e^{i\omega t}$. Therefore, the electric field can be defined as,

$$\mathbf{E}(\mathbf{r}, t) = \sum_{\mathbf{k}} \omega \hat{\mathbf{e}} \mathcal{A}_{\mathbf{k}} \left[i a_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{r}} - i a^*_{\mathbf{k}}(t) e^{-i\mathbf{k}\cdot\mathbf{r}} \right]. \tag{A.7}$$

If we consider the electromagnetic field as a system of independent harmonic oscillators, the Hamiltonian H of the field can be derived as,

$$H = 2 \sum_{\mathbf{k}} |\mathcal{A}_{\mathbf{k}}|^2 \omega^2 |a_{\mathbf{k}}(t)|^2. \tag{A.8}$$

The pair of canonical variables $q_{\mathbf{k}}(t)$ and $p_{\mathbf{k}}(t)$ can also be defined using $a_{\mathbf{k}}(t)$ and $a^*_{\mathbf{k}}(t)$ as,

$$q_{\mathbf{k}}(t) = a_{\mathbf{k}}(t) + a^*_{\mathbf{k}}(t), \tag{A.9}$$

$$p_{\mathbf{k}}(t) = -i\omega [a_{\mathbf{k}}(t) - a^*_{\mathbf{k}}(t)], \tag{A.10}$$

such that the Hamiltonian can be defined as a function of $q_{\mathbf{k}}(t)$ and $p_{\mathbf{k}}(t)$ as,

$$H = \sum_{\mathbf{k}} \frac{1}{2}\left[p_{\mathbf{k}}^2(t) + \omega^2 q_{\mathbf{k}}^2(t)\right] = \sum_{\mathbf{k}} H_{\mathbf{k}}. \tag{A.11}$$

Here, $H_{\mathbf{k}}$ is the Hamiltonian of the $\mathbf{k}$-th harmonic oscillator. Note the commutation relation $[q_{\mathbf{k}}(t), p_{\mathbf{k}'}(t)] = \delta_{\mathbf{k}\mathbf{k}'}$, where the Kronecker delta function $\delta_{\mathbf{k}\mathbf{k}'}$ returns 1 for $\mathbf{k} = \mathbf{k}'$ or 0 for $\mathbf{k} \neq \mathbf{k}'$.

By considering the quantized electromagentic field, we replace the canonical variables $q_{\mathbf{k}}(t)$ and $p_{\mathbf{k}}(t)$ with their quantum mechanical counterparts, the "position" and "momentum" canonical conjugate operators, $\hat{q}_{\mathbf{k}}(t)$ and $\hat{p}_{\mathbf{k}}(t)$, respectively, which can be used to describe the $\mathbf{k}$-th quantized mode of radiation. The pair of non-Hermitian operators, the annihilation operator $\hat{a}_{\mathbf{k}}(t)$ and creation operator $\hat{a}_{\mathbf{k}}^{\dagger}(t)$, can be defined in terms of $\hat{q}_{\mathbf{k}}(t)$ and $\hat{p}_{\mathbf{k}}(t)$ as,

$$\hat{a}_{\mathbf{k}}(t) = \frac{1}{\sqrt{2\hbar\omega}}[\omega\hat{q}_{\mathbf{k}}(t) + i\hat{p}_{\mathbf{k}}(t)], \tag{A.12}$$

$$\hat{a}_{\mathbf{k}}^{\dagger}(t) = \frac{1}{\sqrt{2\hbar\omega}}[\omega\hat{q}_{\mathbf{k}}(t) - i\hat{p}_{\mathbf{k}}(t)], \tag{A.13}$$

where $\hbar$ is the reduced Planck constant. Note the commutation relation $[\hat{q}_{\mathbf{k}}(t), \hat{p}_{\mathbf{k}'}(t)] = i\hbar\delta_{\mathbf{k}\mathbf{k}'}$.

The number operator $\hat{n}_{\mathbf{k}}$ counts the degree of excitation in a harmonic oscillator, with the relation $\hat{n}_{\mathbf{k}} = \hat{a}_{\mathbf{k}}(t)^{\dagger}\hat{a}_{\mathbf{k}}(t)$. Alternatively, in a discrete-variable system, the number operator counts the number of photons in the system, where each photon adds an extra degree of excitation. The annihilation operator can be applied to a state to reduce the degree of excitation by one ($-\hbar\omega$), while the creation operator can be applied to a state to increase the degree of excitation by one ($+\hbar\omega$).

The quantum mechanical Hamiltonian $\hat{H}$ of a harmonic oscillator system can be calculated by summing all of the system modes using $\hat{a}_{\mathbf{k}}(t)$ and $\hat{a}_{\mathbf{k}}^{\dagger}(t)$ as,

$$\hat{H} = \sum_{\mathbf{k}} \hat{H}_{\mathbf{k}} = \sum_{\mathbf{k}} \frac{1}{2}\hbar\omega[\hat{a}_{\mathbf{k}}(t)\hat{a}_{\mathbf{k}}^{\dagger}(t) + \hat{a}_{\mathbf{k}}^{\dagger}(t)\hat{a}_{\mathbf{k}}(t)] = \sum_{\mathbf{k}} \hbar\omega[\hat{a}_{\mathbf{k}}^{\dagger}(t)\hat{a}_{\mathbf{k}}(t) + \frac{1}{2}] = \sum_{\mathbf{k}} \hbar\omega[\hat{n}_{\mathbf{k}} + \frac{1}{2}]. \tag{A.14}$$

That is, the Hamiltonian is a sum over the photon number operators of each mode, multiplied by the energy in each photon, then adds the vacuum fluctuations. An expression for a quantized electric field $\hat{\mathbf{E}}(\mathbf{r}, t)$ can then be derived as an operator using $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^{\dagger}$ as,

$$\hat{\mathbf{E}}(\mathbf{r}, t) = \sum_{\mathbf{k}} \hat{\mathbf{e}}_k \mathcal{E}_{\mathbf{k}} \left[i\hat{a}_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{r}} - i\hat{a}_{\mathbf{k}}^{\dagger}(t) e^{-i\mathbf{k}\cdot\mathbf{r}} \right], \tag{A.15}$$

where $\mathcal{E}_{\mathbf{k}}$ represents a mode-specific weighted coefficient. The electric field operator is usually split into two components as,

$$\hat{\mathbf{E}}(\mathbf{r}, t) = \sum_{\mathbf{k}} \hat{\mathbf{e}}_k i \mathcal{E}_{\mathbf{k}} \hat{a}_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{r}} + \sum_{\mathbf{k}} \hat{\mathbf{e}}_k (-i) \mathcal{E}_{\mathbf{k}} \hat{a}_{\mathbf{k}}^{\dagger}(t) e^{-i\mathbf{k}\cdot\mathbf{r}} = \hat{\mathbf{E}}^{(+)}(\mathbf{r}, t) + \hat{\mathbf{E}}^{(-)}(\mathbf{r}, t), \tag{A.16}$$

such that $\hat{\mathbf{E}}^{(+)}(\mathbf{r}, t)$ and $\hat{\mathbf{E}}^{(-)}(\mathbf{r}, t)$ are functions of $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^{\dagger}$, respectively. Note that the subsequent analysis in this appendix will focus on a single harmonic oscillator, thus, the subscript $\mathbf{k}$ will be dropped.

The Fock state $|n\rangle$ represents a mode of n photons or, alternatively, the n-th degree of excitation of a harmonic oscillator, being the eigenstate of $\hat{n}$ with an eigenvalue of n. As such, the variance (fluctuations) of a Fock state equates to zero. We can define $|n\rangle$ by the successive application of $\hat{a}^{\dagger}$ as,

$$|n\rangle = \frac{(\hat{a}^{\dagger})^n}{(n!)^{1/2}} |0\rangle, \tag{A.17}$$

where $|0\rangle$ is the vacuum state. When the Hamiltonian is applied to $|n\rangle$, the energy of the mode is given as $\hat{H}|n\rangle = \hbar\omega(n + 1/2)|n\rangle$, where the term $\hbar\omega(n + 1/2)$ represents the eigenenergy of the state $|n\rangle$.

Coherent states $|\alpha\rangle$ are normalized, quasi-orthogonal eigenstates of the annihilation operator,

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \tag{A.18}$$

where the eigenvalue α can be written in terms of an amplitude and phase as $\alpha = ae^{i\phi}$. A coherent state can be defined as a superposition of different excitation states n and the complex amplitude α as,

$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \tag{A.19}$$

The average number of photons in a coherent state is calculated as $\bar{n} = \langle\alpha|\hat{n}|\alpha\rangle = |\alpha|^2$. Further, coherent states, unlike Fock states, have a non-zero variance (fluctuations) as they represent a superposition of different Fock states, where,

$$(\Delta n)^2 = \langle\alpha|\hat{n}^2|\alpha\rangle - \langle\alpha|\hat{n}|\alpha\rangle^2 = |\alpha^2| = \bar{n}. \tag{A.20}$$

The probability distribution $P(n)$ of finding n photons in $|\alpha\rangle$ is given by a Poisson distribution,

$$P(n) = \langle n|\alpha\rangle\langle\alpha|n\rangle = \frac{\bar{n}^n}{n!}e^{-\bar{n}}, \tag{A.21}$$

which approximates the photon number distribution of laser light.

A quantum state can be represented in the phase space as a function of the annihilation and creation operators by considering the quadratures of an electromagnetic field. The Hermitian quadrature operators can be calculated as a function of $\hat{a}$ and $\hat{a}^\dagger$ as,

$$\hat{X} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger) \propto \hat{q}, \tag{A.22}$$

$$\hat{P} = \frac{1}{2}(\hat{a} - \hat{a}^\dagger) \propto \hat{p}. \tag{A.23}$$

The commutator of $\hat{X}$ and $\hat{P}$ evaluates to $[\hat{X}, \hat{P}] = i/2$, showing that $\hat{X}$ and $\hat{P}$ do not commute. The probability distributions of measuring $\hat{X}$ and $\hat{P}$ in the phase space are Gaussian, where their standard deviations ΔX and ΔP are defined by,

$$\Delta X = \sqrt{\langle 0|\hat{X}^2|0\rangle - \langle 0|\hat{X}|0\rangle^2} = \frac{1}{2}, \tag{A.24}$$

$$\Delta P = \sqrt{\langle 0|\hat{P}^2|0\rangle - \langle 0|\hat{P}|0\rangle^2} = \frac{1}{2}. \tag{A.25}$$

The relation between the standard deviations of $\hat{X}$ and $\hat{P}$ is then $\Delta X \Delta P \geq \frac{1}{4}$, where a value of 1/4 represents the minimum uncertainty state (the shot noise limit).

As coherent states are simply vacuum states which have been displaced on the $\{X, P\}$-axes in the phase space, a *displacement* operator must be introduced to shift the probability distribution from an origin of zero, as,

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle, \tag{A.26}$$

where the displacement operator can be defined as,

$$\hat{D}(\alpha) = e^{\alpha \hat{a}^{\dagger} - \alpha^{*} \hat{a}}. \tag{A.27}$$

The displacement from a vacuum state of a coherent state $|\alpha\rangle$ can be seen in Figure A.1, where the magnitude of the displacement is equal to $|\alpha|$ at an angle ϕ from the X quadrature axis.

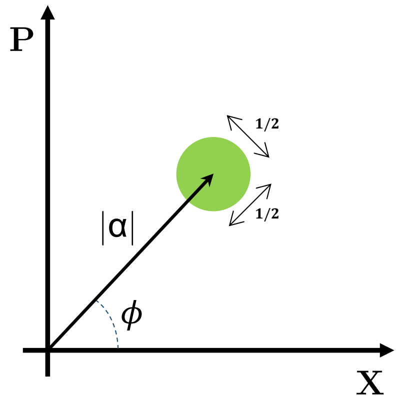

Figure A.1: Phase space representation of a coherent state $|\alpha\rangle$, with the amplitude $|\alpha|$ and phase ϕ, showing the X and P quadrature variances of $1/2$.

Coherent states are used as the basis for encoding secure key information in Gaussian modulated coherent state continuous-variable quantum key distribution protocols, as analyzed throughout this thesis.

Appendix B

Phase Screen Model Validation for Optical Beam Propagation Across Atmospheric Channels

Atmospheric turbulence causes turbulent eddies of varying sizes and velocities to develop, of varying refractive index structure parameters, resulting in fluctuating interactions with optical beams propagating through them. Atmospheric free-space optical (FSO) experiments are time-consuming and complex to run, therefore, modeling the effects of atmospheric turbulence on optical beam propagation is essential for developing new optical communication methods and technologies. Note that the nomenclature used in this appendix applies solely to the content within this appendix.

Analytical models (e.g. [155], [156]) have been derived to describe turbulence-induced phase errors and intensity variations of an optical beam passing through a channel, providing a first approximation of the effects of atmospheric turbulence. Taking the analytical models a step further, phase screen models have been developed to simulate the distortion of optical beams through atmospheric channels (e.g. [91], [114]), with the aim of providing higher fidelity approximations than analytical models. This work focuses on the use of a Fresnel split-step phase screen model, based on [90], [113].

The work [113] developed a phase screen model using Monte-Carlo simulations with a Fresnel split-step propagator. Essentially, a series of phase screens are used to represent a turbulent atmospheric volume by assuming that an optical beam travels as if in a vacuum until interacting with a phase screen at its center. The screen perturbs the phase and intensity of the beam wavefront as it passes through it. Turbulence induced phase perturbations $\Delta\Phi_{ps}(x, y)$ at each phase screen are derived in Cartesian coordinates $\{x, y\}$

using the Fourier series [113],

$$\Delta\Phi_{ps}(x,y) = \sum_{n=-\infty}^{\infty}\sum_{m=-\infty}^{\infty} a_{nm}\, \exp[i2\pi(f_{x_n}x + f_{y_m}y)], \tag{B.1}$$

with f_{x_n} and f_{y_m} being discrete frequencies in the $\{x, y\}$ plane. The complex Fourier series coefficients a_{nm} are sampled from Gaussian distributions with zero mean and unit variance, multiplied by a sample frequency grid and the square root of the modified Von Kármán refractive power spectral density (PSD) $S_{\phi,mVK}(\kappa)$, given by [113],

$$S_{\phi,mVK}(\kappa) = 0.49 r_0^{-5/3}\left[\frac{\exp(-\kappa^2/\kappa_m^2)}{(\kappa^2+\kappa_0^2)^{11/6}}\right], \tag{B.2}$$

for the Fried parameter r_0 and the angular spatial frequency $\kappa = 2\pi(f_x^2 + f_y^2)^{1/2}$ in rad/m, such that $a_{nm} \propto \sqrt{S_{\phi,mVK}}$. Turbulent eddies are defined by their outer scale L_0 and inner scale l_0. The outer scale represents the upper bound of the eddie size, incorporated in Eq. B.2 via the term $\kappa_0 = 5.92/L_0$, which dissipate into smaller eddies, reaching the lower limit of the inner scale, incorporated into Eq. B.2 via the term $\kappa_m = 2\pi/l_0$, where we set $l_0 = 0.005L_0$. Below the inner scale, turbulence dissipates into the atmosphere as thermal energy.

The Fried parameter is a measure of coherence length, over which wavefront phase distortions remain within 1 rad [91]. Assuming that the phase screens are spaced evenly across the channel length L_{ch}, the Fried parameter is calculated as $r_0 = (0.423k^2C_n^2L_{ch})^{-3/5}$. The refractive index structure parameter of turbulence C_n^2 provides a quantitative measure of the fluctuations in atmospheric optical turbulence [91], in units of $\mathrm{m}^{-2/3}$. Higher values of C_n^2 lead to greater beam wavefront distortion. See [113] for a more in-depth explanation of the phase screen model used in this thesis.

We compare our phase screen simulation results with experimental results from the University of Western Australia in Perth, Australia, across an 11.6 km retroreflected horizontal channel. Turbulent refractive index structure parameter and outer scale estimations are calculated using analytical turbulence models fit to the experimental data [136], which are used as inputs to the phase screen model. Validating the phase screen simulation model increases confidence in our results when conducting research into the effects

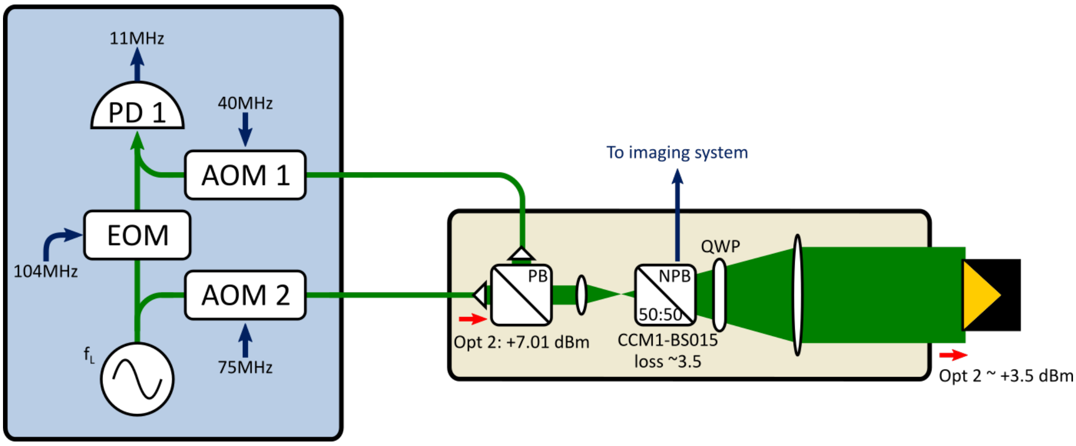

Figure B.1: System schematic of FSO experiments by UWA across horizontal channel. EOM is the electro-optical modulator, PD is the photodetector, AOM is the acoustic-optical modulator, QWP is the quarter waveplate, and Opt represents the optical signal.

of atmospheric turbulence on optical beam propagation in the context of CV-QKD.

Table B.1 details the experimental setup used at UWA. Phase and power fluctuations experienced over the atmospheric channel are obtained using the Mach-Zehnder interferometer shown in Figure B.1. The light from a 1552 nm NKT X15 laser is frequency shifted 75 MHz up by an acousto-optic modulator (AOM) before being launched over the atmospheric channel. After passing over the retroreflected link, the received light is frequency shifted up by a 40 MHz AOM. A heterodyne beat is then formed with the short arm of the Mach-Zehnder interferometer, which has been phase modulated at 104 MHz by an electro-optic modulator. The phase modulation on the short arm of the interferometer creates sidebands evenly spaced by 104 MHz. The heterodyne beat between the first upper sideband (f_L+104 MHz) and the returned optical signal (f_L+115 MHz) form a beat centered on 11 MHz and modulated by the atmospheric phase and amplitude fluctuations.

The free-space terminal operates bi-directionally, launching and receiving the retroreflected optical beam with the same primary optic (⌀45.7 mm clear aperture). All fibers in the interferometer are polarization-maintaining, and the launch and receive collimators are oriented orthogonally. A quarter waveplate later in the optical chain ensures that the retroreflected light is orthogonally polarized to the transmitted light. This allows a polarizing beamsplitter to direct the retroreflected light to the second collimator. A 50:50 non-polarizing beamsplitter is used to split off a portion of the retroreflected light for an imaging system used for alignment and imparts a $\sim$3.5 dB loss in each direction. The collimated beam launched over the free-space link is Gaussian with a 35.10 mm e^{-2} diameter. The CCR at the far end of the link has a clear aperture of 63.5 mm.

Table B.1: Horizontal channel experimental parameters.

Parameter	**Value**
Channel length (L_{ch})	11.6 km
Laser wavelength (λ)	1552.52 nm
Transmitter diameter (D_t)	45.7 mm
Receiver diameter (D_r)	45.7 mm
Retroreflector diameter (D_{ccr})	63.5 mm
Transverse wind speed (v_y)	2.6 m/s

Estimation of channel conditions can be undertaken by comparing the experimental phase error PSD with theoretical turbulence models [136]. PSDs describe how power in temporal (or spatial) data is distributed across frequency components, which can be estimated via discrete Fourier transformation (DFT) [157].

Specifically, we estimate C_n^2 by comparing the phase error PSD from the aforementioned experiments with the Kolmogorov phase error PSD model [155], [158] for turbulent atmospheric channels, then estimate the outer scale L_0 using the estimated C_n^2 and the Greenwood-Tarazano phase error PSD model [155], [159]. Phase errors represent the "zeroth-order" disturbances of the beam as it passes through the atmosphere.

The single-sided Komogorov turbulence model phase error PSD, determined as a function of C_n^2, describes the phase error spectrum for a plane wave propagating across a one-way channel, calculated as [155],

$$S_{\phi,Kol}^{\rightarrow}(f) = 0.0033k^2 C_n^2 L_{ch} v_y^{5/3} f^{-8/3}, \tag{B.3}$$

where f is the Fourier frequency in Hz and v_y is the tangential wind speed in m/s. Note that we assume that the wind is moving only in the y-axis. Similarly, the single-sided phase error PSD for the Greenwood-Tarazano turbulence model, determined as a function of C_n^2 and L_0, is given by [155],

$$S_{\phi,GT}^{\rightarrow}(f) = 5.211k^2 C_n^2 \frac{L_{ch}}{v_y} \int_0^{\infty} \left[q_y^2 + \left(\frac{2\pi f}{v_y}\right)^2 + \frac{\sqrt{\frac{2\pi f}{v_y^2} + q_y^2}}{L_0} \right]^{-\frac{11}{6}} dq_y, \tag{B.4}$$

where q_y is the spatial frequency orthogonal to the beam path, $q_y = 2\pi f_y/v_y$. Note that while Eqs. B.3 and B.4 are derived for a plane wave, ignoring the aperture averaging and subtleties associated with Gaussian beam propagation, the same scaling applies to the phase PSDs for Fourier frequencies below the aperture diameter for Gaussian beams [155].

However, given that the beam propagates through a two-way channel, retroreflecting back from the CCR, phase errors are reciprocal, thus the total phase errors become $\Delta\phi^{\leftrightarrow}(t) = 2\Delta\phi^{\rightarrow}(t)$, such that two-way the phase error PSDs $S_{\phi}^{\leftrightarrow}$ become,

$$S_{\phi}^{\leftrightarrow}(f) = 4S_{\phi}^{\rightarrow}(f), \tag{B.5}$$

for $f < 1/T$ for the round trip time $T = L_{ch}/c$, where c represents the speed of light.

Figure B.2 shows the experimental phase error PSD results. First, to estimate C_n^2, we use a non-linear least squares fit for the Kolmogorov PSD to the experimental data within the frequency range $f_{Kol} \in [0.09\text{Hz}, 0.50\text{Hz}]$, seen as the solid black line in Figure B.2. Then, to estimate the outer scale, we use a non-linear least squares fit for the Greenwood-Tarazano PSD for the experimental data within the range $f_{GT} \in [0.005\text{Hz}, 0.50\text{Hz}]$, where the characteristic downward inflection is present at lower frequencies, plotted as the dotted blue line in Figure B.2.

The average C_n^2 for the horizontal channel is estimated to be $C_n^2 = 1.51 \times 10^{-15}$ m$^{-2/3}$, while the outer scale is estimated to be $L_0 = 20.29$ m (using the parameters in Table B.1).

In order to compare the phase screen model to the UWA experimental data, approximately 200,000 instances of a Gaussian beam propagating through the channel described in Table B.1 are simulated using the phase screen model. Given that the phase screen model does not include the time-domain, the measurements are assumed to be taken during different coherence times. Therefore, we sampled the experimental phase error data for independent coherence times τ_0, calculated as,

$$\tau_0 = 0.314\frac{r_0}{v_y^{5/3}}, \tag{B.6}$$

where the coherence time was found to be $\tau_0 \approx 0.04$ s, based on the estimated $C_n^2 = 1.51 \times 10^{-15}$ m$^{-2/3}$.

To validate the phase screen model, a comparison of the beam power distortion across

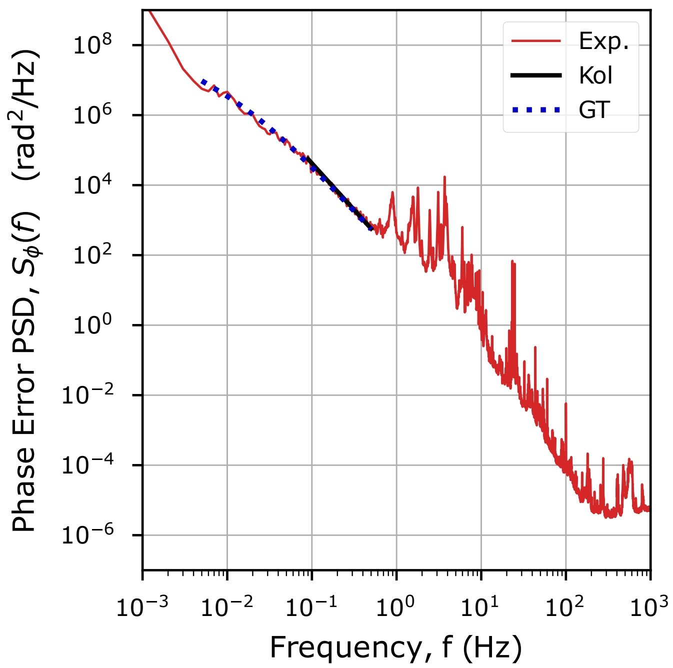

Figure B.2: Phase error PSD for horizontal FSO experiments (red line) and phase screen simulation (blue line). Kolmogorov PSD fit to $f_{Kol} \in [0.09\text{Hz}, 0.50\text{Hz}]$ (solid black line) for refractive index structure parameter C_n^2 estimation, and Greenwood-Tarazano PSD fit to $f_{GT} \in [0.005\text{Hz}, 0.50\text{Hz}]$ (dashed black line) for outer scale L_0 estimation.

the channel is undertaken for both the phase screen model and experimental data (as undertaken in [74]). The scintillation index (SI) σ_I^2 represents the normalized intensity I variance of a beam wavefront after passing through an atmospheric volume,

$$\sigma_I^2 = \frac{\langle I^2 \rangle}{\langle I \rangle^2} - 1, \tag{B.7}$$

where $\langle\rangle$ represents the ensemble average for each measurement across the $\{x, y\}$ plane of the receiver aperture. Alternatively, an approximation of the SI can be made by comparing the normalized total power across the receiver diameter $\mathcal{D}_R$, calculated as,

$$P = \iint_{\mathcal{D}_R} |I(x, y)|^2 \, dxdy, \tag{B.8}$$

for the average power across all measurements $\langle P \rangle$, resulting in $P/\langle P \rangle$.

Figure B.3 gives the probability density function (PDF) for the normalized power $P/\langle P \rangle$ for both the horizontal FSO experimental data (red filled histogram) and the phase screen model simulation results (blue hollow histogram). As can be seen, the normalized power distributions closely align between the phase screen model and experimental data, indicating that the simulated intensity distortions closely match the experimental results, supporting the validity of the phase screen simulation model used in Chapters 3 to 5 in this thesis.

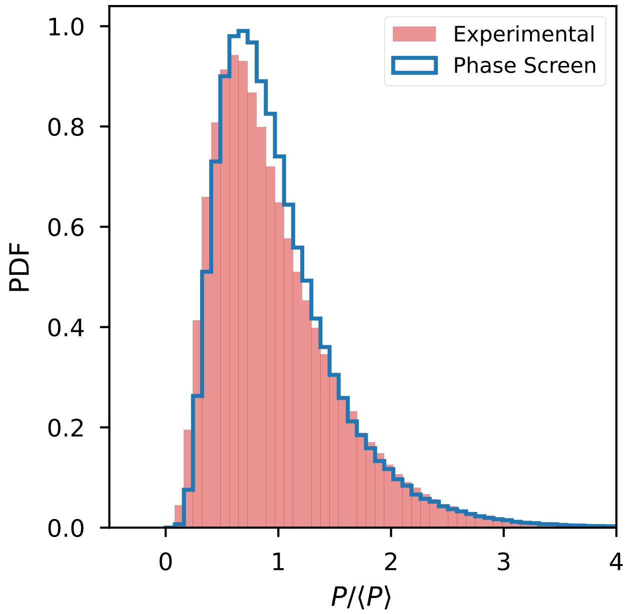

Figure B.3: Normalized power PDF comparison for horizontal FSO experimental results (red filled histogram) and phase screen simulation results (blue hollow histogram), using the estimated C_n^2 and L_0 values for the phase screen model.

Appendix C

Preliminary Experiments on Relative Wavefront Errors in Free-Space Optical Communication

In this appendix, we present the results from a series of preliminary laboratory experiments to expose the presence of relative wavefront errors (WFEs) in an experimental setup, which approximates a continuous-variable quantum key distribution setup. A weak optical signal and strong local oscillator (LO) are generated as fundamental Gaussian modes using the same laser at 1550 nm, then attenuated to $P_s = 0.16$ μW and $P_{lo} = 0.27$ mW, respectively. The signal and LO are then frequency-shifted by 1 kHz and polarization-multiplexed, before transmission across a free-space optical channel (see Figure C.1). Note, although the weak signal is not a true quantum signal — we make the reasonable assumption that any relative WFE found in our experiment will not be negated when the photon number in the weak signal is reduced. A current is run through a nichrome wire, generating heat using electrical power in the 46 cm long channel for four different conditions, with increasing turbulence strength: Case 0 (no power), Case 1 (power = 5.4 mW), Case 2 (power = 12.2 mW), and Case 3 (power = 21.5 mW). The signal and LO are split at the receiver, aligned in polarization, then measured using a multi-plane light converter (MPLC).

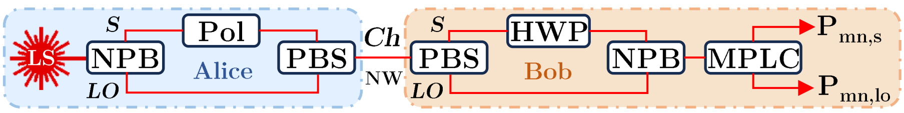

Figure C.1: Experimental setup. LS is a laser source, NPB is a non-polarizing beamsplitter, Pol is a polarizer, PBS is a polarizing beamsplitter, Ch is the channel, NW is the nichrome wire, HWP is a half-waveplate, MPLC is a multi-plane light converter, S is the signal path and LO is the LO path. $P_{mn,s}$ and $P_{mn,lo}$ are the signal and LO powers.

The MPLC decomposes the signal and LO electric fields into the Hermite-Gaussian

(HG) basis using a series of phase masks, coupling each mode into a single-mode fiber in the HG_{00} mode (see [119]). We measure the first eight modes (at a sample rate of 15 kSa/s) using individual photodetectors, so that the voltage measured, V_{mn}, is proportional to the power in each mode, P_{mn} (i.e. $V_{mn} \propto P_{mn}$). The signal and LO form a heterodyne beat, so the signal powers $P_{mn,s}$ and LO powers $P_{mn,lo}$ can be demultiplexed [160].

To quantify relative WFEs, we take the difference ΔP_{mn} between the normalized signal mode power $P_{mn,s}/P_s$ and the LO mode power $P_{mn,lo}/P_{lo}$ in each mode,

$$\Delta P_{mn} = (P_{mn,s}/P_s) - (P_{mn,lo}/P_{lo}).$$

If the wavefronts of the signal and the LO are the same, then $P_{mn,s}/P_s = P_{mn,lo}/P_{lo}$, so $\Delta P_{mn} = 0$, as is commonly assumed. Otherwise, there is evidence of the existence of relative WFEs.

Time is plotted against ΔP_{mn} for each of the turbulence cases, over a period of 30 s, in Figure C.2. It can be immediately seen that *relative WFEs are present* between the signal and the LO ($\Delta P_{mn} > 0$) for all cases. Given that relative WFEs are present in Case 0, they are likely caused by imperfections or inaccurate calibration of optical hardware [5]. It can be also seen in Figure C.2 that ΔP_{mn} is highest for the HG_{01} mode, followed by the HG_{00} mode, then the HG_{10}, HG_{11}, and HG_{02} modes (layered on top of each other). We also see that the HG_{20}, HG_{21}, and HG_{12} modes approach zero (layered on top of each other). In general, the fluctuations in ΔP_{mn} for each mode remain relatively constant for Case 0, indicating that the WFEs are relatively constant.

From Figure C.2, the same ordering of the modes, from the lowest to the highest ΔP_{mn} values, is found for the turbulent Cases 1-3. However, it can be seen that the ΔP_{mn} fluctuations become noisier as the turbulence strength increases, which we quantify in Figure C.3 using the variance of ΔP_{mn}. Although the initial WFEs are likely caused by imperfections or imprecise calibration of the optical hardware, the increase in variances of ΔP_{mn} across all modes, as turbulence increases, indicates that the WFEs are affected by the turbulence itself. In addition, we find that the WFEs in each mode are statistically significant by calculating a two-sample Kolmogorov–Smirnov test [161] using the distributions $P_{mn,s}/P_s$ and $P_{mn,lo}/P_{lo}$. Our results show that the differences in all

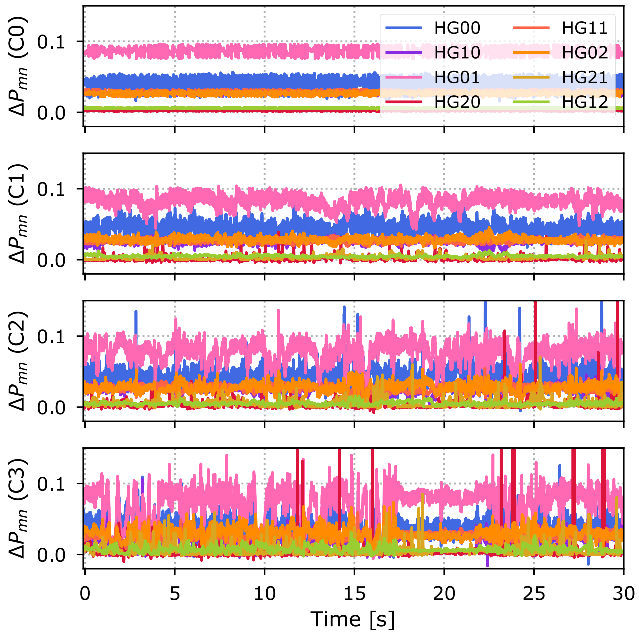

Figure C.2: Time versus ΔP_{mn} for turbulence Cases 0-3 (where C represents Case).

distributions are statistically significant, with a 99.99% confidence, which supports the existence of relative WFEs.

We note that our experimental setup may not represent all CV-QKD setups, as we adopt path differences for the signal and LO, and apply a small wavelength perturbation for co-measurement of the signal and LO using the MPLC. However, our experiments highlight how relative WFEs may occur in real CV-QKD setups. This result is fundamental to CV-QKD across atmospheric channels; if imperfections in optical hardware and turbulence cause fluctuating relative WFEs, then correcting for them could prove vital in attaining positive key rates across free-space channels. The deleterious effects of relative WFEs on free-space CV-QKD would be particularly pronounced in the future satellite-Earth links required for a global Quantum Internet. To mitigate the effect of WFEs in CV-QKD, this thesis explores a machine learning-based solution to correcting WFEs in Chapters 5 and 6, which can lead to lower excess noise and higher key rates across atmospheric channels.

In this appendix, we have provided preliminary experimental evidence of the presence of relative wavefront errors between a polarization-multiplexed strong classical local oscillator and a weak optical signal after passing both through turbulent channels. In addition, we demonstrated how the turbulence strength affects these relative wavefront errors. The

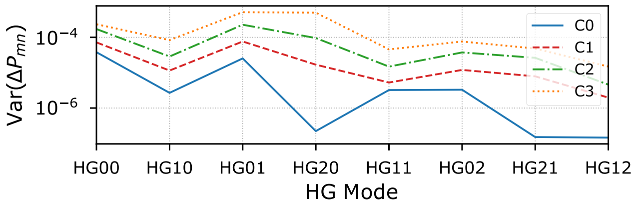

Figure C.3: HG mode versus ΔP_{mn} variance for turbulence Cases 0-3 (where C represents Case).

presence of relative wavefront errors in continuous-variable quantum key distribution setups has the potential to negatively impact secure key rates, necessitating the development of wavefront correction methods to establish a global Quantum Internet, where we explore machine learning-based wavefront correction algorithms in Chapters 5 and 6.

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